User Manual **MN152258EN**

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Power Distribution Monitoring & Control powered by PXG1000 User Guide



Powering Business Worldwide

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1. Introduction

The Power Distribution Monitoring & Control (PDMC) powered by PXG1000 provides a graphical web interface to monitor, control, and diagnose Eaton switchgear from a safe location. It provides features required to maintain a switchgear assembly such as:

- Rich interface to monitor parameters of all devices and study the trends of those parameters
- Ability to configure and monitor alarms for devices
- Remotely open/close and enable/disable Arc flash Reduction Maintenance Systems (ARMs) on circuit breakers through control mode
- Configure, monitor, and control for various Main and Tie Breaker source arrangements

2. Connecting the PXG1000 hardware to a PC

When connecting the Gateway for the first time, the network 1 (LAN B) and network 2 (LAN A) ports will be configured in private mode by default with static IP configuration as below:

Network 1 (LAN B) port: static (manual):

- ipv4_address: 169.254.1.1
- subnet_mask: 255.255.0.0
- gateway: 169.254.0.0

Network 2 (LAN A) port: static (manual) private network:

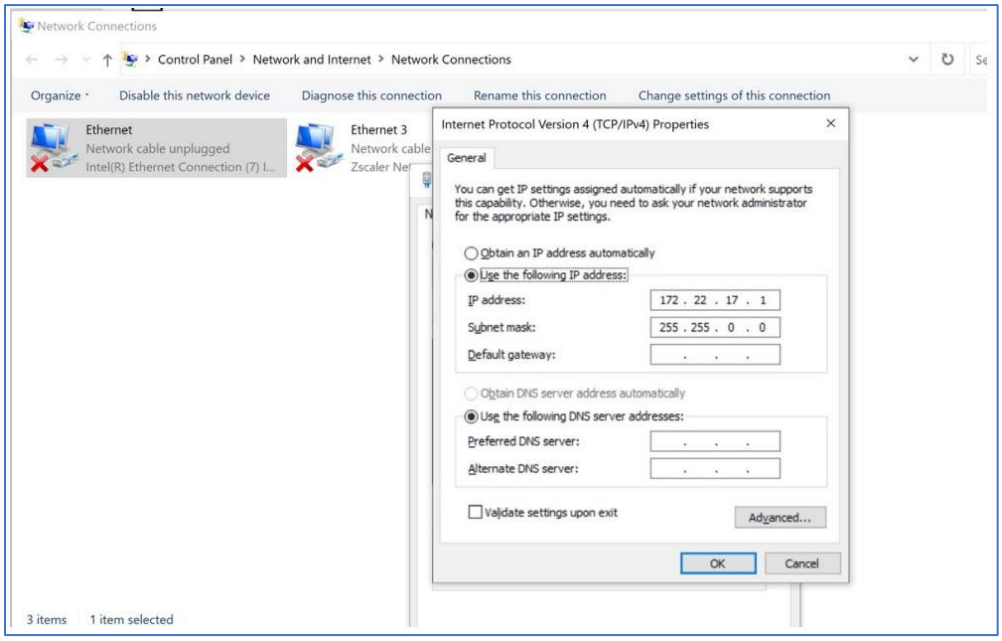
- ipv4_address: 172.22.17.254
- subnet_mask: 255.255.0.0

The default admin password is the 'Last 6 characters of the product_serial_number + last octet of the LAN B (Network 1) MAC address'. For example, for the product serial number: KSD0300581 and LAN B (Network 1) MAC address: C4:00:AD:C0:C4:0A, the password would be 300581+0A = 3005810A

To load the application user interface, the user should connect a PC/laptop to the gateway's network 2 (LAN A) port. The IP address of PC should be in similar subnet of network 2 (LAN A) (i.e., 172.22.17.x). User can change the network 1 (LAN B) IP settings.

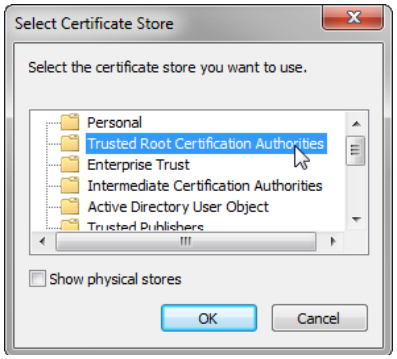
Note: Note: When changing the IP address of the gateway, we recommend recording the new address for future access.

Configure the IP address and Subnet mask settings for PC's ethernet port from **Network & Internet Settings > Ethernet > Change Adapter Options > Select Ethernet port (1/2/3) > Select protocol as, Internet Protocol Version 4 (TCP/IPv4) > Properties > Edit** as shown below:



Installing the certificate

1. Double-click the certificate file to launch the certificate installation wizard.
2. Click **Install certificate**.
3. On the Welcome dialog box, click **Next**.
4. Select Place all certificates in the following store and then click **Browse**.
5. Select **Trusted Root Certification Authorities** from the list and then click OK. "Trusted Root Certification Authorities" store is not the default, administrator privileges (or elevated/additional privileges) may be required to manage certificates.
6. Upon completing the certificate import wizard dialog box, click **Finish**.
7. You should see an alert box stating that the import was successful. Click **OK**.
8. Close and restart the browser. You can now access the PXG1000 using the HTTPS protocol.



3. Configuring Device Communications

Devices are connected, configured, and tested at the factory before shipping. New devices can be added, or existing devices can be moved through the **Setting > Devices** screen.

- You can configure the PDMC's Modbus serial ports, Modbus TPC/IP port, on the **Settings > Devices > Device Tree** tab. Choose Edit to change any of the port configurations. You can access the port settings by choosing the appropriate gear icon.
- When adding devices with a lineup having Enhanced graphic license, use **Setting > Devices > Elevation**. If the user doesn't have enhanced graphic license, devices can be added on the **Settings > Devices > Device Tree** tab.
- Before connecting the PXG1000 to a twisted pair Modbus, make sure that you observe the proper polarity when wiring the cable to one of the black & orange ten-pin connectors provided. The PXG1000 RS-485 ports are clearly marked for the D+ (B+), D- (A-) and GND (common) connections. The PDMC modeling tool can provide estimated communication performance estimates for the individual communication ports. The ports have the following device limits

COM1 port	32 devices, supports Modbus RTU devices
COM2 port	32 devices, supports Modbus RTU devices.
Network 2 (LAN A)	20 devices, supports Modbus TCP devices.
Network 3 (LAN B)	60 devices, supports Modbus TCP devices when Private Network is enabled.

3.1 Modbus Serial Ports

You can set Modbus Ports 1 and 2 individually through their respective gear icons. Set the serial port parameters for your Modbus RTU network. You should only change the time out settings if necessary.

3.2 Network 2 and Network 3 (Ethernet)

Most of the Network 1 (LAN B) and Network 2 (LAN A) settings are configured on the **Settings > Network** tab. With Edit Mode enabled choose the gear icon to access the network settings. The Network 1 (LAN B) IP address setting can be viewed on the Device Tree page. When Network 2 (LAN A) is set to Private Network, you will be able to view and change the Network 2 (LAN A) IP address here. Devices added to the Network 2 (LAN A) private subnetwork are accessed by the PDMC via Modbus TCP on this private subnet. These devices will not be accessible from the main Network 1 (LAN B) subnet. Don't change the time out setting unless this is necessary. When Network 2 (LAN A) is set to Private Network the first 3 octets of the Network 1 (LAN B) and Network 2 (LAN A) IP addresses must differ by at least 1 digit.

3.3 Adding Devices

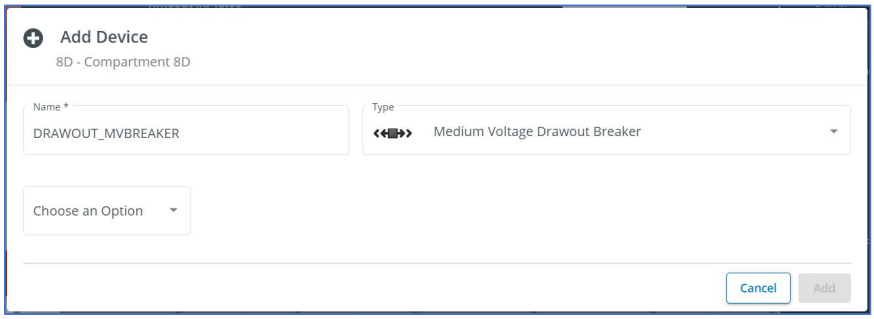
After you've properly configured the serial ports and connected the PXG1000 to your serial networks, you can begin adding any supported devices currently on that serial network. Adding devices, like port configuration, is done on the **Settings > Devices > Device Tree** tab. The preferred location for adding devices is the **Settings > Devices > Elevation** screen (Enhanced license required). It is possible to add devices on the Device Tree tab, but compartment information must still be entered to add a device. This is easier on the Elevation screen.

3.4 Initial Setup

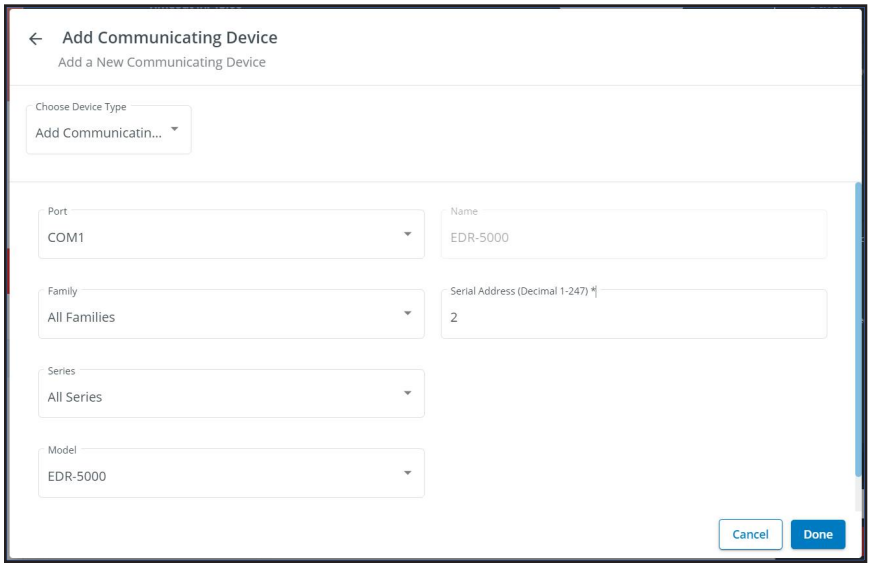
The PDMC helps you when adding devices, guiding you through the initial setup steps.

To add a Modbus RTU device, if Enhanced Graphic license is available, use below steps:

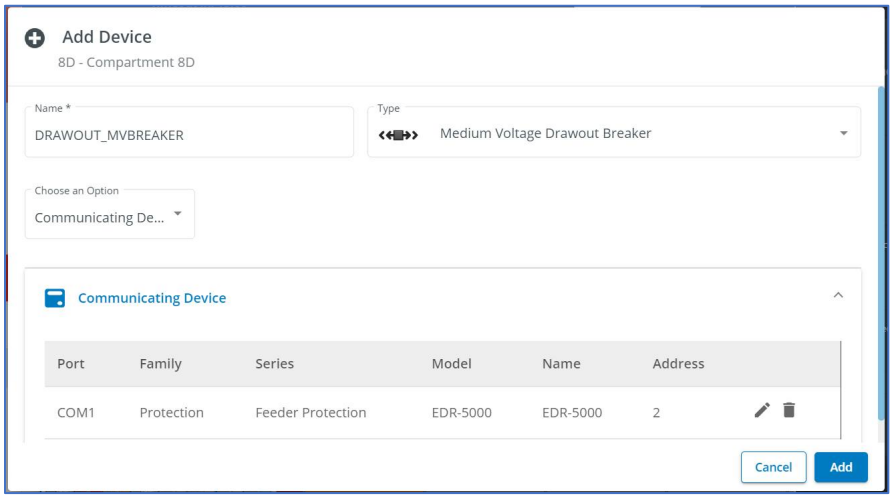
1. Choose Edit on **Settings > Devices > Elevation Screen**.
2. Add section if not already added.
3. Sidebar will be opened on click of compartment in which device needs to be added.
4. Choose “Add Device” button.



5. Select Type, enter name and select “Communicating device” option from “Choose an Option” dropdown.
   - On “Add Communicating Device Modal”
   - Select the device type under Family.
   - Select the Model Series.
   - Select the device Model.
   - Enter a meaningful name for the device.
   - Enter the device address: 1-247 decimal for Modbus. If you enter an address that’s already in use, the box outline will turn red, and you won’t be able to save the device configuration until you select a unique address.
   - If Enable Waveforms is available, decide whether you wish to have the PDMC capture these.
   - Click “Done”



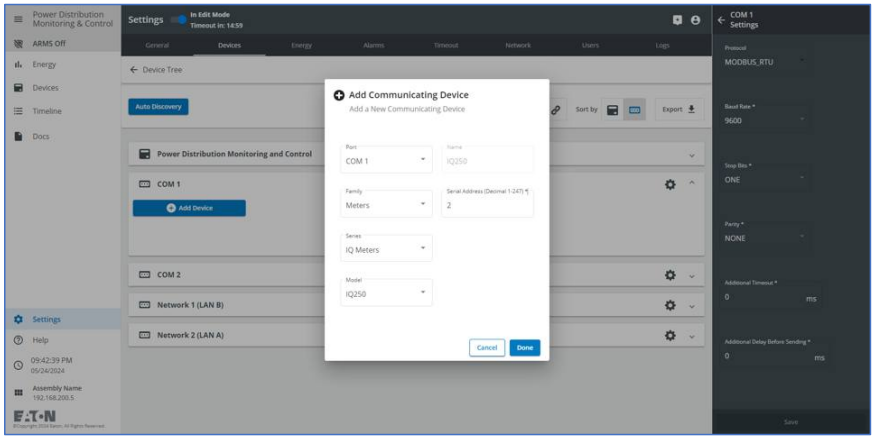
6. On “Add Device” Modal, click “Add” button to save the configuration.



The following examples assume that the devices are on the appropriate networks and that the ports are properly configured. If Enhanced Graphic license is unavailable, add device from **Settings > Devices > Device Tree**.

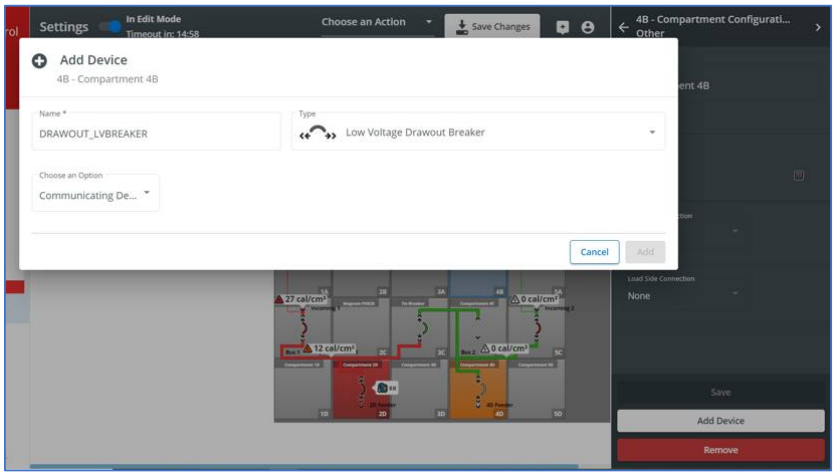
Example 1: Adding an IQ 250 Meter

1. Connect the IQ 250 Meter to either COM1 or COM2.
2. Choose the **Settings > Devices Tree** tab.
3. Choose Edit.
4. Choose Add Device under the appropriate COM port column.
5. Select Meters under Family.
6. Select IQ Meters under Model Series.
7. Select IQ250 under Model.
8. Set the Serial Address (1-247 decimal) to match the address you set on the meter.
9. Choose “Done”.

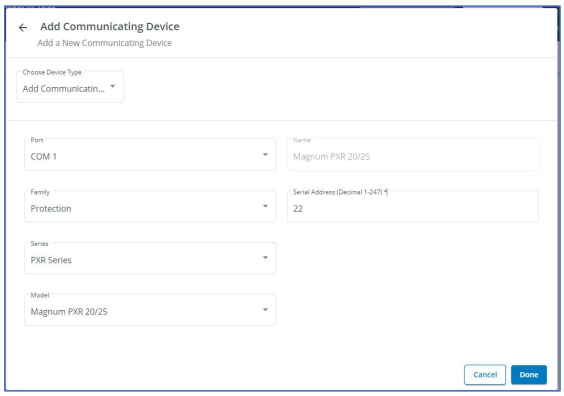


Example 2: Adding a Magnum PXR trip unit to LVA lineup from Settings > Devices > Elevation (Enhanced Graphic license is required)

1. Connect the Magnum PXR trip unit to either COM1 or COM2.
2. Choose Edit on **Settings > Devices > Elevation Screen**.
3. Add section if not already added.
4. Sidebar will be opened on click of compartment in which device needs to be added.
5. Click “Add Device” button.
6. On “Add Device” Modal, select Type, enter name and select “Communicating device” option from “Choose an Option” dropdown.



7. On “Add Communicating Device Modal”
   - Select appropriate port.
   - Select Protection under Family.
   - Select PXR under Series
   - Select Magnum PXR 20/25 Model
   - Set the Serial Address (1-247 decimal) to match the address you set on the Magnum PXR
   - Click “Done”



8. On “Add Device” Modal, click “Add” button to save the configuration.

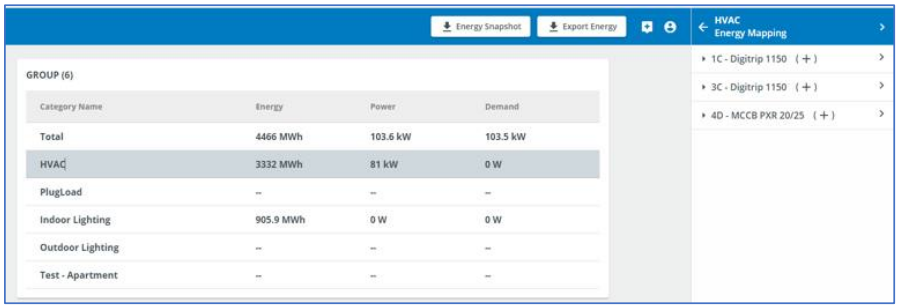
4. Energy

Power Distribution Monitoring & Control accumulates energy reported by devices assigned to 8 different categories.

- User can view aggregated values for Energy, Power, and Demand channels for each category on the Main Energy screen.
- After clicking on category name, sidebar gets opened which displays the list of devices and their respective channels contributions to accumulated Energy, Power, and Demand.
- On Click of More button, Trend data of all the Energy Grouping channels can be seen.
- The trend configuration can be changed for below:
  - Actual data for Energy channel
  - Average, Min & Max data for Power and Demand channels can be trended.
  - Trend data can be exported for all trended channels for selected time interval and time duration.
  - The trend data (of 15min interval) for energy grouping is retained for 3 years.



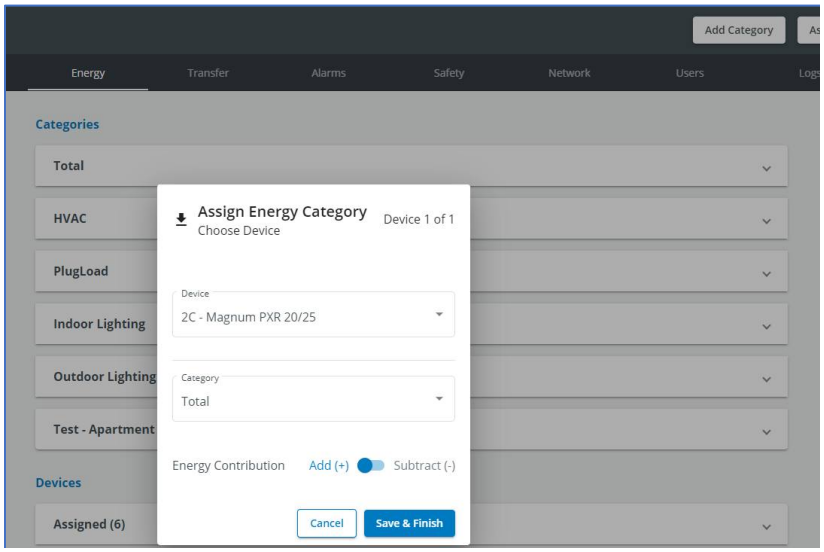
- The energy snapshot can be exported in .csv format. This will include the accumulated Energy, Power and Demand for all categories captured at that instant.
- The energy grouping configuration is included in gateway configuration which is exported in .cfg file.



4.1 Settings Energy

From **Settings > Energy**, each device can be assigned an energy category. The energy contribution can either be added or subtracted.

- Devices available for assignment must have one of the energy parameters available.



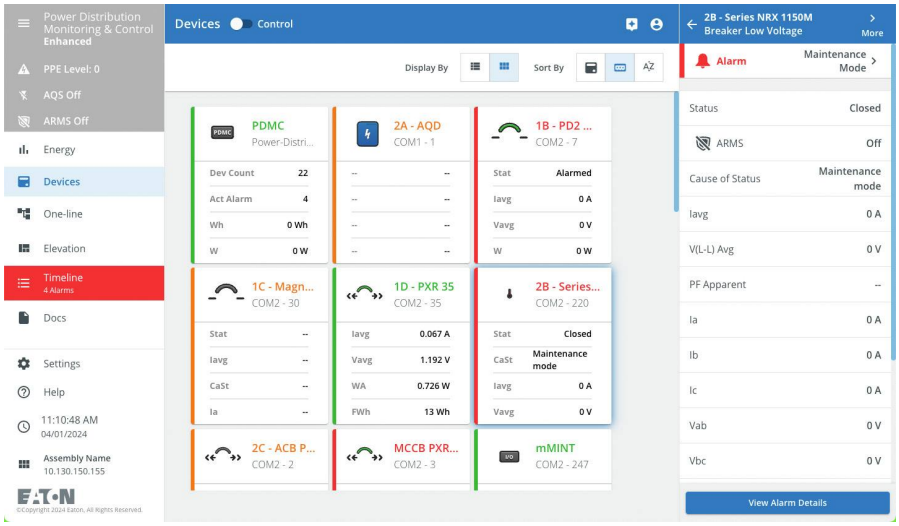
- The accumulated energy data is sent upstream via MODBUS TCP, BACnet, SMTP and MQTT publisher (The energy data is sent to Campus View gateway using MQTT protocol).
- There are five standard energy categories (as per electrical standard ASHRAE 90.1) to monitor electrical energy. The following categories names can be changed:
  - Total electrical energy
  - HVAC systems
  - Interior Lighting
  - Exterior Lighting
  - PlugLoad
- User can add custom category apart from standard ones and can add up to three custom categories:
  - UserCategory_1
  - UserCategory_2
  - UserCategory_3

5. Devices (Card View)

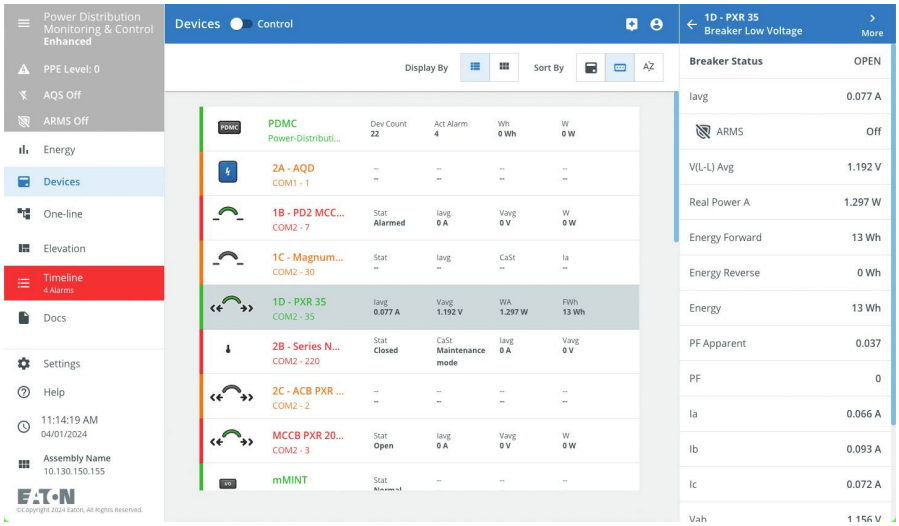
The PDMC offers a card view of all the configured devices with top 4 channels in it. All devices can also be shown as a list view. Devices can be sort by Device Model, COM Port or Name.

- Action bar holds major two options:
  - Display as (List view / Card view)
  - Sort By (Device Model / COM port / Name)
- Color of the left side strip and device name of the card shows the state of the device:
  - **Red** color signifies that the device is in the alarmed state.
  - **Orange** color signifies that the device is in a non-communicating state.
  - **Green** color signifies that the device is in the communicating state.
- When a device is selected in either Card or List view it provides a sidebar with the top 16 channels of a device, its ARMs status (in case of breaker), and any active alarms.
- The 'More' button in the sidebar navigates to device details view with device health, parameters, alarms, trends etc.

5.1 Card View



5.2 List View

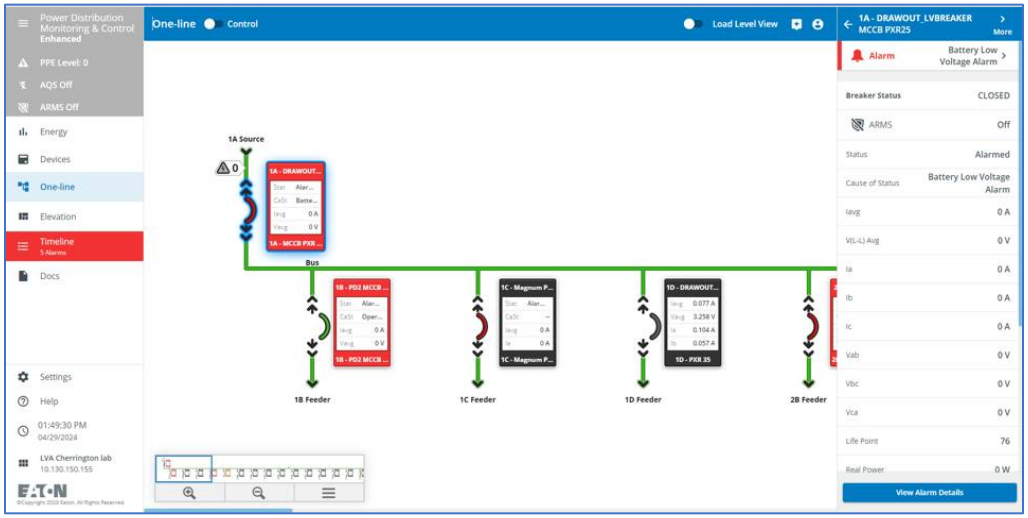


- Control mode allows an authorized user to perform control operations, such as Opening/Closing breakers and Enabling/Disabling ARMS, remotely.*
- Control operations can be performed from the pocket at the bottom of the device sidebar.

6. One-line View

The One-line view is a graphical representation of the single line power flow diagram for the switchgear lineup. It shows the electrical connection of the configured power system devices.

- Each device is represented by a symbol and a widget showing the top 4 parameters for the device.
- The color of the widget outline shows the state of the device:
  - A widget highlighted in **Red** color signifies that the device is in the alarmed state.
  - A widget highlighted in **Orange** color signifies that the device is in a non-communicating state.
  - A widget highlighted in **Black** color signifies that the device is in the communicating state.
- When a widget is selected it provides a sidebar with the top 16 parameters for the device, its ARMs status (in case of breaker), device status, and any active alarms.



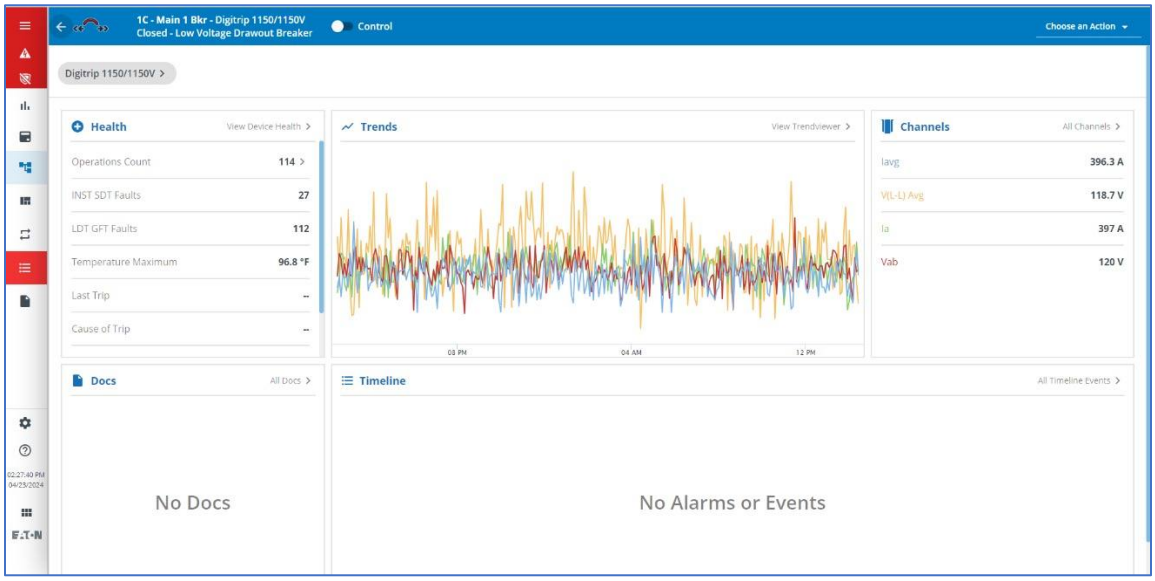
- The “More” button at the top of the sidebar navigates to a Device Details Stack view with Device Health card, Channels card, Timeline card, Trends card, Document card. The “Choose an Action” Dropdown in the action bar provides access to all the supported Control Commands for the device.

6.1 Load Level View

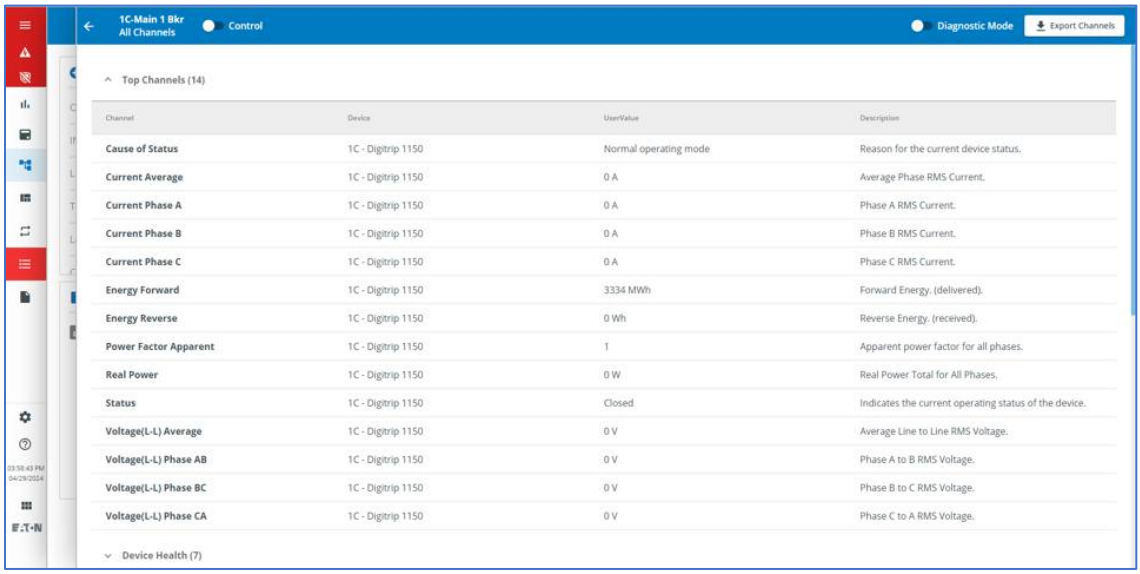
The Load Level View on the One-line screen provides visualization of the average measured current compared to the High Load Alarm and the Long Delay Pickup settings.

1. On click of Load Level View Toggle the Widget will flip to show a graph of the average current compared to the pickup setpoints.
2. Widget shows the Long delay pickup (Ir) value In Progress bar filled with gray background.
3. Widget shows the IAvg value in the Progress bar filled with Green/Yellow/Red Background:
   - Green – Iavg < High Load Alarm limit
   - Yellow – Iavg > High Load Alarm and < Ir
   - Red – Iavg > Ir (Long Delay Pickup Setpoint)
4. Load level values on widgets are visible only for communicating devices.
5. For load level widget view sidebar shows below values -
   - Iavg
   - Peak Current Demand (if available)
   - Long delay pickup (Ir)

- High load Alarm setpoint



- The 'More' button on the Channels card navigates to All Channels Stack view. All Channels from the device. Channels are organized into groups by electrical measurement category. The Export Channels button generates a pdf file containing a list of all the device channels with a snapshot of the values.



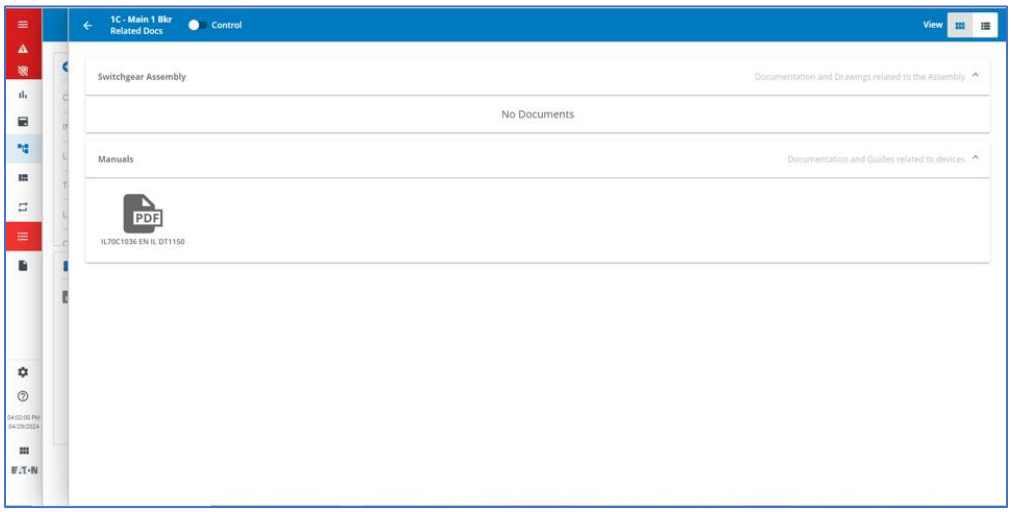
- The 'More' button on the Trends card navigates to Trend Viewer Stack view
- Choose Add Channel to select a channel from the device. Channels are arranged in categories. Clear a check box to remove a trend line for that channel. To zoom In, select the Zoom button from Action bar and Click the Zoom Out button to return to the default view. Export Trends saves a .csv file with trend data from that device. For Export Trends you can either export data from the top 16 channels or all channels.

To pan, select the Pan button and then click the left or right arrows. There are a few other controls:

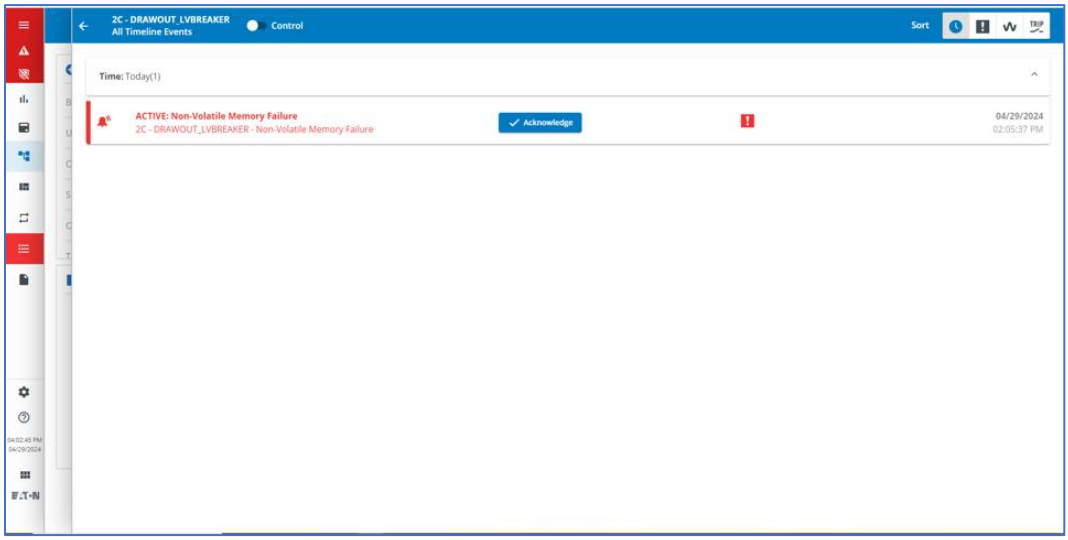
- Use the calendar control to select the date or time range.
- Place the cursor over any point on the graph to see its value and time stamp.

- Click the Export Chart button to save a .png file snapshot of the graph to your local file system.

- The 'More' button on the Docs card navigates to Document Stack view



- The 'More' button on the Timeline card navigates to the Timeline Stack where alarms associated with the selected device are present. Selecting an alarm will open the Alarm Details page showing the detailed alarm history.



7. Elevation View

- The **Elevation** view shows the front view of the switchgear lineup with the status of breakers and buses.
- A **Red** colored compartment signifies that the device is in an alarmed state and an Orange colored compartment indicates that the device is in a non-communicating state.



- If a component, such as a compartment, section, device, or wire, is clicked, the user is provided with a sidebar displaying detailed information about that component.

7.1 Downstream Assemblies Feature

1. On click of the Downstream Assembly toggle, breakers which are feeding PDMC gateways will show the downstream assembly symbol such as switchgear in the respective compartment.
2. If user clicks on switchgear symbol, sidebar will get open which will show details of the downstream assembly PDMC gateway.
3. Navigation to Downstream Assembly
   - User can navigate to PDMC downstream assemblies. On sidebar bottom Login button will be available. On click of that confirmation pop up will be opened. Click Ok button to confirm the navigation to downstream assembly.
   - Once user is navigated to downstream assembly, “Back to Main Assembly” button will be available in the left side menu panel, to navigate back to the feeding assembly. User will be auto logged in to Main assembly.
   - On session timeout of downstream assembly user will be auto logged out from downstream assembly and will be navigated back to main assembly with auto login.
4. Configure Downstream Assembly
   - For LV/MV assembly type:
     - On “Add Device” Modal, once communicating device has been added user can select “Add Load” option.
     - On “Add communicating Device” Modal, select family, series, and model.
     - Downstream assemblies can be added only on Network 2 & Network 3 port.
     - Enter valid device IP address and unit id as 2. Click “Done” button.
     - On “Add Device” Modal, click “Add” button to save the configuration.
   - For LVMCC Assembly type:
     - Add an option as “Assembly load” on “Add load” step in wizard flow.
     - If user has selected “Assembly load”, then in the next step user will get an option to configure downstream assembly.

7.2 Control Feature

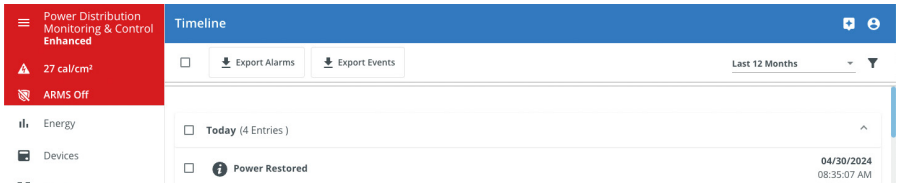
- Control mode allows an authorized user to perform control operations, such as Opening/Closing breakers, racking breakers in and out, and Enabling/Disabling ARMS, remotely.
- Control mode can be entered from either the One-line or Elevation view.
- When **Control Mode** is enabled, it is indicated by slanted blue lines.
- Control operations can be performed from the pocket at the bottom of the device sidebar.

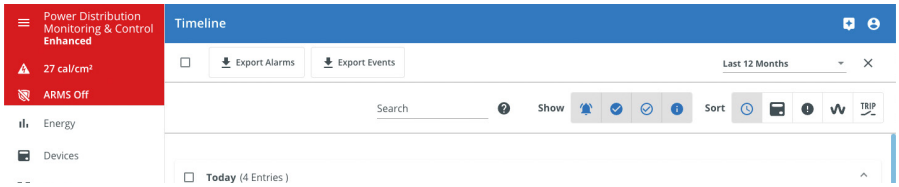
If any breakers are in ARMS mode, the ARMS On indication will appear in the upper left Safety button, a blue shield with the word **On** will appear beside the breaker symbol, and the bus below the breaker with ARMS enabled..



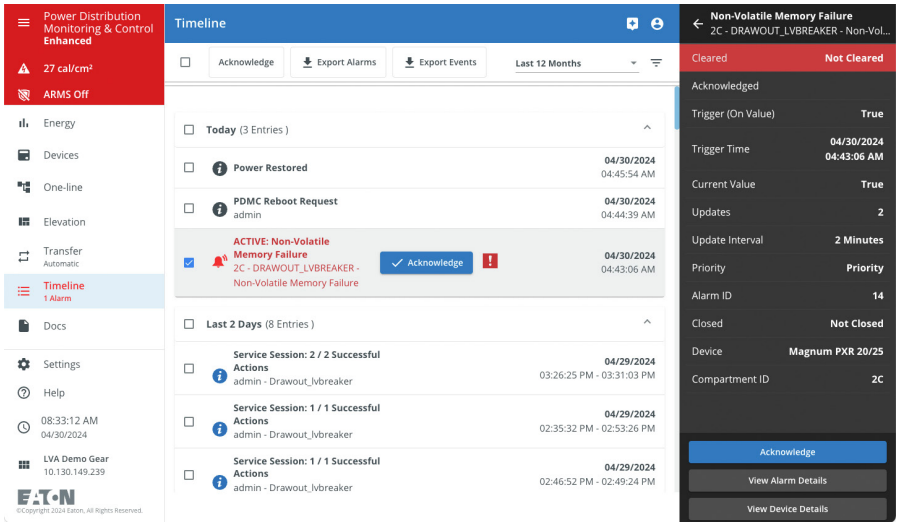
8. Timeline

- The timeline tab provides a consolidated view of active / unacknowledged / acknowledged alarms and control operation events.
- The timeline tab, if highlighted in **Red** color, shows the number of active alarms present in the configured lineup.
- An alarm entry, if clicked, opens a sidebar that provides more information and allows the user to acknowledge the alarm.
- Multiple alarms can be acknowledged at the same time. This is accomplished by clicking the **"Acknowledge"** button that becomes visible in the Action bar if multiple unacknowledged alarms are selected.
- Options for sorting and filtering alarms are available by selecting the filter funnel on the top right in the action bar. The description of the icons can be viewed by hovering over the question mark for help. There is also a search option to search for specific device or alarm data.



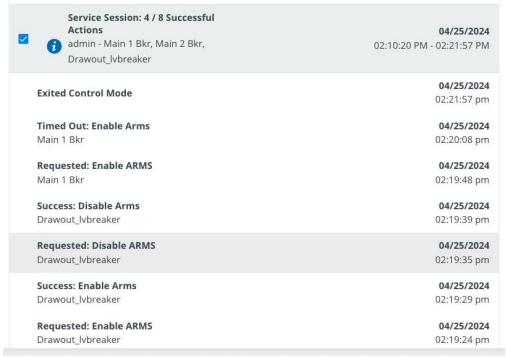


- Waveform and Trip logs are also captured in the Timeline. To view a waveform from a Trip event, navigate to the **Alarm Details** of the trip logged event.

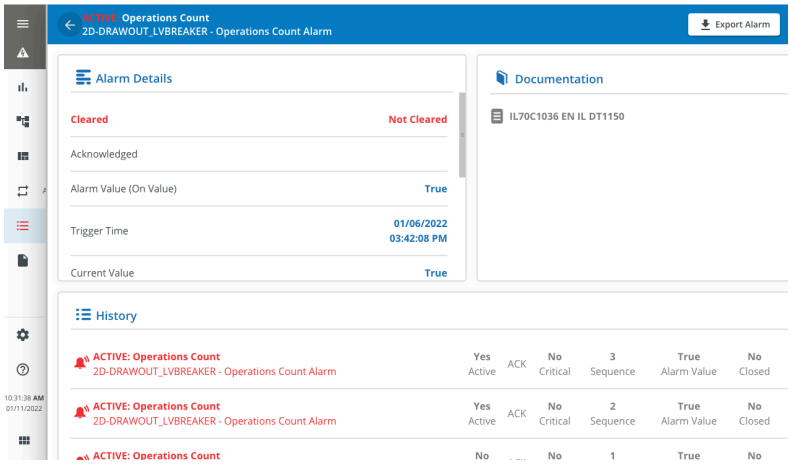


- When the user performs Control Operations such as Enabling ARMS, Open/Close Breaker, and remote racking, all such operations get logged as an 'Event' on the Timeline screen.

- When this **“Event”** entry is clicked it further expands to show the all the actions i.e. Entering the Control Mode, Request Open Breaker, Close Breaker Request.



- Additional details of an alarm, such as type of alarm, type of device, device location, time of occurrence, value at occurrence, history, etc., can be viewed on the Alarm Details page.
- The PDMC shows alarms when devices indicate something is wrong. Devices with alarms are shown as **Red** in the Timeline Tab. If there are any open alarms on the PDMC, there will be an alarm callout at the top of the page indicating the number of open alarms (shown above). Priority alarms are shown with an exclamation mark in red box.
- The Timeline tab shows all alarms from the selected time range (the default is ‘Last 12 Months’) plus all open alarms.
- Users acknowledge alarms to indicate they’ve seen the problem. You can acknowledge alarms individually using the acknowledge button. Alarms are active when the condition that caused the alarm is still present. Even if an active alarm has been acknowledged, it still shows under Active alarms until the condition goes away.
- While alarms are initially sorted by date, you can also sort them by device or priority. The current time range for alarms in the list is shown next to the calendar button. You can filter the list to show a specific date range. You can further filter the list to show active alarms, acknowledged alarms, or unacknowledged alarms. Any open alarms will always be shown, regardless of when they occurred.
- The **Export** button on the Alarms tab downloads a CSV file containing an alarm log. You can limit the date range for the alarm or the number of alarms in the file. When an alarm is selected, you can use the Choose an Action button in the sidebar to export just the information about that alarm as a CSV file. Use Alarm Details to view a pop out with all information about the alarm, including its history.
- You can acknowledge alarms in groups using the check boxes, then clicking **Acknowledge** at the top. The top check box will select all. Past alarms are always available for viewing by selecting to show all alarms and All Dates.

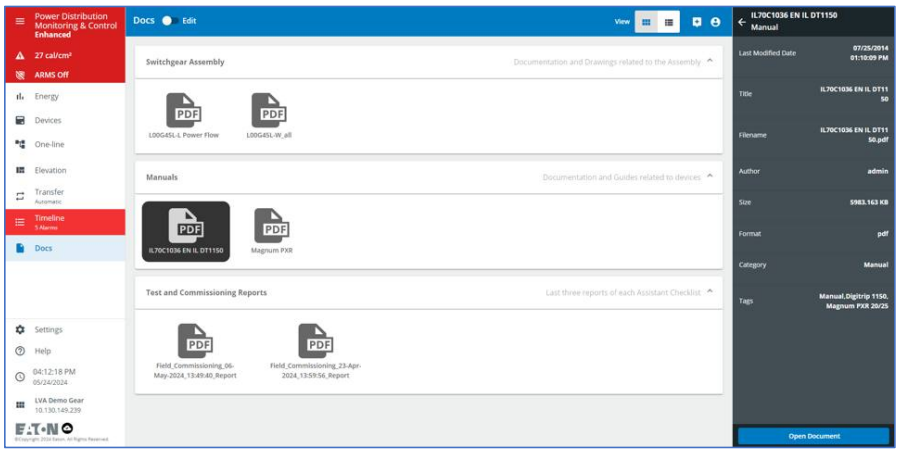


- The list of alarms can be exported using the **“Export”** button in the Action bar of the Timeline screen.

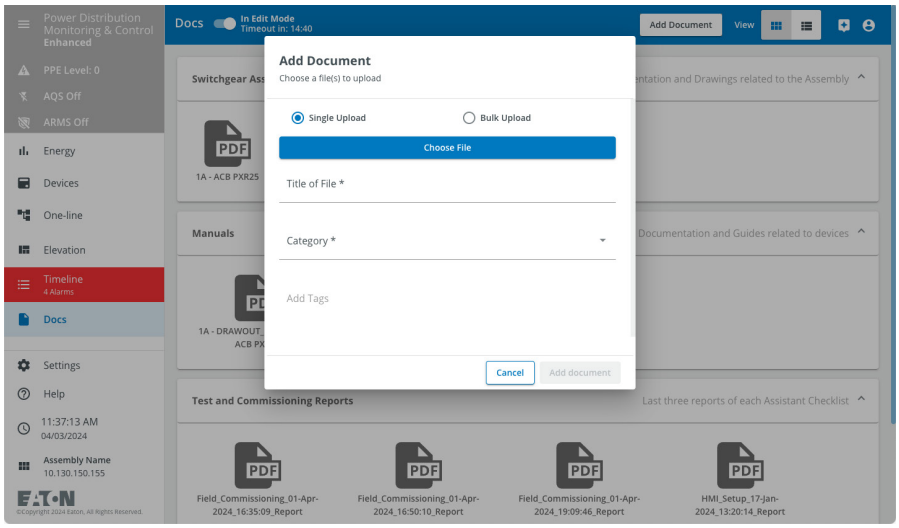
9. Docs

The Docs tab provides switchgear documentation including drawings and device manuals.

- A document can be viewed by either a double click or by clicking on the open document from the sidebar.
- In edit mode, the sidebar provides document details along with allowing the user to export and remove the document.



- The user can upload a single document or bulk documents in edit mode by clicking on **Add Document** Button in the Action Bar or within a specific section.



9.1 Bulk Upload

Bulk upload of documents feature is available from the Add Document modal on the Docs screen.

The Bulk upload feature has 2 pre-requisites

1. CSV File: This is a metadata for all documents that user wants to upload.

A	B	C
name	category	tags
MN401I2-N Unit Schedule.pdf	Switchgear	Unit Schedule
MN401I2-P Unit Schedule.pdf	Switchgear	Bill of Material
MN401I2-R Front Layout.pdf	Switchgear	Layout Diagrams
MVA Power Flow Diagram.pdf	Switchgear	Power Flow Drawing
EDR 5000 Manual.pdf	Manual	Protection, EDR-5000
il01301064_digitrip 1150.pdf	Manual	Protection

- **name**: It will contain name of the PDF document along with its extension.
- **icategory**: This column will contain any one of the two categories: Manual or Switchgear.
- **tags**: This is a comma separated list which will be recognized as custom tags for documents. Below is the list of defaults tags for the user to select from.

2. Documents: Documents the user wants to upload available in a single folder on the user's computer.

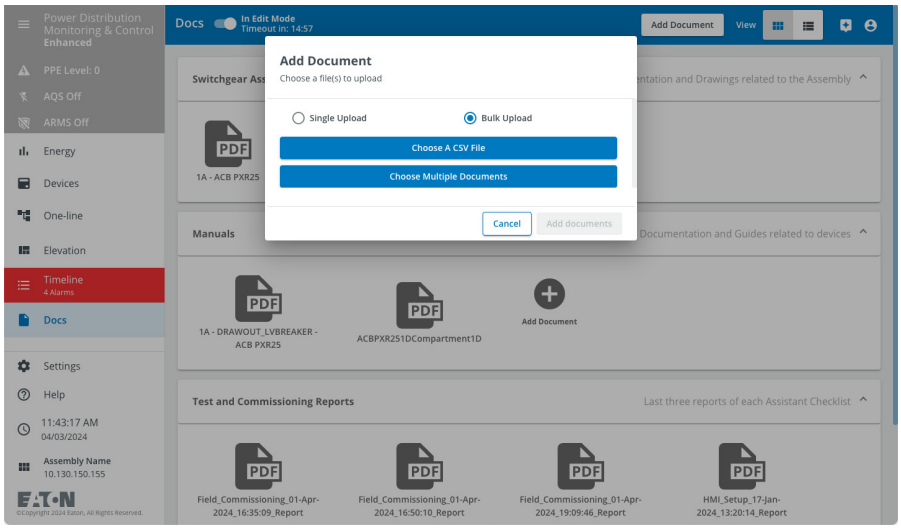
Device Tag List

IQ220/IQ320	Digitrip 520MC	IQ35MA2	IQ-500
IQ230/IQ330	Digitrip 810	IQ130	MP-3000
IQ Analyzer 6000/6200	Digitrip 910	IQ140	MP-4000
IQ Analyzer 6400/6600	Digitrip 1150	IQ150	MPCV Relay
IQ Data Plus II	Digitrip OPTIM 550	IQ230M/IQ330M	Series NRX 520I
IQ DP-4000	Digitrip OPTIM 750	IQ250	Series NRX 1150I
IQ Energy Sentinel	Digitrip OPTIM 1050	IQ260	Universal RTD Module
IQ Power Sentinel	FP5000	PM3 Modbus	ATC-400
IQ MES II	FP6000	PXM1000	ATC-600/800
DG1 Series VFD	Digitrip 3000	PXM4000	MR2 (Remote Racking)
MVX9000	Digitrip 3200	PXM6000	M-Max
SPX9000	Digitrip MV	PXM8000	H-Max
SVX9000	EDR-3000	PXM2250	Universal RTD Module II
C441V Ethernet 24VDC I/O	EDR-5000	PXM2260	ETR-4000
IQ35MA1	EGR-5000	PXM2270	ETR-5000
PX Dashboard Transfer	Power Xpert Branch Circuit Monitor	PXM2280	S611 Soft Starter
ELC Programmable Logic Controller	Power Xpert Multi-Point Meter	PXM2290	S811+ Soft Starter
EGR-4000	C440 Overload Relay	EMR-4000	ATC-300+
PXR 20	C441 Motor Insight Overload Relay	EMR-5000	ATC-900
PXR 25	C445 Motor Management Relay	Series NRX 520M	AutoVAR 300
TC-50-Mod	EMR-3000	Series NRX 1150M	InsulGard
TC-100	Addressable Relay II	Addressable Relay II (Inv)	BIM II
ELCPV28NNDR	Solar Inverter		

Bulk Upload Workflow:

Log in to a PDMC with the user having 'Configuration file save/restore' permission.

3. Click on **Docs** in the Left Pane.
4. Click **Edit** toggle to enter edit mode.
5. Click **Add Document** button in the action bar or from the 'Switchgear Assembly' & 'Manuals' section.
6. Enable bulk upload from 'Add Document' modal and the below screen will be displayed.

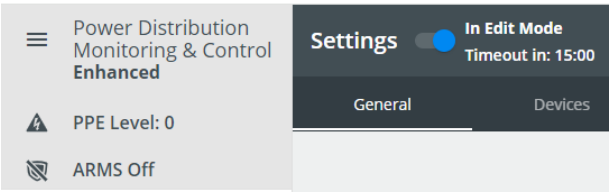


7. Select the required .CSV file and the documents to be uploaded.
8. Click **Add Document** button. ('Adding Document...' notification is displayed.)
9. 'Document Uploading Status' modal is displayed with the statuses of all the documents which were attempted for upload.
10. Click **OK** button on 'Document Uploading Status' dialog and all the added documents are displayed on documents page/screen.

10. Settings > General

Configuration of the PDMC begins on the **Settings > General** tab.

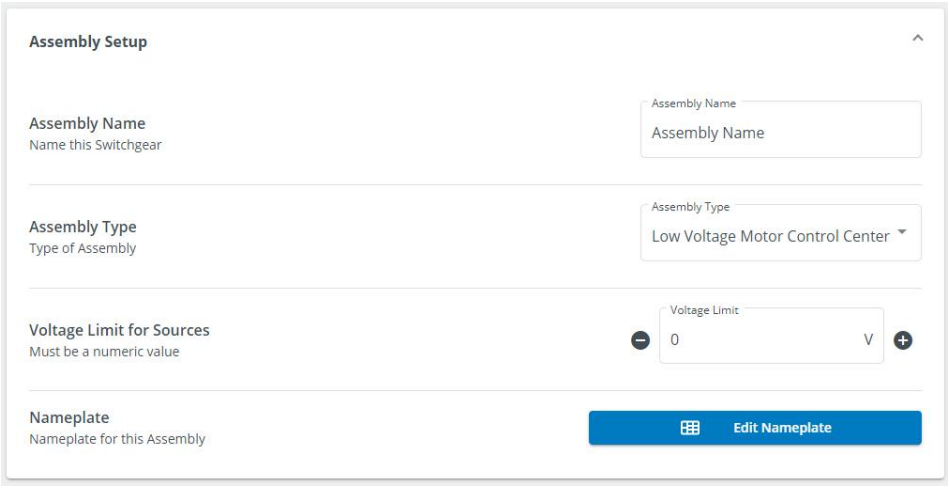
- Enable 'Edit' mode using the slider in the Action bar at the top of the screen.



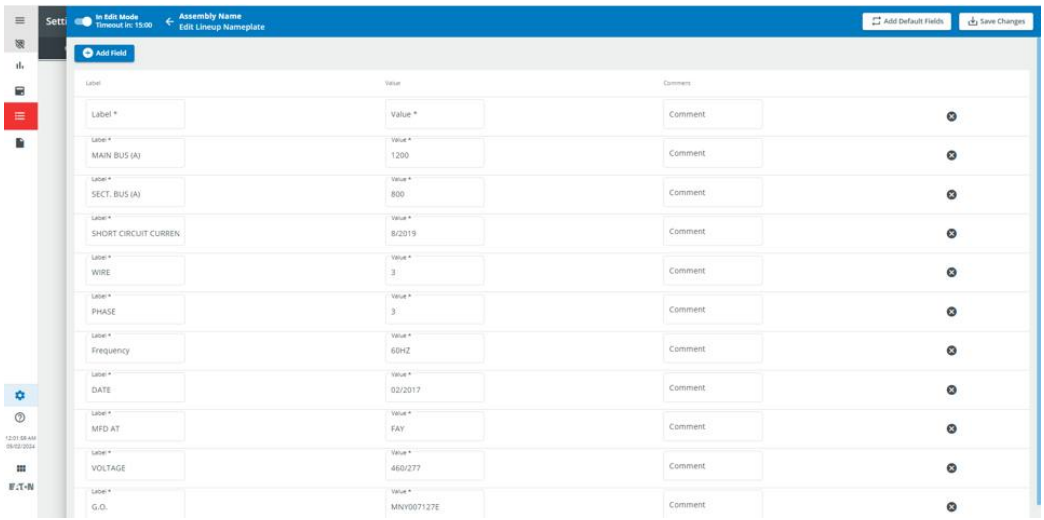
10.1 Assembly Setup

The Assembly Setup section allows the user to enter or edit the Assembly Name, choose the Assembly Type, set the Voltage Limit, and enter the Nameplate information.

- Assembly Type options include: Low Voltage Assembly, Medium Voltage Assembly, Low Voltage Motor Control Center, MEF Special and Switchboards/Panelboards.
- The Voltage Limit for Sources is the threshold voltage used to determine if a source or bus is shown as energized or de-energized on the One-line and Elevation Views.

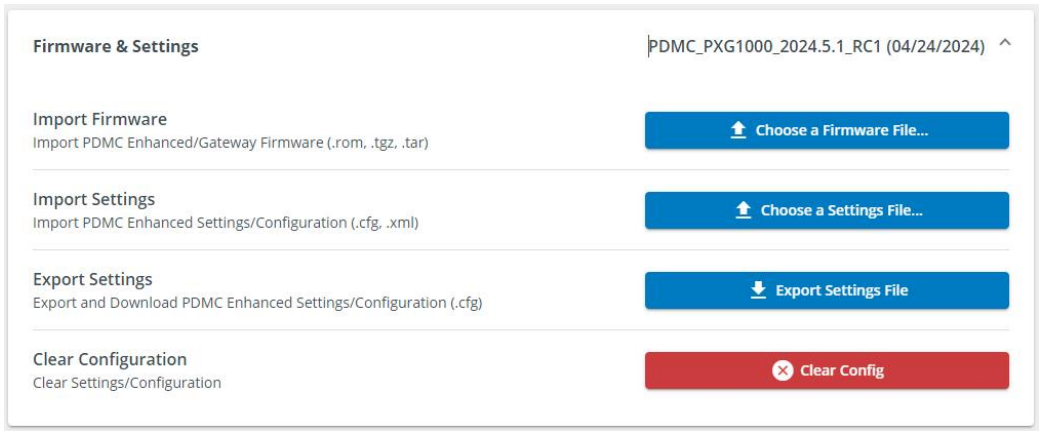


Nameplate for assembly: The PDMC provides a location to record assembly nameplate information. Selecting “Edit nameplate” will open a new stack where the “Add Field” button will provide a line for entering label, value and comment for each nameplate entry.

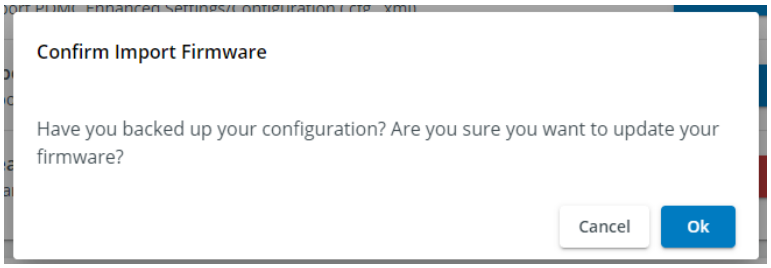


10.2 Firmware & Settings

The Firmware & Settings section provides Firmware upload, Import and Export of Configuration Files, and the options to clear the PDMC.

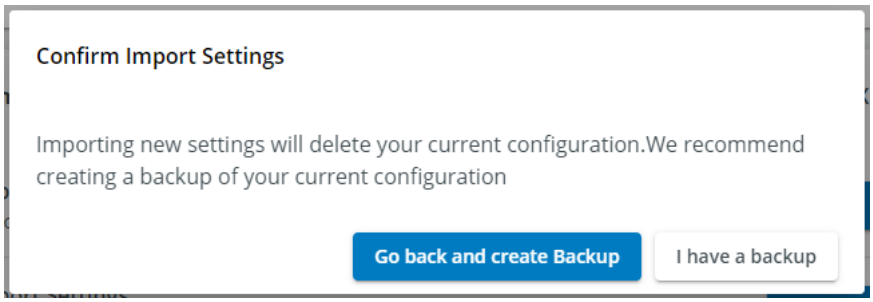


Import Firmware: To upload a Firmware file, click “Choose a Firmware File ...”, browse to the appropriate folder, and select the firmware file. Confirmation modal will be opened. Select “Ok” button to upload firmware file.

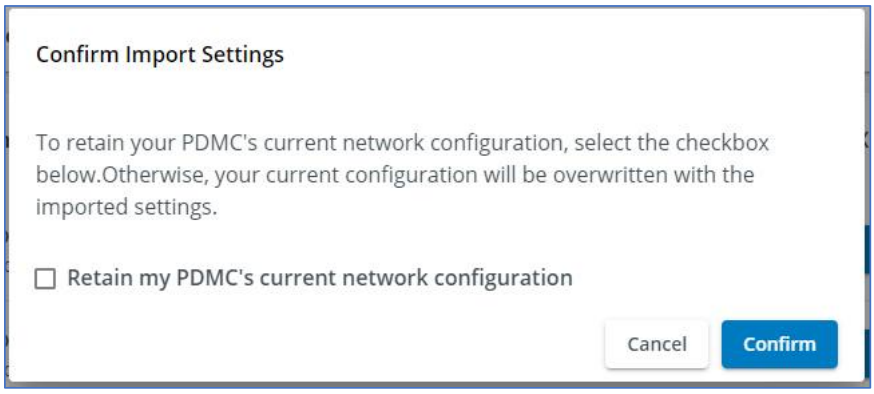


Import Settings: The Import Settings file upload provides Network backup option before either an xml or a cfg configuration file.

- User can take current configuration backup by clicking "Export Settings" button.
  - If user selects "I have a backup" button, another modal will be opened.



  - User can select a checkbox to retain network configuration. This option will be keep LAN A & LAN B settings the same.



Export Settings: Use the Export Settings to save the .cfg configuration file. Always export the settings file before making any changes to the PDMC. The exported file can be uploaded using the Import Settings feature to restore the PDMC configuration at any time.

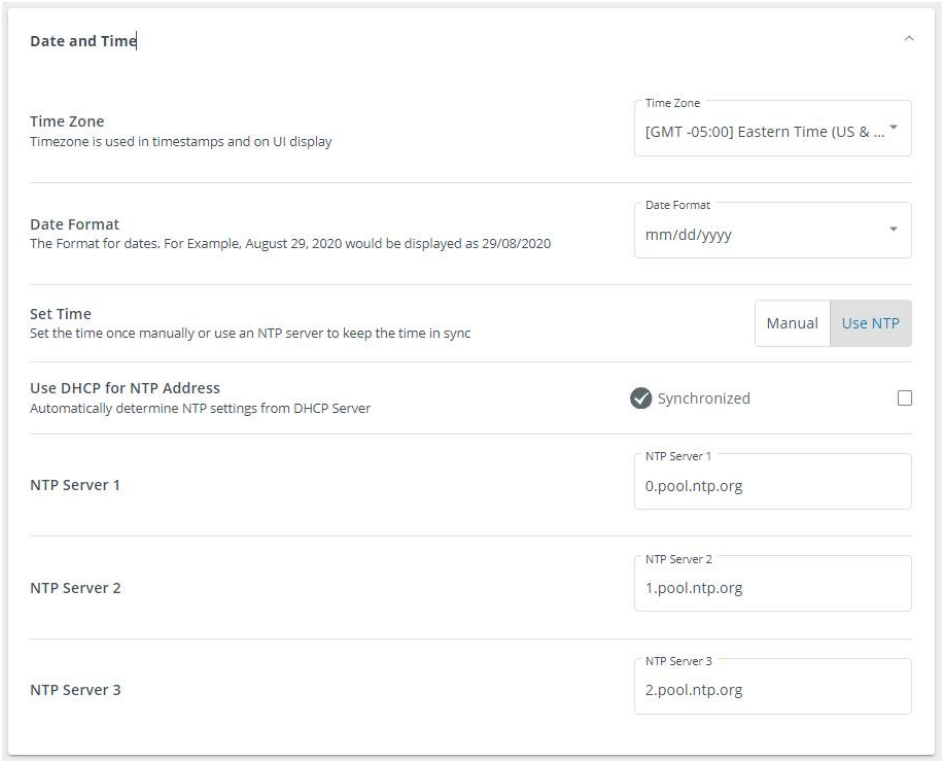
Sr. No	Description	Description
1	Uploading **PXDBP** configuration files on PDMC_PXG1000	Uploading **PDMC Enhanced** configuration file works properly if the Enhanced Graphics license is present in **PDMC PXG1000**. **Note:** INCOM device configuration requires the USB INCOM module.
2	Uploading **PXG1K_PDMC_2023.10.x /PXDBL** configuration files on **PDMC_PXG1000**	Improvements will be added in the **PDMC_PXG1000_2024.5.2** release for properly importing PDMC Basic configuration file.

10.3 Date and Time

The Date & Time can also be Configured and set under the General Settings tab.

If time stamp accuracy on data is important to you, NTP synchronization is the best option. However, you'll either need access to the Internet or you'll need to install an NTP server on your network. The PDMC will periodically check the time and correct itself. Under Manually Set Time and Date, choose Set Time and Date to either set the clock to match your PC or set the time and date yourself.

The PDMC was set to United States format for date and to US Eastern Time (UTC-05:00) by default. Set the time zone and date format. As of this time, English is the only available choice. Changing the time zone will not change the time setting.



10.4 Licenses for Additional Features

The PDMC includes licensed features for

Control: It is required to OPEN/CLOSE breaker, Enable/Disable ARMS, perform Remote Racking and Transfer scheme.

ARMS: It is required to Enable/Disable ARMS.

Device Setpoints: It is required to change setpoints on devices.

Enhanced Graphics: Provides web UI for Elevation, One-line and Incident Energy.

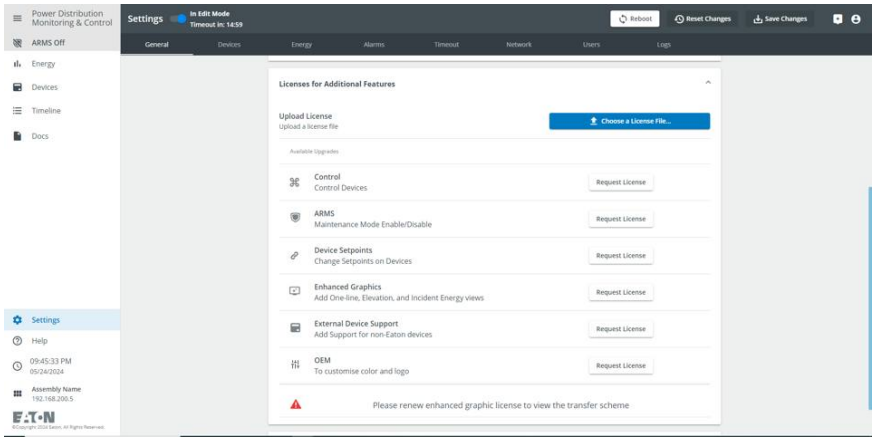
External Device Support: It is required when using non-Eaton devices. Sensor devices are supported without the need for External Device Support license.

OEM: It is required to customize logo and color.

Remote Racking: It is required for Remote Racking of Medium voltage and Magnum breakers.

Transfer Scheme: The Transfer scheme license is required to support configuration and visualization of the Transfer scheme screens.

The Remote Racking and Transfer Scheme features require special hardware, and it is suggested to purchase these items during the initial PDMC order. The Device Setpoints app can be purchased initially or added later. Enhanced Graphics license could be purchased later but it is good to purchase this initially as the configuration and testing can be done at the plant.



Follow these steps to purchase a licensed feature:

- Click the “Request License” button beside the Feature to be purchased and save the file.
- Contact the Customer Integrity Team at **MRSupport@eaton.com** and email the exported license request file. CIT will provide the instructions to pay for the licensed feature.
- Once payment is received a license key file will be provided for the specified PXG1000.
- Click “Choose a License File ...” and upload the license key file to enable the licensed feature.

10.5 Additional General Settings

Many additional aspects of the PDMC can be configured and customized from the **Settings > General** tab.

- The Display and Extras section provides selection of:
  - The default screen when logging in.
  - Selection of which tabs to show or hide in the navigation bar
  - Wide scrollbars
  - Show Load Grouping, Energy Grouping, Devices tab, Virtual Meters, Document View
- Unit conversions and Default System Colors Configuration are also available.

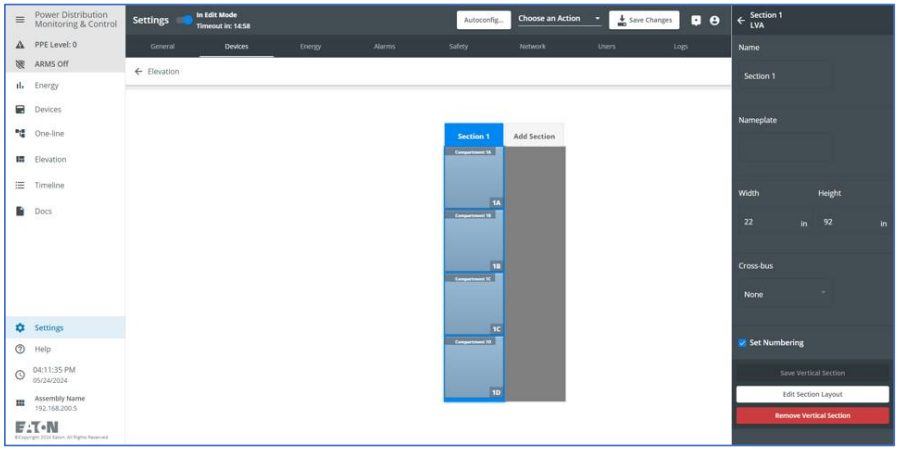
- The user may choose to show temperature data in Celsius or Fahrenheit.
- Customer Support Information is defined here.
- A custom system use notification can be added.
- Brand label: To view this collapse user require to install Brand Label license.
  - Configuration:
    - Enter the received OEM keycode in the text box and click on “Save Changes” button at the top and on “OEM Customization” modal select “Ok” button.
    - It will take approx. 1 minute to apply, and application will be refreshed and navigate to Login screen. PDMC will be updated with OEM name, logo and color on Login screen and left navigation menu seen after Login.
  - Delete Existing OEM keycode:
    - Enable Edit mode, click on “Delete Keycode” and select “Ok” on confirmation modal.
    - Application will be refreshed and navigate to login screen. It will restore the original Eaton Logo with default color on Login screen and left navigation pane seen after Login.

11. Settings > Devices

11.1 Settings > Devices > Elevation

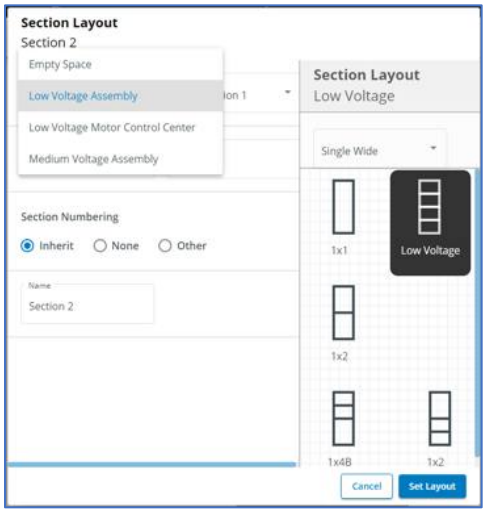
The graphical switchgear lineup representation shown in the One-line and Elevation tabs is configured on the **Settings > Device > Elevation** tab. All configuration requires Edit Mode to be enabled.

- **"Import a configuration file"** or **"Clear configuration"** using the **"Choose an Action"** dropdown in the Action bar.
- When creating a lineup manually, click on the **"Edit"** mode slider to view the **"Add section"** button for adding new sections and compartments.

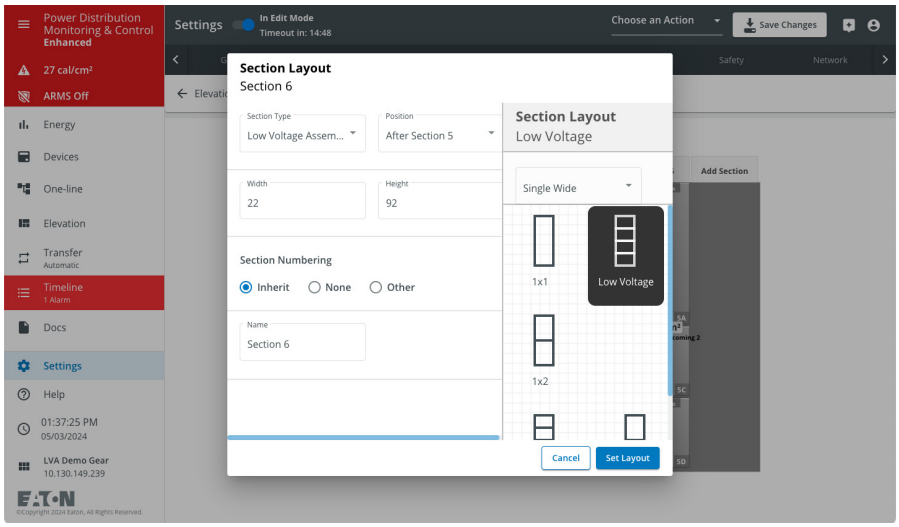


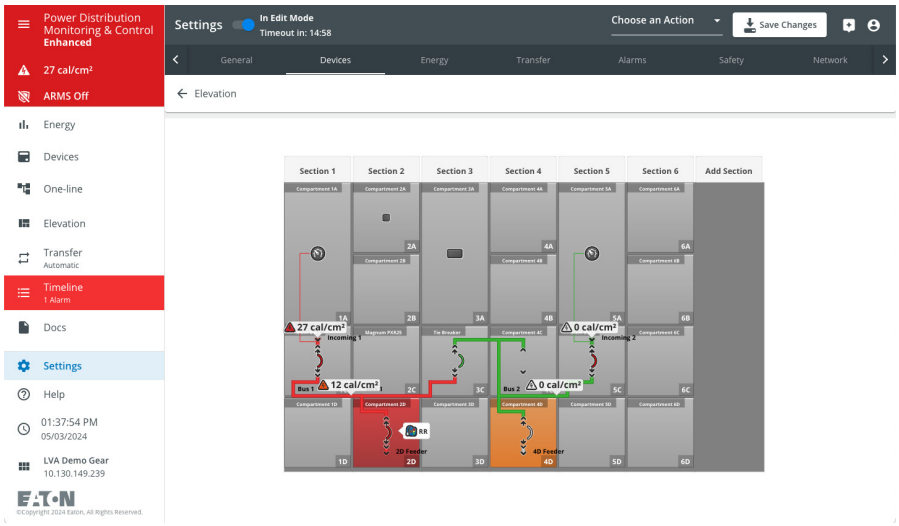
- Click **"Add Section"** and select the desired section type, position, and compartment layout from the Section Layout modal

Note: Section Type will default to a specific type based on the assembly selected. You can select another option if desired.

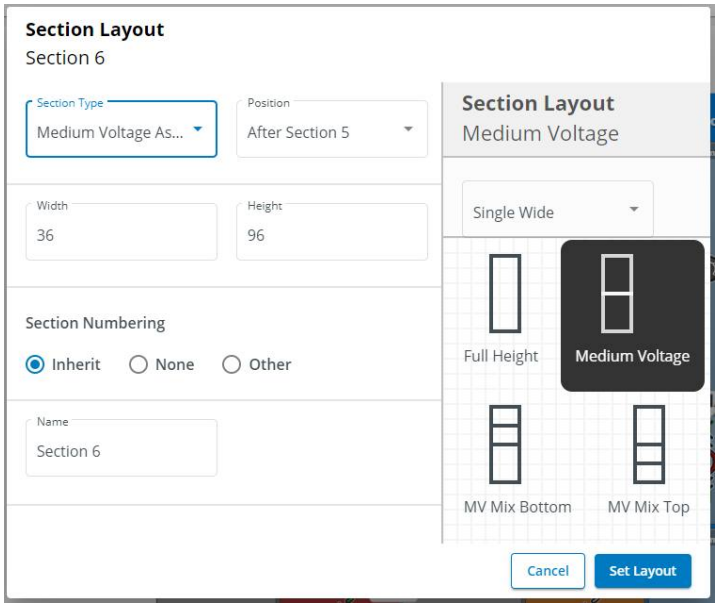


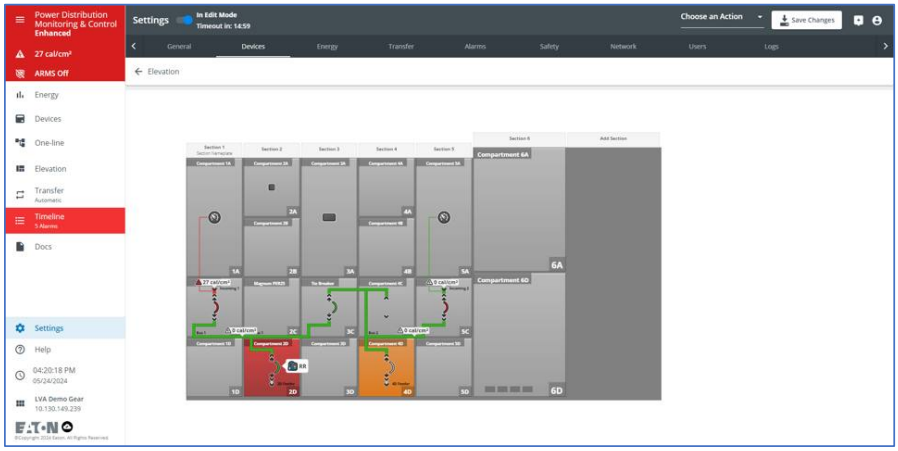
Example of Low Voltage Assembly (Section 6 added)



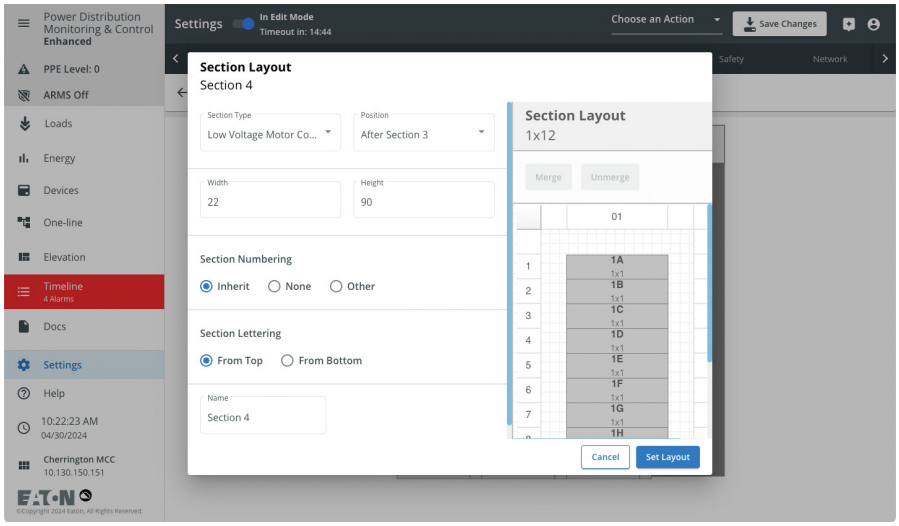


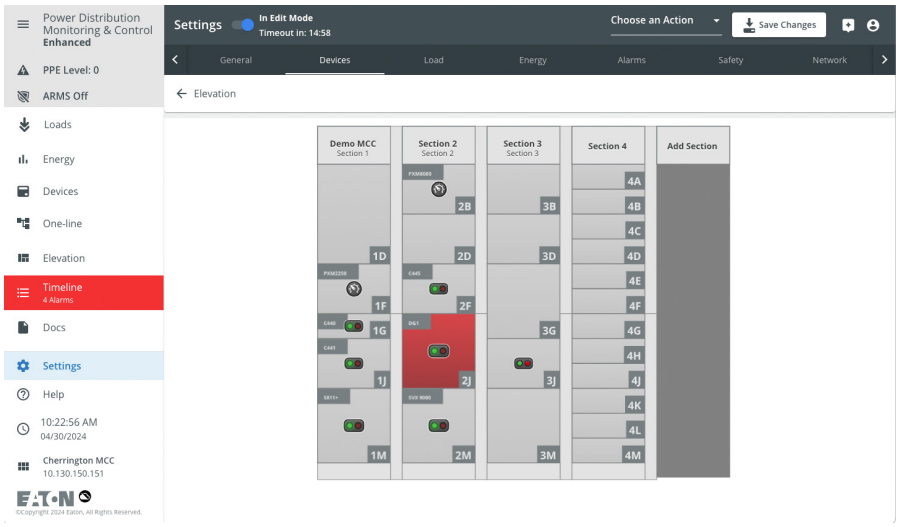
Example of Medium Voltage Assembly (Section 6 added)





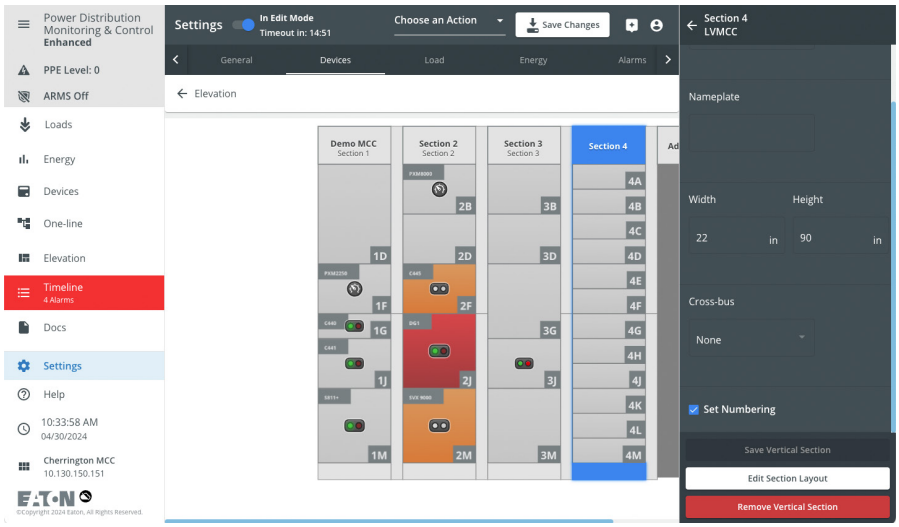
Example of Low Voltage Motor Control Assembly (Section 4 added)

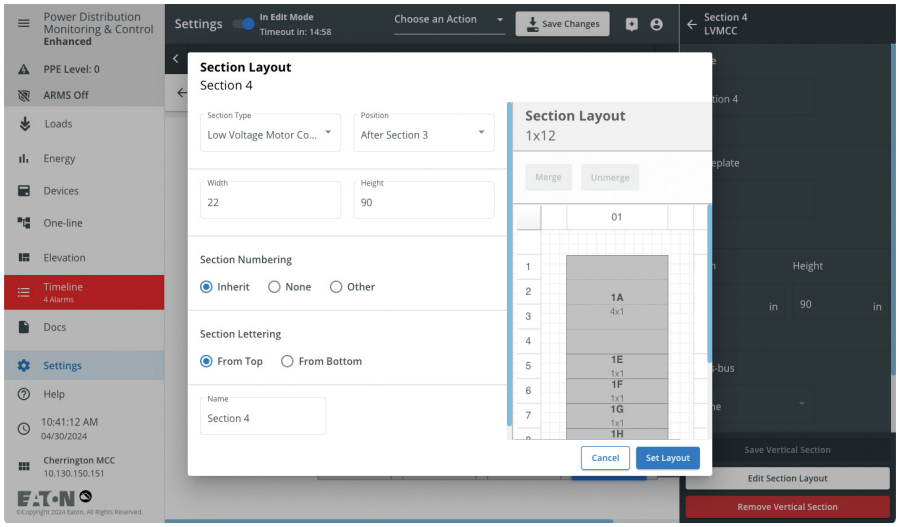




Example of Low Voltage Motor Control Assembly (Section 4 added)

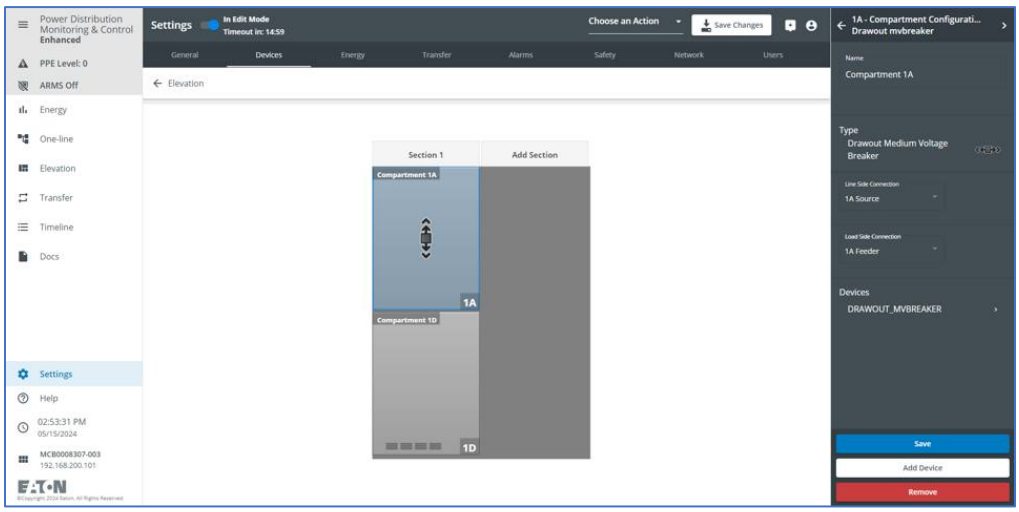
During initial setup, or after you have added it, you can edit the layout of the sections. Below is an example of an MCC section that has **four x units** merged to create **one four x unit high compartment / bucket**.



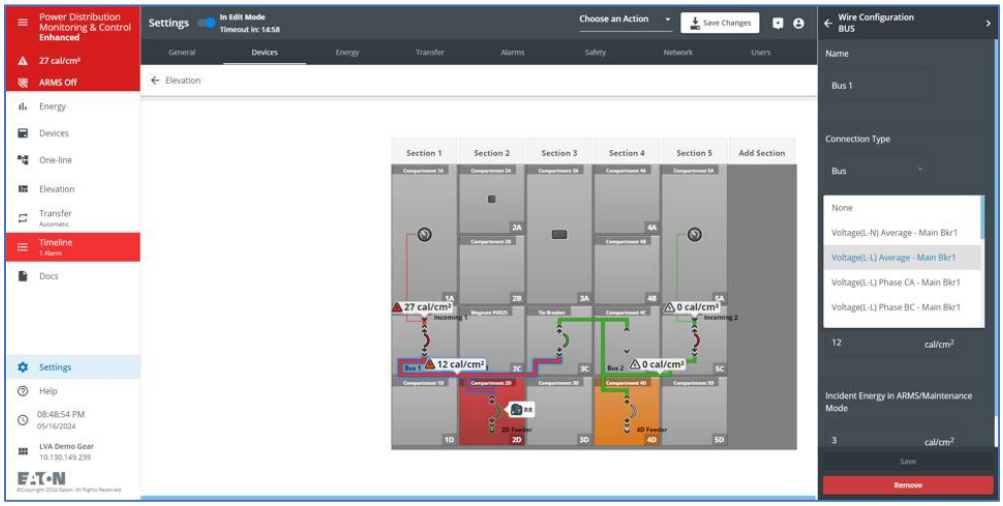


- To add devices in any compartment, select the compartment and click on the 'Add device' button on the compartment configuration sidebar.
- "Wiring" of devices in a compartment can be done using 'Line side' and 'Load side' connection dropdowns. The line side connection is the upstream power source and the load side connection delivers power to the downstream devices.

11. Settings > Devices



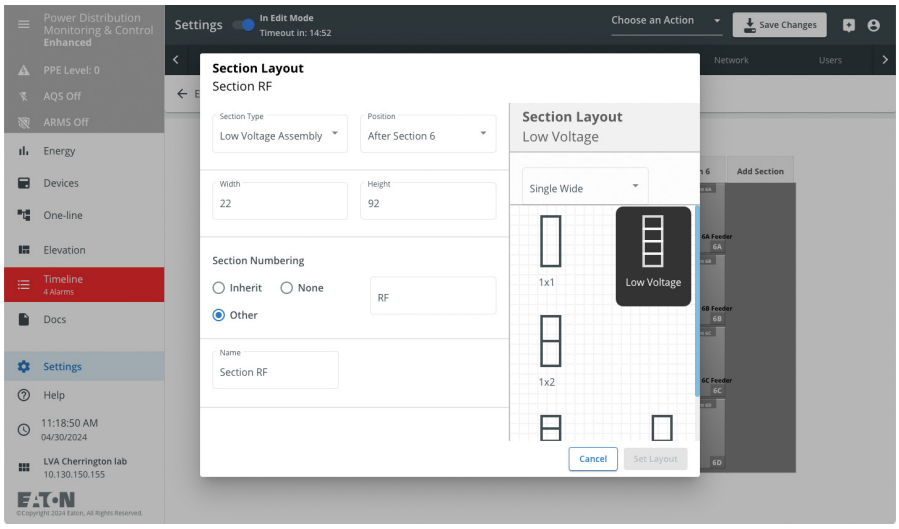
- On click of the 'More' button in the compartment configuration stack, one can configure additional details like compartment type, static information related to compartment, wiring details, etc.
- To check if the connections made in the Elevation view are appearing correctly in the One-line diagram, consult the **Settings > Devices > One-line** tab.
- Source and bus lines can be configured to show energized (red) or de-energized (green).
  - Select the bus line on the **Settings > Elevation** screen to open the Bus sidebar.
  - You can edit the name and connection type and select the voltage channel from a connected device to monitor to determine if the bus should be shown as energized or de-energized.

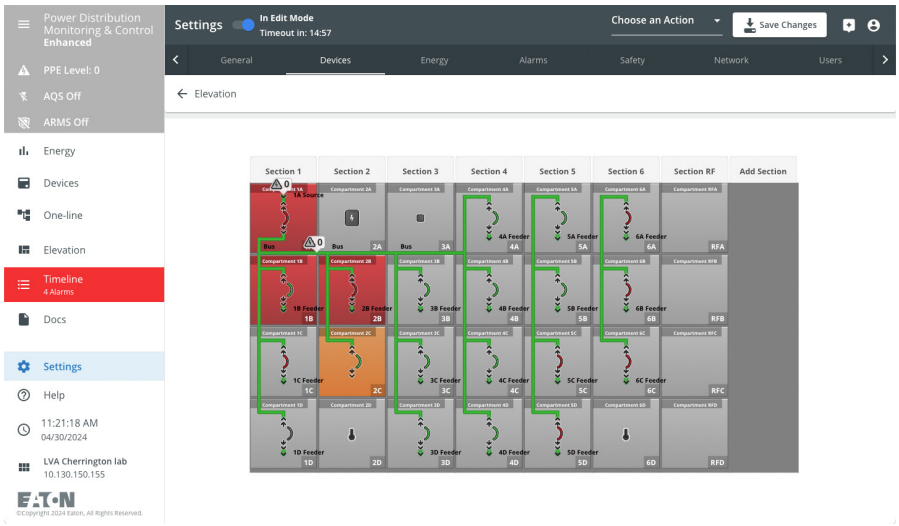


  - The selected voltage channel is compared to the "Voltage Limit for Sources" entered on the **Settings > General** screen under Assembly Setup.

11.2. Custom Compartment Identification

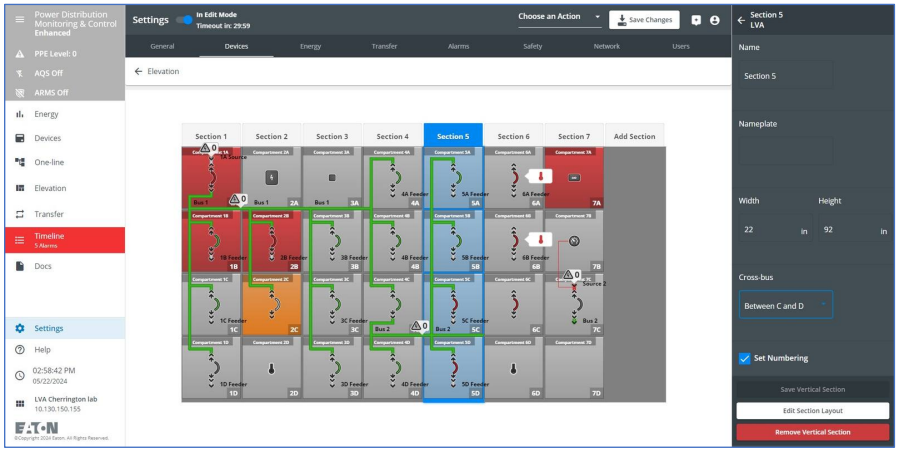
If you need to give the compartments in a new section a custom designation, choose Other for Section Numbering on the Add Section modal, and then enter a two-letter designation in the field. Below is an example where the user created a new section named “RF”.

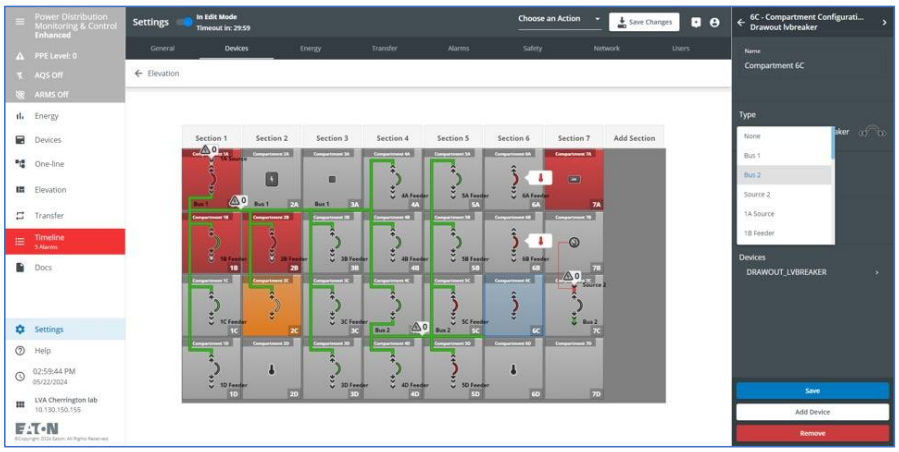


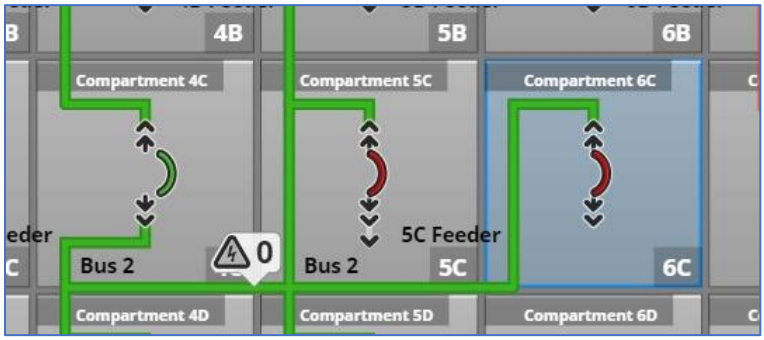


How to Create a Cross-Bus Connection

In **Step 1** below, the user creates a cross-bus connection between **C** and **D** on **Section 4**. Then, in **Step 2**, the user selects the compartment in **Section 5** and chooses **Bus** from the **Line Side Connection** option. The resulting connection is seen in the screens below.





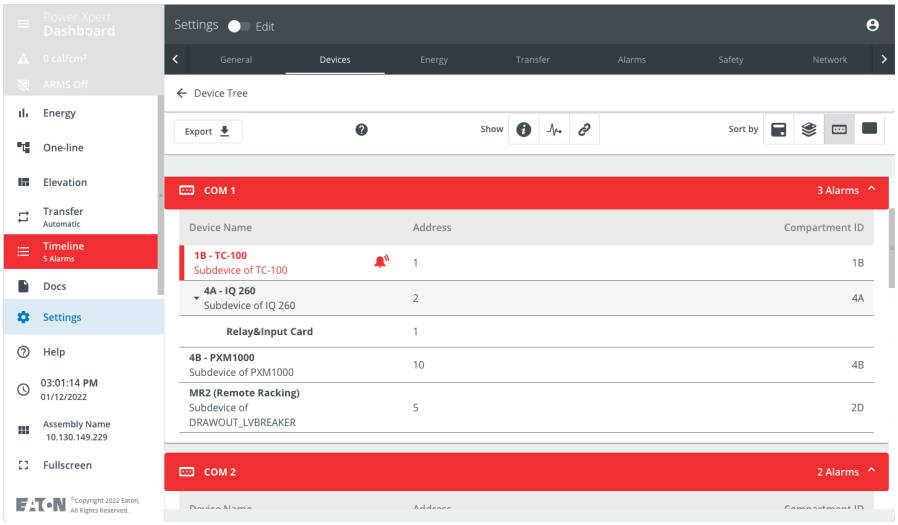


11.3 Settings > Devices > Device Tree

The **Settings > Devices > Device Tree** tab shows all the devices attached to the PDMC. These are grouped under their communication port. Devices are color coded based on status:

- **Red** shows that there are alarms from one or more channels for that device.
- **Orange** means that the device is not communicating.
- **Purple** means that a device is disabled.
- **Black** means the device is communicating.

Details of the PDMC PXG1000 Communication Ports, Devices, Inventory Information, Communication Statistics and Link Information associated to the devices can be viewed and configured from this tab. The view of the devices can be sorted on basis of Device Model, Composite Type, COM Port and Sections



The names of individual devices can be edited in the sidebar.

Additional information about the devices is shown when the Show buttons are selected:

- Inventory information such as model name, serial number, and firmware version is displayed
- Communication Statistics are displayed
- Linked setpoint files for the devices are displayed

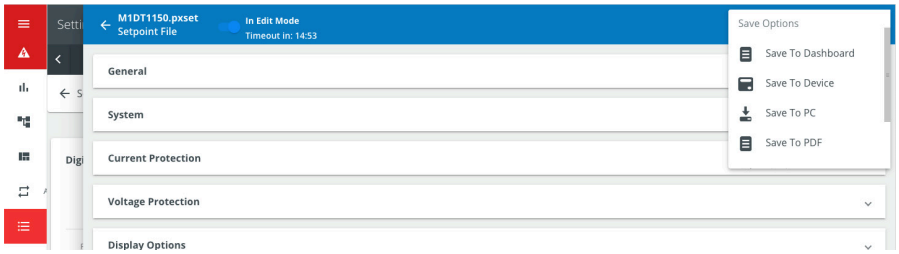
The System Inventory and Communication Statistics can be exported and sent to CIT for use in debugging communication issues.

11.4 Settings > Devices > Setpoint Files

The tab provides a means to manage device setpoint files and link them to devices. This feature is applicable for Trip units. In this feature setpoints from Trip units can be Viewed anytime and modified when a device is communicating. In this feature setpoint files can be created, imported, viewed, modified & saved anytime. Viewing and editing of device setpoints can also be accessed from the Device Details screen from the main navigation menu and from **Settings > Devices > Device Tree**, selecting a device, then clicking the More icon at the top of the sidebar. In either case the Edit and View Setpoints functions may be grouped in the Choose an Action dropdown

The Setpoint supported devices include Digitrip 1150, Digitrip 520MC, Digitrip OPTIM 1050, Digitrip OPTIM 750, Digitrip OPTIM 550, ACB PXR20, ACB PXR25, Digitrip 810, Digitrip 910, MCCB PXR20, MCCB PXR25, MCCB PXR20D.

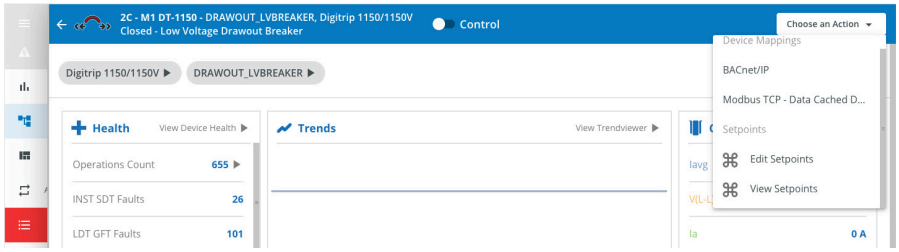
This feature requires the Setpoints License and Setpoint Permission. Device setpoints can be viewed for all the listed devices. Setpoints can be changed for certain devices like the Digitrip 1150 and PXR25 trip units. Options for saving the setpoint file include Save To PDMC, Save to Device, Save as Setpoints File on the HMI or user PC, and Save to PDF for a human readable list of the device setpoints.



The PDMC setpoints file is compatible with the PXPM, and the setpoint files are interchangeable.

To modify setpoints of Trip units follow these steps:

1. Navigate to One-line.
2. Select a device to open the sidebar
3. Select the More button at the top of the sidebar to open the Details stack.
4. Select the “Choose an Action” dropdown and select View Setpoints or Edit

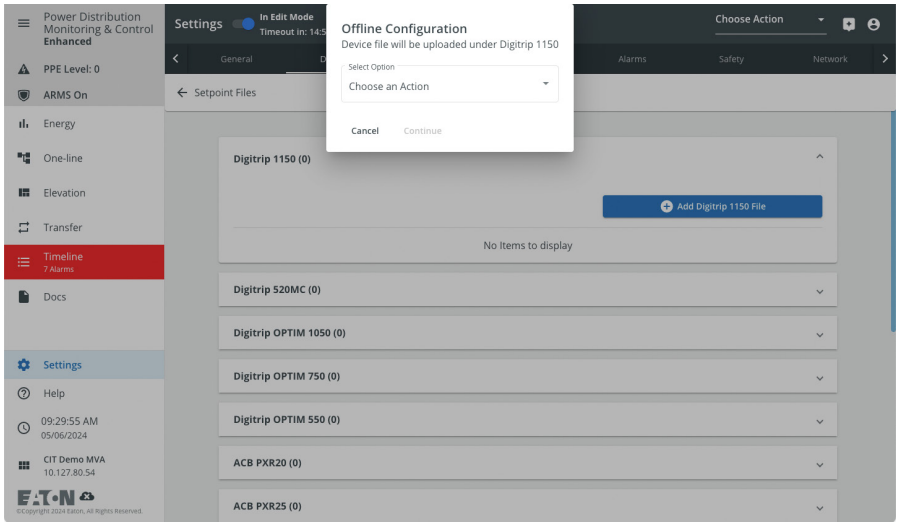


11. Settings > Devices

On Setpoint stack, user can edit and configure the settings related to the device. This feature can also be used to configure setpoints for a model of Trip unit even though no Trip units are communicating in lineup.

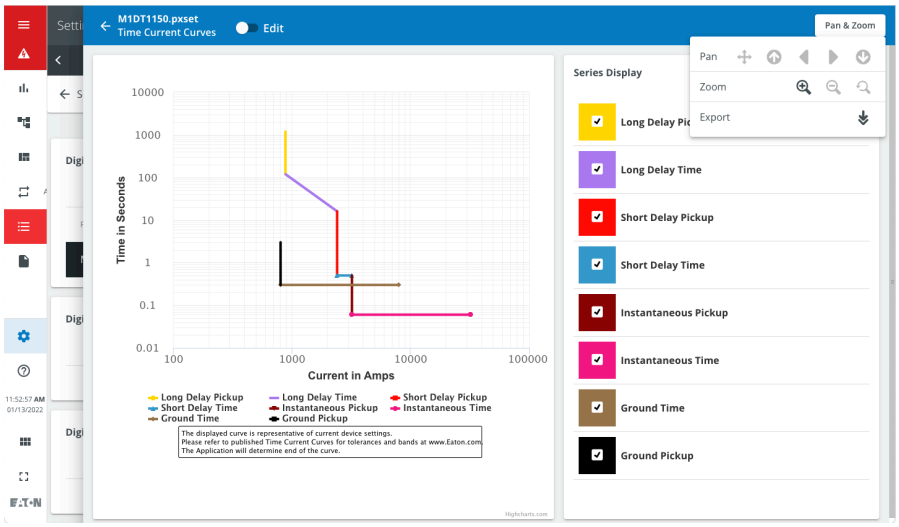
To access this feature, follow these steps:

1. Go to **Settings > Device > Setpoint**
2. Add new file
3. Select desired Device model and series.



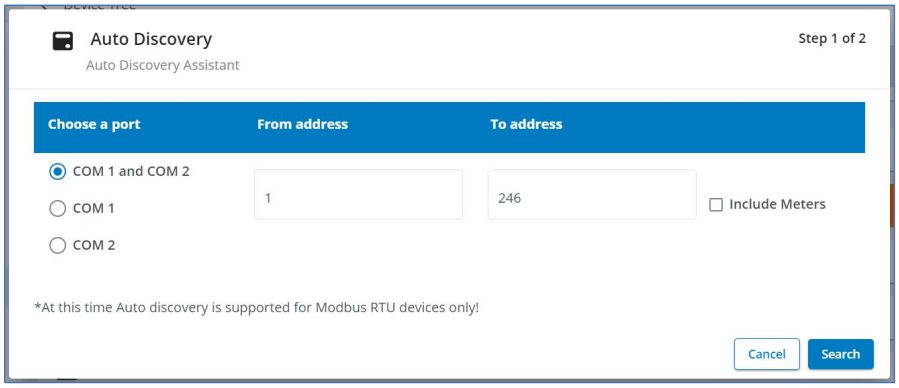
The Link Feature allows the user to link a Setpoint file with a connected communicating device of the same family and style.

Users can view the setpoints differences between a device and linked setpoint file and download setpoints from file to device and vice versa. The protection Time Current Curves can also be viewed.

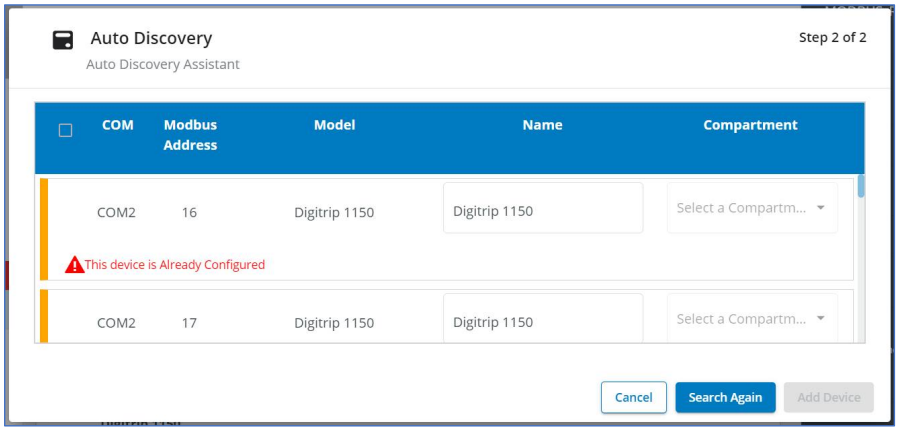


11.5 Auto Discovery

- Auto discovery will launch a wizard that allows user to search devices connected to Modbus RTU ports of the Gateway. User can give range of Modbus address to discover devices. If no range has been set, then it will try to discover in full range of 1-246. Device communication will be disabled when auto discovery process has started and will get restored to communicating state after discovery process is completed.
- If **Include Meters** checkbox is not selected, then meters will find as unidentified devices and needs to be configured manually.



- During auto discovery process, progress will be displayed on UI with address range. User can Cancel auto discovery process any time using Cancel button.
- Once the scan is done, user can see list of devices found under the given range in a tabular format with columns such as – COM port, Modbus Address, Model and Name (Editable). Compartment column will be displayed only if the Enhanced graphic license **(See setion 10.4 Licenses for Additional Features)** is already installed on the gateway.



- If the device is already configured in the gateway, an error message will be displayed – "This device is Already Configured."
- If a new device is found with already existing Modbus address, then a warning message will be displayed – "This address is already in use by another device. Adding this device will Replace Existing device." If user wants to replace new devices with existing device of the similar Modbus address, then that can be done.
- To add DT1150 device, then mMint device must be already configured on the gateway. If it is not present, then the error message will be displayed as – "Digitrip 1150 must be configured as a sub-device of an mMINT. Please configure the mMINT first." Digitrip device gets added as a sub device of the mMint.

Compartment selection (Enhanced Graphic License)

- If enhanced graphic license is present, and user has already added sections and compartments from **Settings > Elevation**, then user can select the compartment for each device. Compartments which already has a device will not be shown in the dropdown.
- If compartment is not selected for a device during auto discovery, then these devices will be added as an orphan device. To see these devices on oneline/elevation user must add these devices as an existing device from **Settings > Elevation (See section 10.4 Licenses for Additional Features)** and add wire to it.
  - If the same compartment is selected by another device and it will get added as an orphan device with below error message:

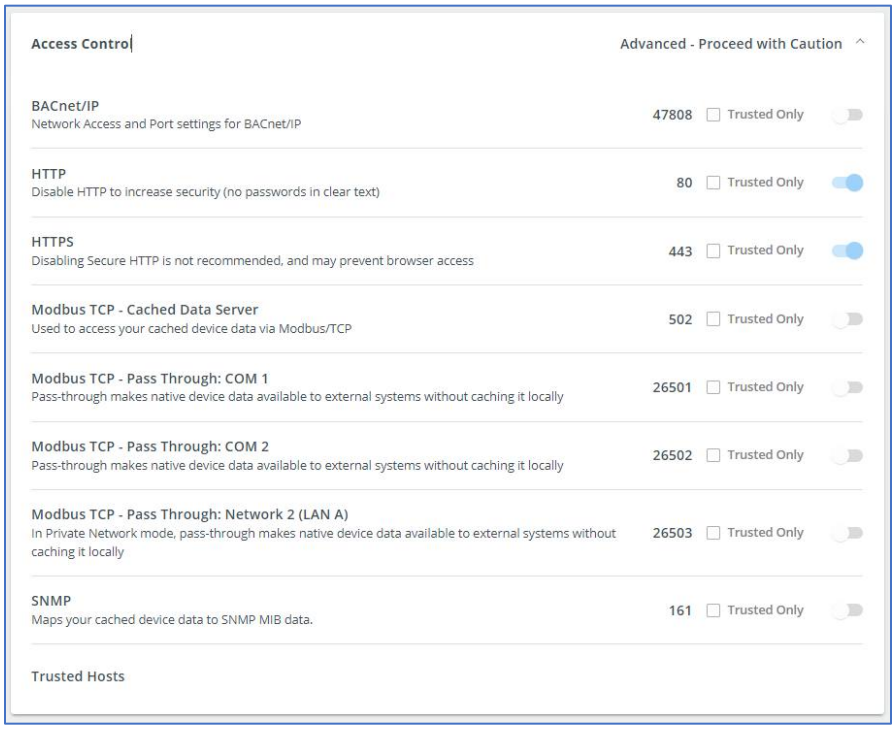
    "This compartment already contains a configured device. Choose another compartment or configure on **Settings > Elevation**." **(See setion 10.4 Licenses for Additional Features)**
  - Compartment types will be default as follow:
    - PD breaker, PXR trip units: V breaker
    - Protective relays: MV breaker
    - All Meters: Meter compartment

12. Settings > Network

Ethernet network configuration is provided on the **Settings > Network** tab. Configuration of the IPv4 Network 1 (LAN B) Connection, Name Servers, Access Control, Certificate Management, Network 2 (LAN A), SMTP server, Device Mapping and PDMC Campus connection are available.

12.1 Access Control

As per factory default settings, ethernet communications defaults to HTTPs using 443 port. HTTP and Modbus TCP ports (80 and 502, respectively) are disabled by default.



12.1.1 Disabled HTTP and Enabled HTTPS After Firmware Update

You can enable, disable and change these ports.

12.1.2 HTTP and HTTPS Settings are Disabled

You can also set each port so that it can only be used to access the PDMC PXG1000 through trusted host names or IP addresses. While using trusted hosts increases security, you must be careful. You can set the PDMC PXG1000 to use trusted hosts for HTTP and HTTPS access, and if your PC isn't at an address on the list you could be "locked out" via Ethernet.

Note: You must enable Trusted Only for at least one Access Control type in order to save machine names or IP addresses you enter in the sidebar.

To add a trusted host:

1. Choose one of the Trusted Only boxes. This must be selected to save your trusted hosts.
2. Click on Add Trusted Host/IP Address Button.
3. Enter the name or address in the Modal.
4. Click Add.



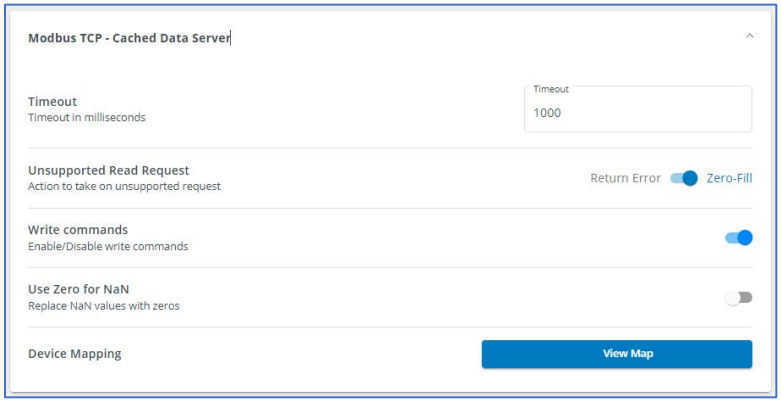
12.2 Modbus TCP Server



The PDMC can also function as a Modbus TCP server, providing channel data as Modbus registers from Modbus devices. The device addresses and Modbus maps for these devices can be downloaded directly from the PDMC. Normally, the PDMC provides only channel data supported by its internal device definition: the EDS file. However, the PDMC also has a Modbus "pass through" mode, allowing a Modbus client to connect directly to the device and use the device's native Modbus register set. You can set pass through for COM1, COM2, and Network 2 (LAN A) ports. Use of pass-through mode should be limited to commissioning and occasional use for issue investigation. Use of the pass-through mode during normal operation may have an impact on bandwidth available for normal PDMC communication with the devices.

12.2.1 Modbus TCP Server Setup

Modbus TCP Server setup is in Settings, on the Network Access tab. Choose Edit to access the configuration settings. You enable the server under the Access Control group (Modbus TCP must be enabled to see Modbus TCP Server Configuration). You can also change the TCP port. To enhance security, enable Trusted Hosts and then add the IP address of each Modbus client using Add Trusted Host/IP Address. Under Modbus TCP Server Configuration, you can set up the way the server responds to client requests. For example, you can allow the server to respond to write commands, where appropriate, or not to allow them.

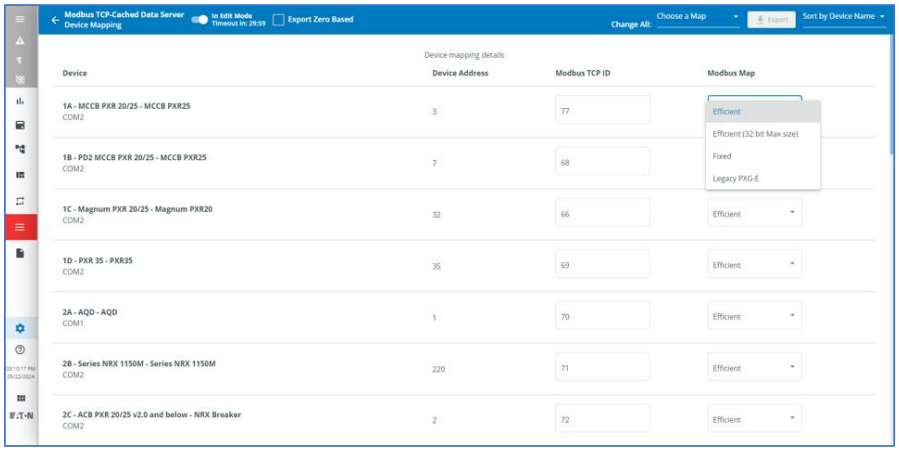


12.2.2 Viewing and Modifying the Modbus Map

Under Modbus TCP Cached Data Server, the View Map button produces a window that lists the various devices attached to the PDMC with their Modbus TCP ID. You can sort this list by device name, the native serial address set on the device, or the TCP ID. You can download the contents of the list as a CSV file by choosing Export. The PDMC maps data from all scanned devices, regardless of native protocol, to self-created Modbus TCP registers.

In Edit Mode, you can set the Modbus TCP ID for each device. You can also select the Modbus map type for all devices at once, or for each device individually. The possible map types are:

- Legacy PXG-E matches the maps from previous versions of the PXG-900 and PXDBP firmware.
- Fixed reflects the native device map, which may have gaps in the register numbering.
- Efficient provides a register mapping that is optimized to have no gaps. The maximum data value size is 64 bits (4 registers).
- Efficient (32-bit Max size) provides a register mapping that is optimized to have no gaps. The maximum data value size is 32 bits (2 registers).



With Edit mode disabled, the download link beside each device provides the Modbus register map for channels in the device configuration file. Before downloading the map, decide if you want a zero-based register map and then select the

check box.

The map contains all the information you should need, including:

- The register offset.
- The number of registers for the channel.
- The register type.
- Possible values when there is a limited set (such as for Boolean and some integer values).
- Whether you can write to the register.
- Provides recommended Modbus Function code for register.
- The units for the data.
- Whether the channel data is trendable.

To help you reconcile the register map with the device channels shown in the PDMC, the CSV file also contains:

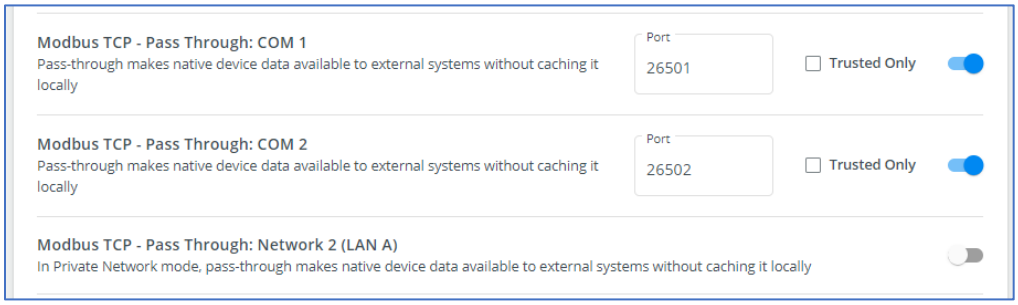
- The device channel name
- The PDMC category, such as Power, Current, or Demand.

12.3 Pass Through

12.3.1 Modbus

On the Network Access tab, you can enable pass through mode for either Modbus RTU port as well as Network 3 (Ethernet). Doing so allows you to access the full, native Modbus map for the device. These maps are not provided through the PDMC, so you will need to access them from the device documentation.

Note that each Modbus pass through has its own port address. Thus, you can access the full register map for a connected device by simply referencing this port in your TCP connection. When using pass through mode, you must use the actual Modbus Serial Address for the device. This is also listed on the Device Mapping page and in the device list available through the Export button.



12.4 BACnet IP Server

12.4.1 Setting up a BACnet/IP Server

Note: Edit must be active to make these changes. The Base ID setting won't be reflected in the Device Mapping until you save and exit edit mode.

If you enable BACnet/IP. you can present the connected devices as virtual devices on a BACnet/IP virtual network. Routed Network Number You must provide the Routed Network Number that will be assigned to the virtual network. Possible values are 1 through 65534. Ask someone in your facility maintenance group that is responsible for maintaining the BACnet network infrastructure for guidance in selecting a Routed Network Number. The Routed Network Number must be unique from all other BACnet network numbers within the BACnet global network.

Caution: If you choose a Routed Network Number that is already in use, communications problems will result.

12.4.2 Base ID for Auto-Assign

This is the base ID value used by auto-assign (if enabled) to assign the gateway and virtual device instance numbers.

12.4.3 Auto-Assign Gateway ID

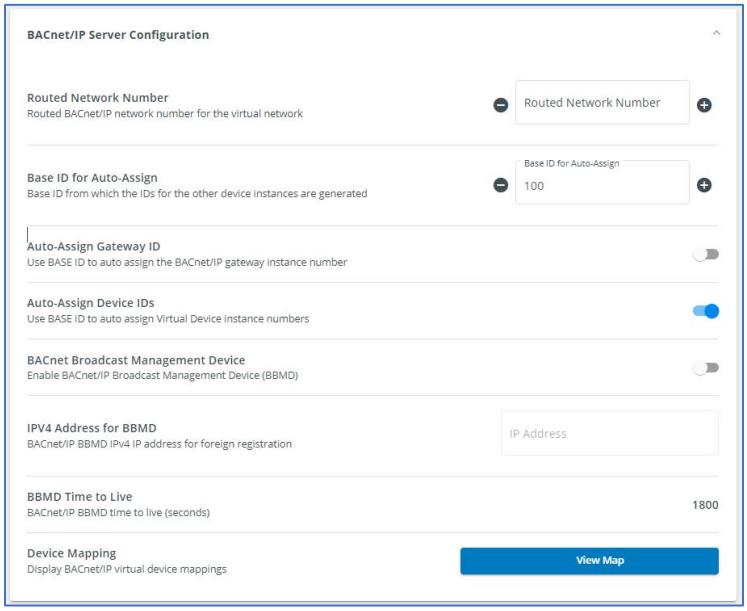
If enabled, the ID for the gateway is automatically assigned. If you set a Base ID, the gateway will be the base value plus one. If disabled, you can manually set the ID by clicking View Map.

12.4.4 Auto-Assign Device IDs

The various virtual device IDs will be assigned based on the Base ID (if specified) and incremental from the ID assigned to the gateway. If disabled, you can manually set the IDs by clicking View Map.

12.4.5 BACnet Broadcast Management Device

If you're using a BACnet Broadcast Management Device (BBMD) to connect your meter to another subnet in your BACnet/IP network, you must also define the BBMD IP address (IPV4 only) to register with the BBMD on the remote subnet. You can also set the BBMD Time to Live value in seconds.

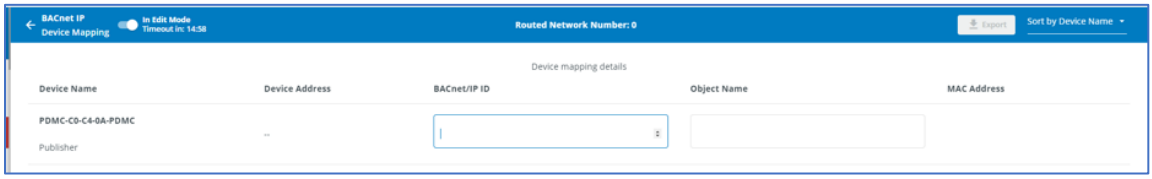


12.5 Mapping

You can view and download the mapping information for your virtual network and virtual devices by clicking View Map (the download arrow is available only in non-edit mode). If you are manually setting IDs, you can do that in the Device Mapping dialog box. You can also download an EPICS .tpi file for each virtual device.

Click Edit to change the name of virtual device (Object Name) and, if you're manually setting IDs, to set the ID numbers. You can sort devices by any of the column types. To download an EPICS file for a device, click the download arrow at the right of the device row.





12.6 SNMP

Simple Network Management Protocol (SNMP) is an application layer UDP (User Datagram Protocol) protocol and is part of TCP/IP protocol suite. SNMP used to manage and monitor network-connected devices in Internet Protocol networks which have SNMP enabled.

- SNMP defines a standard for recording, storing, and sharing information about network devices. SNMP facilitates network management, troubleshooting, and maintenance.
- SNMPv1/SNMPv3 settings are used to enable SNMP and configure the general SNMP protocol settings.
- The Devices that support SNMP includes Meters, Routers, Printers, Switches, Gateways.
- Below are the important parts of SNMP:
  - SNMP Manager
  - SNMP Agent
  - MIB - Management information Base (Used by SNMP Manager to translate the data from SNMP agent into user understandable format)
- Commands: SNMP manager supports multiple commands, the most used are (We support all these below):
  - GET: Ask for Information at Any Time - To get status information from the agent, the manager can issue Get and GetNext messages to request information for a specific variable. Once a Get or GetNext message is received, the agent will issue a GetResponse message to the manager. It will contain either the information that was requested or an error explaining why the request cannot be processed.
  - TRAP: The Most Common SNMP Message - Trap messages are initiated by the agent and are sent to the manager when an important event happens. This makes Traps perfect for reporting alarms to the manager - rather than wait for a status request from the manager when it gets around to polling the agent.
  - SNMPWALK: Get All the Data - SNMPWALK uses multiple Get-Next requests to retrieve an entire tree of network data from a managed object. These are especially useful when using a SNMP tool to read data.
- Use Case of SNMP: Used in Building management system to Monitor traffic of Alarms from Multiple Gateways.
- An SNMP port is an SNMP communication endpoint that identifies, sends, and receives SNMP data transfers, commands, and messages. The default SNMP ports for sending commands and messages are: UDP port 161 connects the SNMP Managers with SNMP Agents (i.e. polling)
- The default UDP is 161 to which SNMP queries are sent and UDP for traps (for e.g. alarms, important notifications) is 162.
- However, you can configure it so that the agent listens to the requests on a different port too.

Configuration

1. To view SNMP settings collapse section under **Settings > Network**, enable SNMP toggle under **Settings > Network > Access Control**.
2. Enter UDP Port for the SNMP traffic on which the SNMP service needs to be enabled.
3. SNMP section has below fields. Both the protocols SNMPV1 and SNMPV3 can be enabled simultaneously from PDMC i.e., SNMP agent pointing to different Trap recipients.
   - SNMPv1: To configure/view SNMP V1 protocol
   - SNMPv3: To configure/view SNMP V3 protocol
   - MIBS: To View pre generated MIBs
4. In SNMPv1 protocol communication between SNMP Agent and SNMP Manager happens through community strings.
5. To configure SNMPv1 follow below steps:
   - Enable V1. (Checkbox to enable SNMPv1 protocol)
   - Read-Only Community String: with default value as Public (The valid range is 1 to 256 alphanumeric and special characters. The community's name acts as a simple authentication feature to restrict the devices on the network that can request data from the SNMP agent. The name functions as a password, and the request is assumed to be authentic if the sender knows the password.)

- Read-write Community String: With default value as Private. (The valid range is from 1 to 256 alphanumeric and special characters. Setting a community name is like setting a password. Only the requests from the machines that identify themselves with this community name are accepted.)
- Trap Recipient Community String: It is used by the trap receiver to determine which traps are accepted from a device. It is "Public" by default. This parameter is particularly useful to avoid unwanted floods of traps from a malicious source. By configuring the Trap Community String, only the relevant traps are processed by the trap receiver.
- Enable Auth Fail: In order to get "Auth fail", user needs to make sure that the Trap recipient community string is valid, else no Auth failure will be detected

Note: SNMPv1 and SNMPv2c provide only simple authentication and do not address encryption. SNMPv1 should only be used in private networks where security is not a major concern. When Enable Auth Fail if there is mismatch in string values for Manager and Agent, it displays AUTH failed Trap. (A different type of entry will be displayed in the Trap section)

6. SNMPV3: Has SHA Authentication and displays below fields. Unlike SNMPv1 SNMP version 3 supports authentication and encryption. This means that SNMP data cannot be read after capture, and only authorized users can access specific object identifiers.
   - Enable V3.
   - Read only user & Read Write User contain below fields: Username & Passphrase (Enter the name that identifies the SNMPv3 user. Usernames can contain up to 32 alphanumeric characters. If you specify SHA as the authentication type, enter the pass phrase i.e Password phrase configured in SNMP users tool to enable the SNMP agent to authenticate the requests sent by the user. The pass phrase must be between 8 and 32 characters in length.)
   - Use Privacy (When Enabled Encryption will be used: Note: PXG1000 uses AES 128 encryption.)
   - Trap Recipients

- **Setting Up Encryption in SNMP Manager**
  - For both SNMP V1 and V3, check if the UDP port in the manager tool matches with that configured in PDMC application, to ensure the successful communication.
  - For SNMP V1: Set up the Read Only and Read-write Community strings which match with the ones set in PDMC application.
  - For SNMP V3:
    - Create a New User and set up a passphrase in the Manager tool (for both Read-only and Read-Write operations) which will be replicated in PDMC application.
    - Set up the desired level of Privilege for the configured users from NoAuthNoPriv, AuthNoPriv, AuthPriv. The same will be used for configuration in PDMC application.

Note: Note that this User along with its privilege will be used for both Read-write operations and receiving the traps in SNMP manager.

7. Trap Recipients for SNMP V3: It is used to configure the SNMP recipient settings like IP, user, and privilege.
   - HostName /V3 Address: use to configure the IP for SNMP recipient
   - V3 User Dropdown: use to select configured user
   - Authorization Groups: use to Select the Authorization Level

SNMPv3 allows user to combine users into groups of different authorization and access privileges. Each group is associated with one of three security levels:

- noAuthNoPriv: No authentication and no data encryption (no security).
- authNoPriv: Authentication, but no data encryption. With this security level, users send SNMP messages that use the SHA key or password for authentication, but not AES128 for encryption.
- authPriv: Authentication and data encryption. With this security level, users send the SHA key or password for authentication and a AES128 for encryption. For groups that require authentication, encryption, or both, you must define the AES128 keys or passwords on the SNMP Users page.
- MIBS: (Note: These are pre generated files and SNMP manager uses MIB file to fetch the SNMP agent Data through

object identifiers. It acts as a bridge between SNMP agent and Manager, SNMP manager traverses the MIB and translates into user understandable format of what is being read by SNMP Agent.)

On click of **View MIBS** button, a modal should be displayed with the following fields:

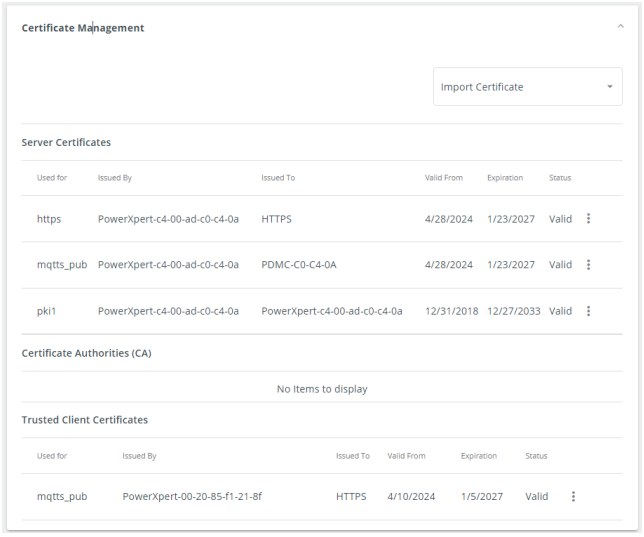
- MIB Name
- Description: contains description of usage of that MIB.
  (Ex: Eaton PCD: MIB: are used for to identify all Gateway Related Parameters.)
- Eaton: PXG MIB: is used to identify Alarms and Traps related Parameters etc.
- Export button: to export the MIB to user's system.

8. MIBS Use case: If user wants to read specific values like current, voltage etc. for connected and communicating devices, they need to upload PCD-MIB in their SNMP tool as SNMP tools by default don't support these channels.

12.7. Certificate Management

Certificate management module is present under **Settings > Network** with certificate services for the following:

- Server Certificate
- Certificate Authorities (CA)
- Trusted Client Certificates



Field	Description
Used For	The Service that uses the certificate.
Issued By	PKI that issued the certificate.
Issued To	Protocol/Data Server that certificate was issued to.
Valid From	Date the certificate became valid. MM/DD/YYYY format.
Expiration	Date the certificate becomes invalid and expires. MM/DD/YYYY format.
Status	Valid state of the certificate.

Important Key Features of Certificate Management:

- View Certificate along with status and validity information
- Generate New self-signed Certificate
- Generate CSR (Certificate Signing request) file
- Import a Certificate from Certificate Authority (CA).

Note: This is required to import MQTT certificate and establish communication with PDMC campus.

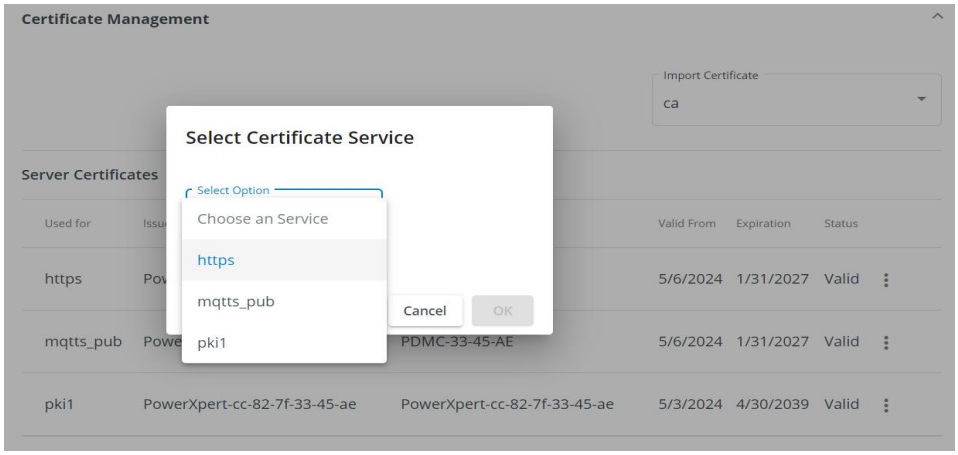
- Export Certificate
- Revoke Certificate. (CA/Client)

Please refer below for more information on these options:

- **View:** Opens up the new screen with Certificate Details with a back button to go back to Certificate Management screen.
- **Certificate Details Screen:** Displays the following details:
  - Clicking on Certificate Issuer
  - Certificate Subject, Validity
  - Details, Import
  - Generate CSR (Applicable only for Server certificate)
  - Export, Generate new self-signed certificate (Applicable only for Server certificates)
  - Revoke
  - Common Name
  - Company Name
  - Division Name
  - City
  - State
  - Country
  - Valid From
  - Expiration
  - Public Key Size
  - Public Key Type
  - Signature Algorithm
  - Serial Number
  - Certificate version
  - Extensions (With option to Copy the text)

Import: To import Certificate for all available services, click on Import certificate, it will open a Modal with below options -

- HTTP
- MQTT
- PKI1



Generate New Self-Signed Certificate (Note: this option is available only for Server Certificates)

Note: Generate New self-signed certificate is applicable only for Server Certificates.

Server Certificates

This feature will revoke old certificate and generate the new Certificate with current date with default Validity i.e., for MQTT and HTTP - 3 Years and PKI1 - 15 years.

Whereas for CA and Trusted Clients options - this feature will revoke/remove the certificate completely.

- Generate CSR (Note: this option is available only for Server Certificates): Clicking on CSR will download a .CSR file to user's system. i.e., a certificate signing request.
- Export: Clicking on Export will Export the Certificate (a .cert file) for the service to the User's system.
- Revoke: Revoke Functionality will Revoke the certificate for Client and Certificate Authority (CA) however for Server option – Revoke would generate a new certificate and replace the existing certificate with today's timestamp.

12.8 Brightlayer

Configuration to connect PDMC to Brightlayer Cloud

Configuring PDMC gateway to fetch IoT data on BLI can be divided into 2 sections

1. Configuring PDMC Gateway to enable IoT
2. Configuring PDMC Gateway on BLI to visualize/read the sent IoT data.

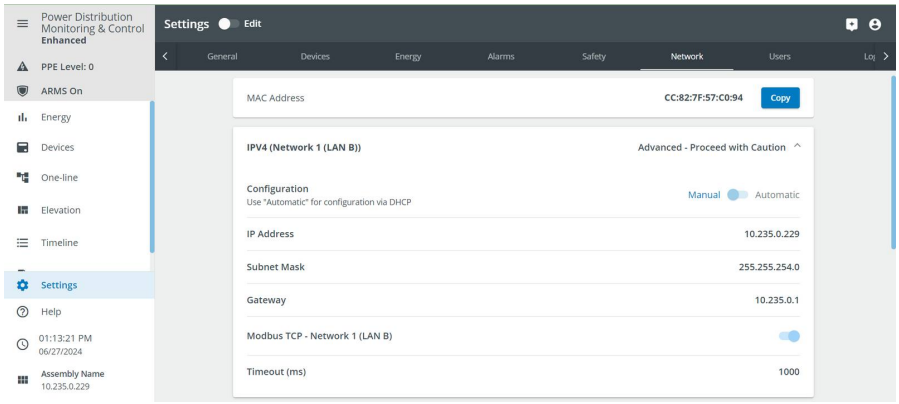
Steps to configure PDMC Gateway to enable IoT connection

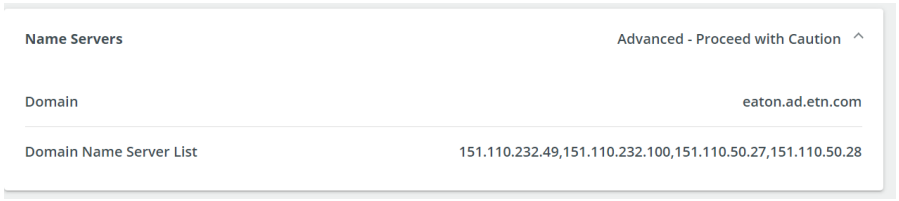
Pre-requisite:

1. LAN/Ethernet is connected to gateway and Gateway UI is up
2. Required devices (For which we need to fetch IoT data) are connected and communicating over PDMC.

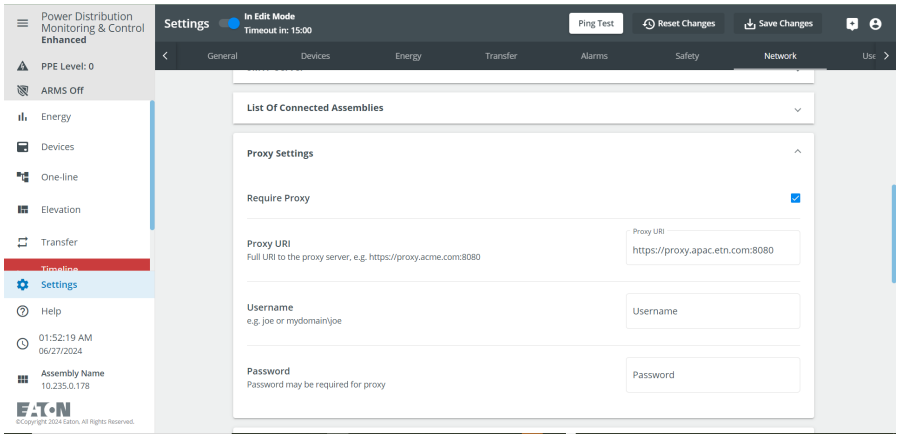
Step 1: Configuring Name Server (DNS)

When Gateway is connected over DHCP mode, the Name Server i.e., Domain Name and Domain Name server list is automatically fetched. However, when we configure IPV4 Network in Manual/Static, we need to provide the Name server details.

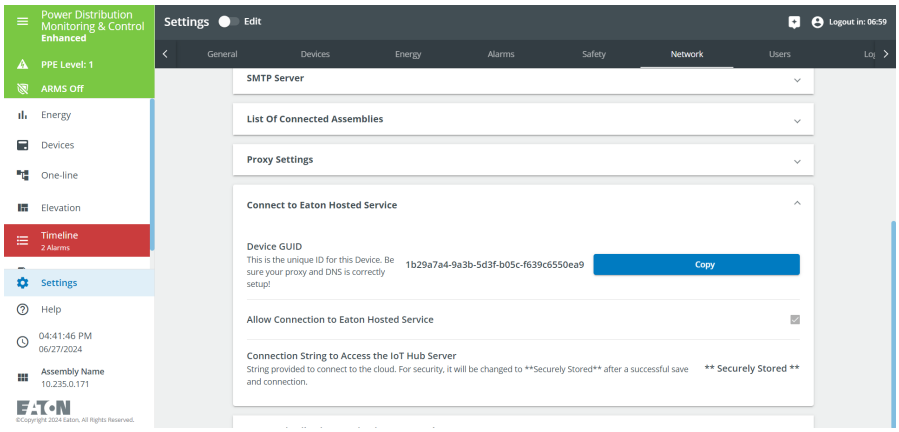




Step 2: Configuring Proxy

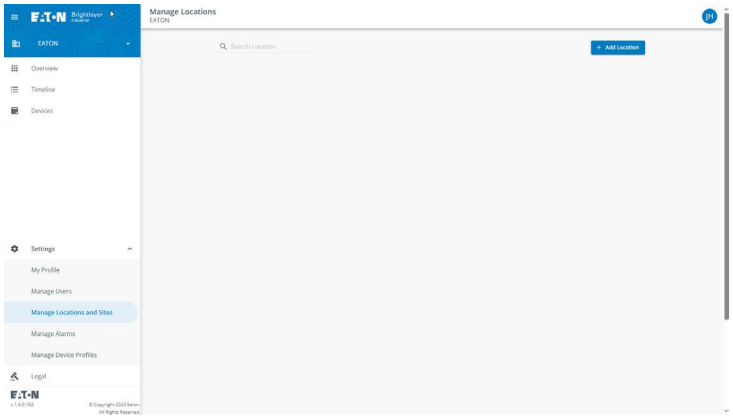


Step 3: Copy the GUID of PDMC Gateway

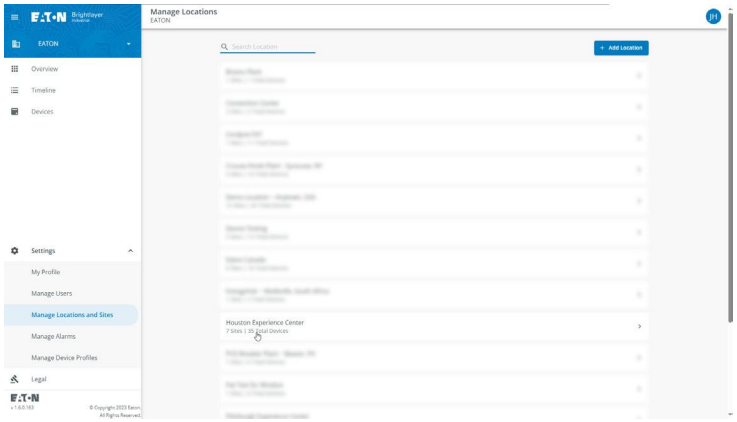


Steps to configure PDMC Gateway on BLI to visualize/read the sent IoT data.

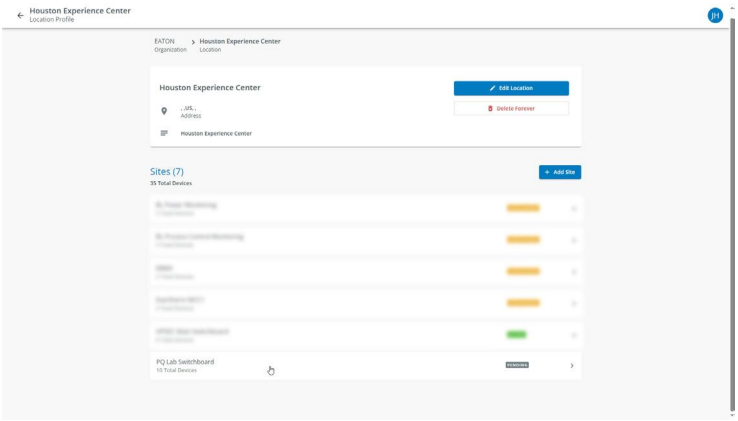
Step 1: Login to the Brightlayer Industrial and Go to Manage Location and Sites



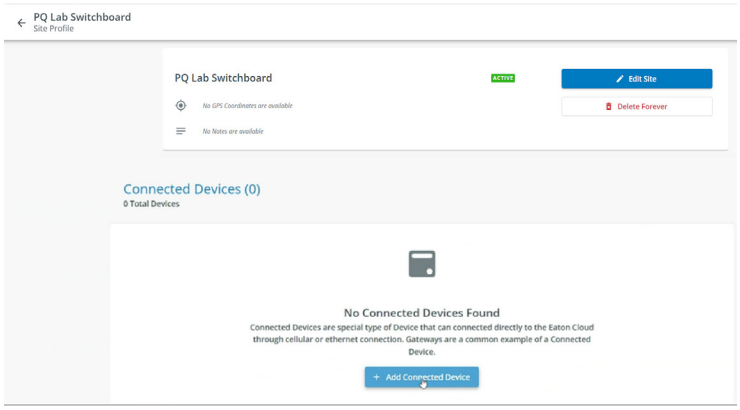
Step2: Choose an existing location or add a new location as per requirement



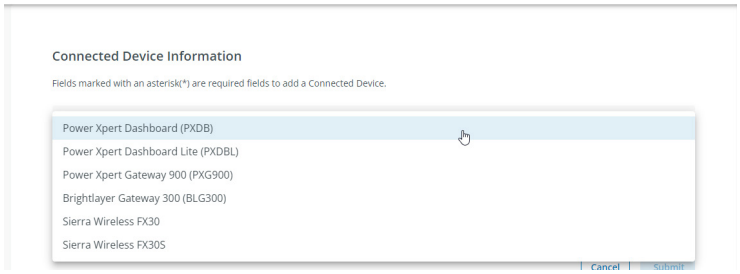
Step 3: Choose from configured Site or add a new site.



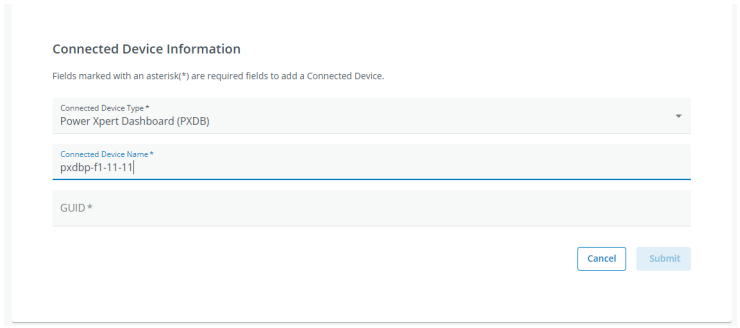
Step 4: Select “Add Connected Device”



Step 5: Select the gateway type “PXDB for PDMCE” and “PXDBL for PDMC”



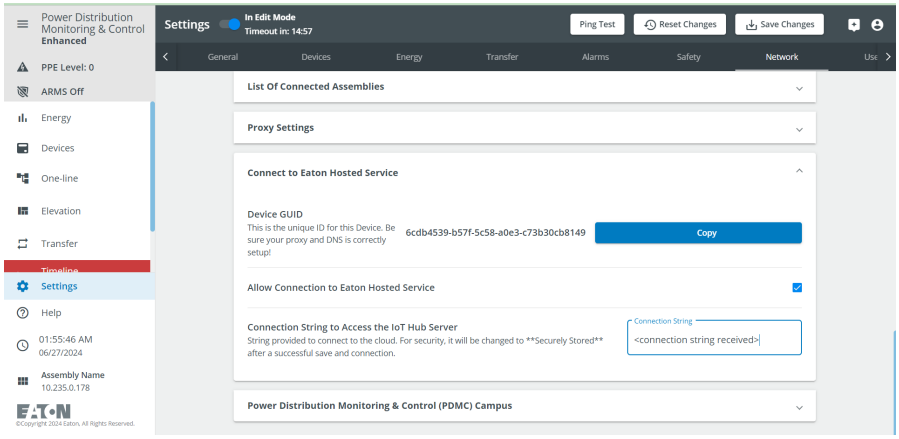
Step 6: Paste (The copied GUID form PDMC > Network > Connected to Eaton Hosted Service) the copied GUID for the Gateway



Step 7: User will receive a Connection string in response.

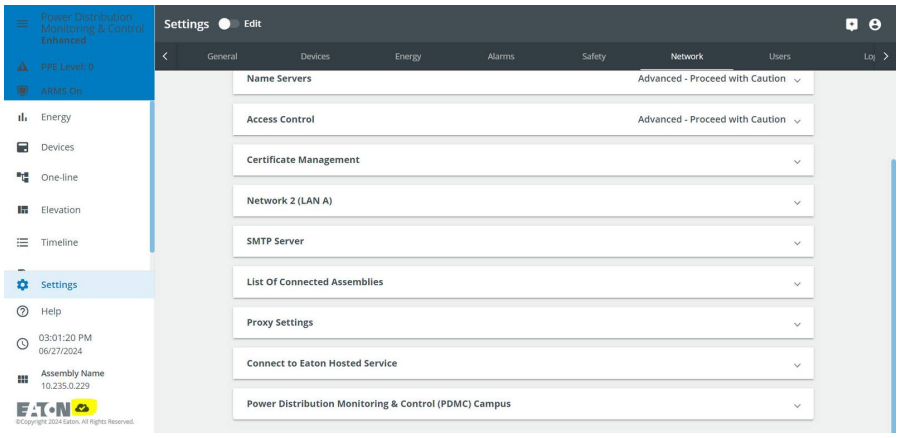
Step 8: Copy the connection string and paste is to PDMC Gateway and check the “Allow Connection to Eaton Hosted Service”

Note: Connection string will be converted to “Secure String” for privacy security



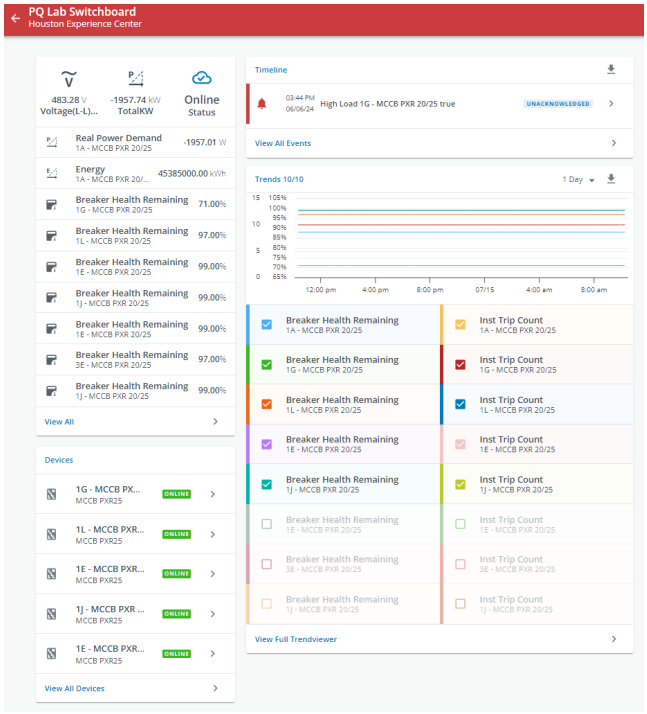
Step 9: Verify IoT connection is successfully established:

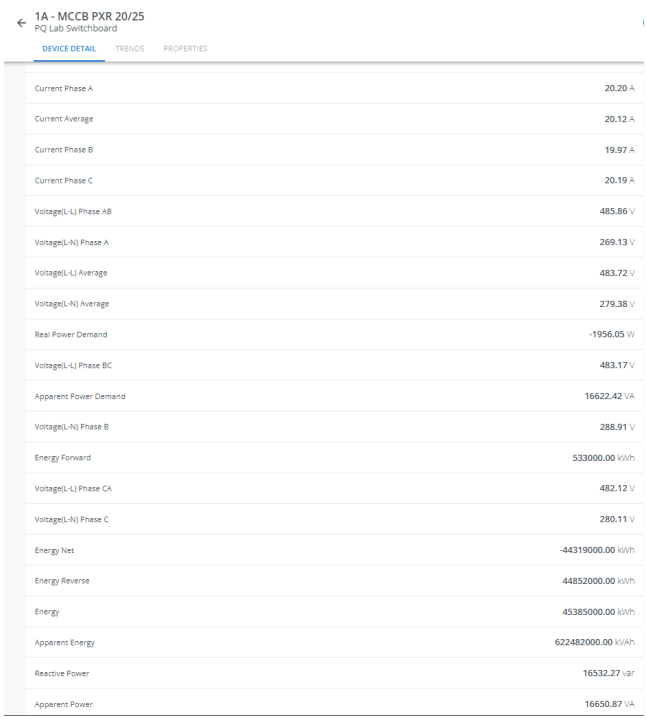
The IoT cloud symbol will show connected



Step 10: Now Go back to Brightlayer Industrial, On the main Overview page we can see the configured PDMC Gateway.

Select the configured PDMC Gateway card. User can explore all the Gateway data fetched via BLC like Connection status, child devices, trended data , alarms etc.

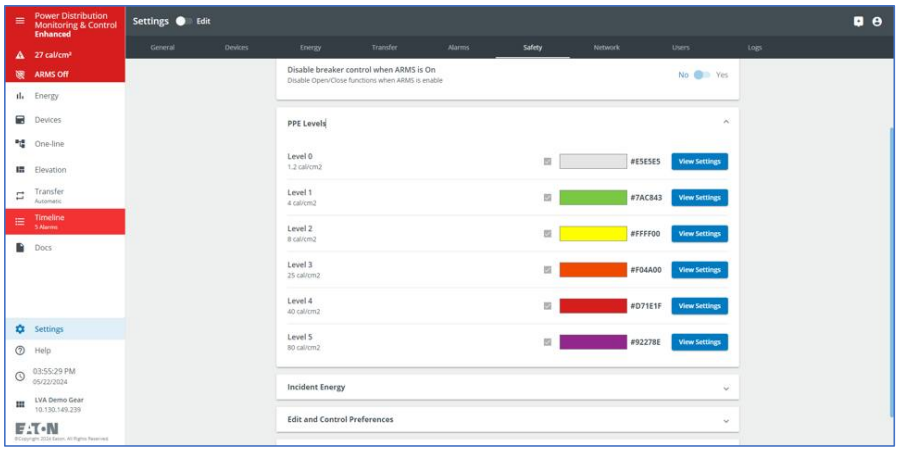




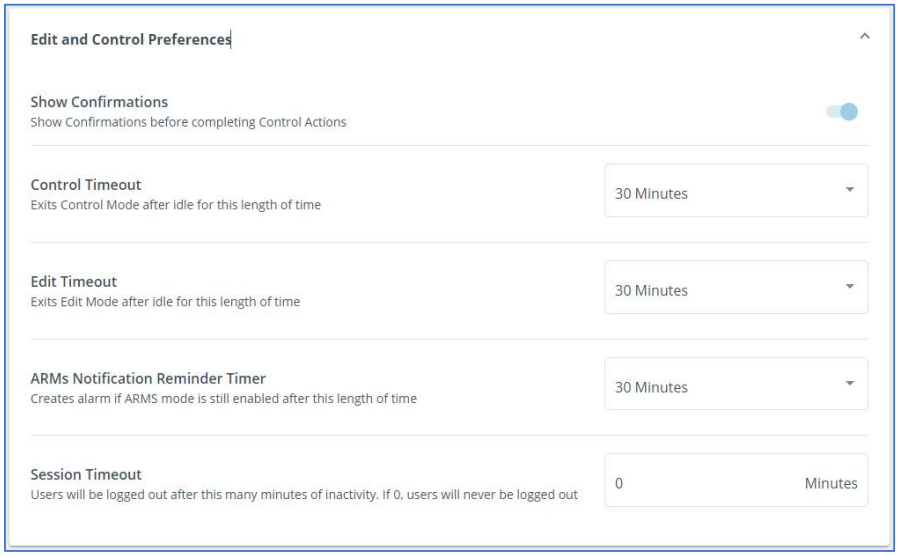
13. Settings > Safety

The **Settings > Safety** tab allows configuration of:

- PPE levels, color, and equipment requirements

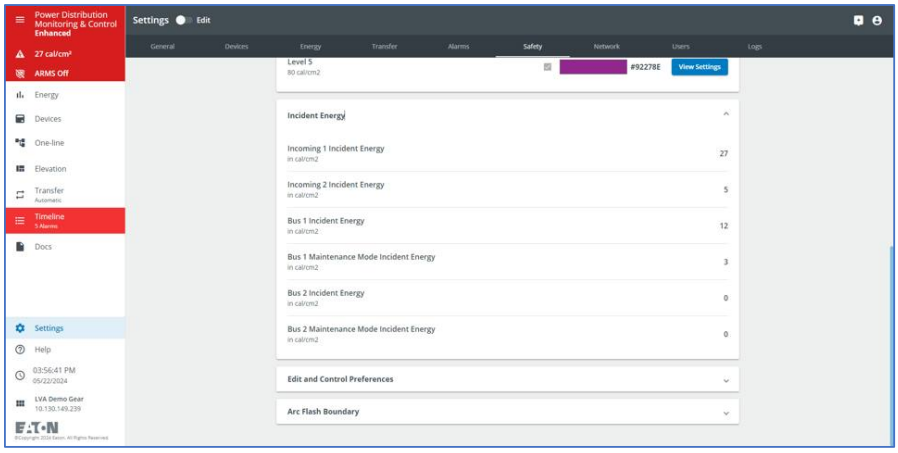


- Edit and Control Preferences which allow user to set Control Timeout, Edit Timeout, ARMs Notification Reminder Timer and Session Timeout

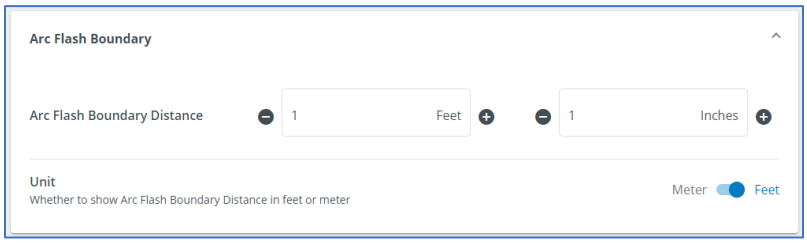


13. Settings > Safety

- Configuration of incident energy levels for Sources and the Bus. There are two incident energy settings for each bus – one for normal operation and one when maintenance mode is active. These values must be set for the incident energy values to display properly based on maintenance mode (ARMS) active.



- Arc Flash Boundary Distance can be entered in Meters or in Feet and Inches

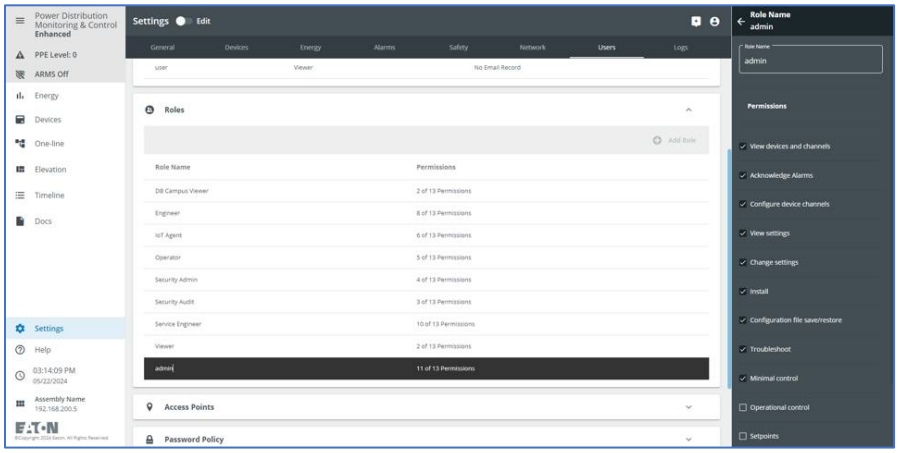


14. Settings > Users

Controlling access to the PDMC is a vital component in any effort to secure it. Many regulatory agencies and standards organizations now recommend or require Role-Based Access Control (RBAC) as part of any access control effort. To support this, the PDMC has a robust set of tools that can be used to create the set of users and role-based permissions you need to comply with security policies in effect at a site. Before jumping into user and role setup, review all policies to ensure a good understanding of the access control requirements for the site. By default, the PDMC comes with three users:

- **Admin:** has access to all functions and can edit anything (admin role). Default password for the admin account is “admin”.
- **User:** can view any information on the tabs but can’t access Settings or edit anything. Default password for user is “user”.
- **dbcampus:** Required to provide navigation from PDMC campus to connected PDMC assemblies. Default password is “Campus*1” and this can be changed by the user for maximum security.

Before doing anything else, change the default account names and logins. Not only are these default users not compliant with RBAC, keeping them is a security vulnerability. It is a best practice to replace these accounts with RBAC compliant ones to meet the needs of your security policy.



14.1 Building Roles

You should define roles that fit your organization’s security policy. Permissions can be appropriately assigned to each type of role. Users should be assigned roles, such as Engineer or Operator, based on their responsibilities. Each user can have only one role.

Roles are typically named for the job function they represent. The default set of roles includes job functions such as Operator and Service Engineer. You can create your own roles or redefine the defaults to fit your own security policy.

To create a role:

1. On the **Users** tab in **Settings**, click **Edit**.
2. Click **Edit**.
3. Click **Add Role**.
4. Enter the **Role Name** in the sidebar. This should describe the job function so that everyone in your organization knows what tasks this role is required to do.
5. Select the various permissions the role requires. Permissions are detailed in the following table.
6. Click **Save**.

Permission Details	
View devices and channels.	View devices and information from their channels. Every user gets this by default.
Acknowledge alarms	Click the Acknowledge button and enter a note about the alarm; e.g., what caused the alarm and what action was taken.
Configure device channels	Enable trending for channels (where trending is available), set alarm triggers and enable or disable individual channels.
View settings	View, but not change, PDMC settings. This is automatically selected if you select Change settings.
Change settings	Change PDMC settings in the various Settings tabs.
Install	Install devices in the PDMC and update the PDMC firmware.
Configuration file save/restore	Save all configuration information to a file and set all PDMC configuration parameters from a configuration file.
Troubleshoot	Remotely reboot the PDMC PXG1000. A role with this permission may be required by Eaton personnel to service/troubleshoot the PDMC PXG1000.
Minimal control	Issue device commands that don't have control capabilities. The following figure shows the set of device commands available for the Eaton EDR 5000 with both this permission and with the addition of the Operational control permission.
Operational control	Issue device commands, such as opening or closing breakers or resetting breaker trips. Also associated with configuring DIM KYZ settings.
View users	View users and their assigned roles. This includes anything that lists users, such as the Audit logs. Also, view the individual permissions assigned to a role.
Manage users	Create or delete users, as well as assign a role to each user. Also, create and delete roles and assign permissions to roles. View users is automatically selected if you select Manage users, as you must be able to view the users to manage them.

14.2 Creating Users

1. On the **Users** tab in **Settings**, click **Edit**.
2. Click **Add User**.
3. Fill in the fields on the sidebar. Note that you can assign only one role to each user.
4. Click **Save**.

14.3 Deleting Admin and User

After you've set up the various roles and users, log in as a user with a role that contains Manage users. You can now delete both the user and admin users and the admin role. Users can only be deleted if they are not currently logged in.

To delete a user:

1. On the **Users** tab in **Settings**, click **Edit**.
2. Click the user that you wish to delete.
3. In the sidebar, click **Remove**.
4. Click **Save**.

You can delete a Role if it is not assigned to any users. If you get an error a message when you attempt to delete it, check if any users are assigned to that role and then reassign them.

To delete a role:

1. On the **Users** tab in **Settings**, click **Edit**.
2. Click the user that you wish to delete.
3. With the User or Role selected, click **Remove** to delete it. If the Remove button isn't active, the role is in use or the user is logged in.
4. Click **Save**.

14.4 Accessing Logs

- You can restrict or provide access to the various Audit Logs using permissions. The following table lists each of the Audit Logs and the permissions required to access them. Any of the listed permissions will allow the user to download that Audit Log file.

Audit Log Name	Any of These Permissions Allow Log Download
Legacy	View users
User	View users
Device	View users, Install, Configure Device Channels, Troubleshoot
Configuration	View users, Install, Configure Device Channels
Session	View users
Command	View users
Update	View users, Install. Troubleshoot, Configuration file save/restore

14.5 Roles and Permissions Cookbook

The following are examples of how to combine permissions to create roles that make sense in your organization.

The examples explain four of the default roles in the PDMC. These default roles may not be a good fit for your security policies; however, discussing what they allow and how the permissions work can help you in picking the right permissions when you create your own roles.

14.6 Security Audit

This role must be able to view the various audit logs for the system as well as verify the set of users and their roles.

In addition, this role must allow the user to verify the settings in the PDMC. However, unlike a true administrator, users with this role only view this information; they cannot edit any settings. You can grant these capabilities through the View users permission. View users allows the user to view the Security tab under Settings. It also lets them see all the logs on the **Settings > Logs** tab.

14.7 Security Admin

This role must be able to create/delete users and roles, as well as change user passwords or other settings. Its permissions (and capabilities) are like the Security audit, but with the added capabilities for user administration. So, in addition, the following permission has been added:

Manage Users: This lets the user not only view the current users and roles (View users is selected automatically when this is selected), but create, edit, and delete them as well.

14.8 Engineer

An engineer must be able to control everything related to the various devices that are connected to the PDMC PXG1000. However, an engineer doesn't need to access any of the security or maintenance features of the PDMC PXG1000. Therefore, that role has the following permissions that are related to working with devices.

Change Settings: This setting permits the user to view Settings (which is automatically selected when Change settings is selected). It also unlocks the various fields in the sidebar within the Network tab.

Install: This lets the user add and edit devices.

Configure Device Channels: As the name implies, this permission allows the user to edit the list of device channels.

It also allows the user to remove a device and configure the Modbus and INCOM ports.

Acknowledge Alarms: The user can click the Acknowledge button and enter notes about the alarm.

Minimal and Operational control: The user can issue all available device commands through the Choose an action menu.

14.9 Service Engineer

In addition to what an Engineer can do, the Service Engineer must be able to troubleshoot and remotely reboot the PDMC PXG1000. This user must also be able to save and restore a configuration file. So, added to the permissions granted an Engineer, the Service Engineer also has **Troubleshoot** (remote rebooting) and **Configuration file save/restore** permissions. Eaton field service and customer support personnel may require a user with this role to help troubleshoot a PDMC PXG1000.

14.10 IoT Agent

An IoT agent role has the following permissions that are related to working with devices.

Change Settings: This setting permits the user to view Settings (which is automatically selected when Change settings is selected). It also unlocks the various fields in the sidebar within the Network tab.

Install: This lets the user add and edit devices.

Minimal and Operational Control: The user can issue all available device commands through the Choose an action menu.

14.11 Password Policy

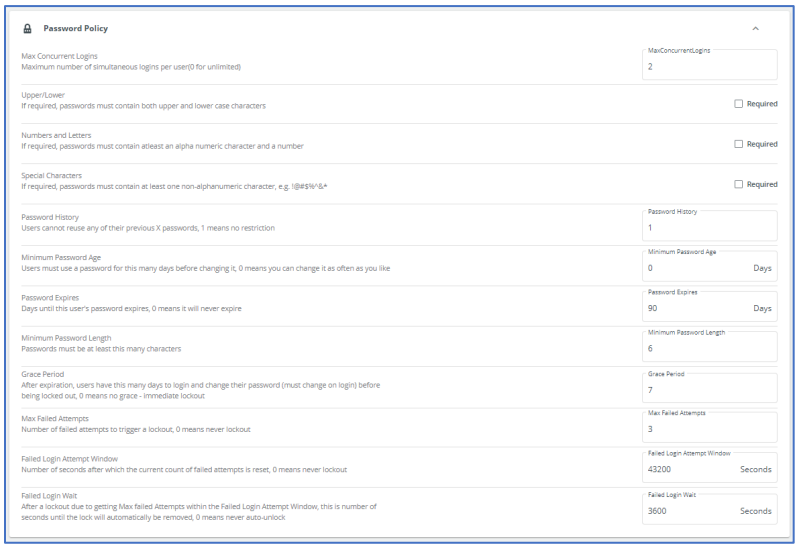
The security provided by restricting access to authorized users is only as strong as uniqueness of the user passwords. The Password Policy settings provide a mechanism for security administrators make it more difficult for unauthorized users to guess the passwords of their users. It also provides a mechanism to control how often users must change their passwords. This single password policy applies to all users in the system.

To change the password policy:

- On the **Users** tab in **Settings**, click **Edit**.
- Expand the **Password Policy** section.
- Change any of the settings
- Click **Save**.

After changes are made, any attempts to make password changes (e.g., from the **Users** tab or **Change Password** dialog) will be restricted by the new policy settings. Existing passwords will continue to work even if they don't meet any new content and length requirements, but all accounts will be immediately affected by the password history, age and grace period settings. New password requirements may be enforced to via **User Password Management** settings. Users will be prompted to change their password at their next login. New password policy settings are also available for users are Max Failed attempts, Failed Login Attempt Window and Failed Login Wait.

Note: PDMC now supports minimum password length of 6.



14.12 User Password Management

The security administrator may use the **User Password Management** settings to view and modify password expirations and lockouts for each user individually. The per user management provides a way to extend the global **Password Policy** choices; the administrator can override the **Password Expires** time or choose a **Expiration Date** instead. By setting a fixed expiration date, the administrator is saying the user's account will be locked at the start

of the fixed date versus just requiring a password change as is the case when the **Password Expires** field is set to a number of days.

This sidebar also provides the ability to immediately lock (or unlock) a user account and the ability to force a user to change the password on next login (vs. waiting for the password to expire).

To change a user's password settings:

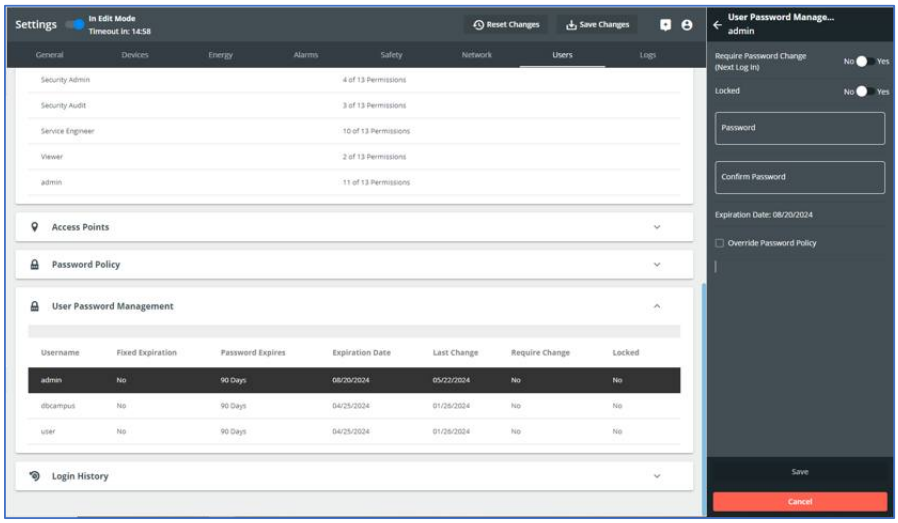
1. On the **Users** tab in **Settings**, click **Edit**.
2. Expand the **User Password Management** section.
3. Click the user that you wish to modify.
4. Change any of the settings in the sidebar.
5. Click **Save**.

Note: When a user's account has become locked due to password expiration, the account must both be unlocked and the password must be changed so that it doesn't immediately get locked again. It's also typical to set the "**Require Password Change**" flag to so the user can replace the password with one of their own.

14.12.1 Password Expiration Changes

If the password expiration is changed from “never” to some finite time on an existing product, existing users who haven’t changed their passwords in a long time may get locked out immediately at that time. That can be a problem, especially if this change happened automatically as part of a firmware upgrade. (The same problem can happen if you change password expiration in the UI). So to mitigate this, the following actions cause a user’s expiration date to be re-evaluated:

1. Changing user’s password.
2. Editing user’s properties.
3. Editing global password policy (whether expiration time or something else like “require special characters”) re-evaluates all users.



15. Settings > Alarms

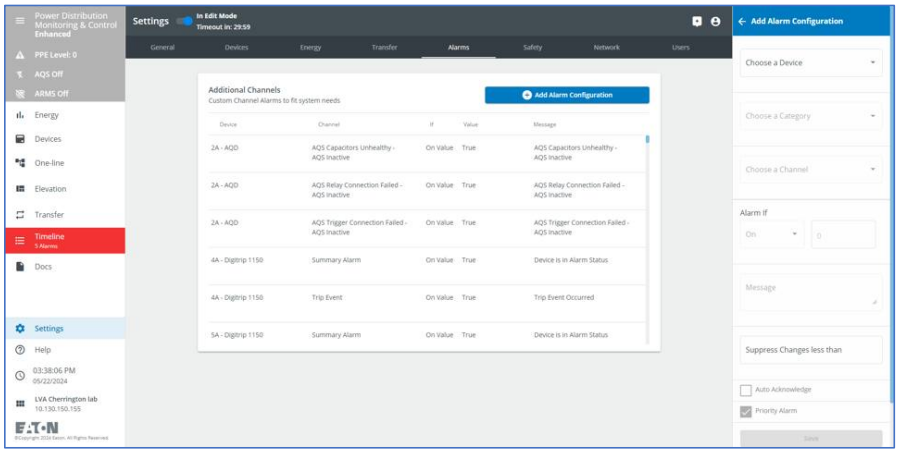
You can set custom alarms for any channel on any connected device directly in the PDMC. Each alarm configuration references values on a single channel and specifies either:

- A High or Low alarm value for analog channels.
- A Between value for power factor channels (explained below).
- An On Value trigger that matches a state in a multi-state alarm.

If you wish to have both High and Low alarms set for a single channel, create an alarm configuration for each. You configure alarms through Settings, on the Alarm Settings tab.

15.1 Creating an Alarm Configuration

1. Click Edit.
2. Click Add Alarm Configuration. A sidebar appears with all of the alarm settings.
3. Select the device from the drop down list.
4. Select the channel type; i.e., Voltage, Current, Power Factor, etc.
5. Select the individual channel.
6. For analog channels other than power factor, choose whether it's a High or Low alarm, then set the alarm trigger value. For Boolean channels, set whether to trigger an alarm on either True or False. Other discrete multi-state channels may have more than two options, e.g. the Status channel on many of the devices. See below for more about multi-state alarms and alarm levels.



Power factor channels have a "Between" setting that applies to both positive and negative power factor. So, if you set power factor to 0.90, any value less than that will raise an alarm for both positive and negative power factor. Like other analog alarms, power factor alarms are automatically assigned a level.

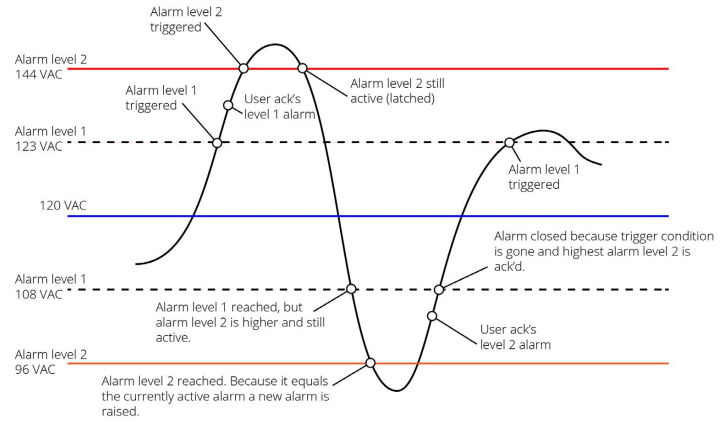
7. Type a custom message that will appear with this alarm.
8. Set Suppress Changes less than to eliminate alarm "chatter" (rapid oscillation above and below the trigger point) by setting a dead band value.
9. Choose if the alarm should be acknowledged automatically if the value falls outside the trigger range.
10. Choose if the alarm will be a Priority Alarm. Priority alarms are shown with an exclamation point and you can filter on these in the Alarms tab.
11. Save the Alarm Configuration when you're done. If you're going to create a similar configuration, choose Save and Duplicate Alarm. You can then just edit the fields in the copy to create a new configuration.

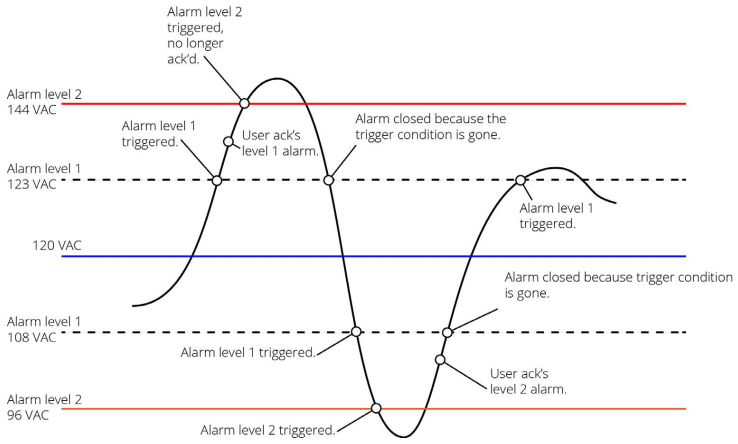
15.2 Alarm Levels

Every channel can have multiple alarm triggers (not just High and Low). Each time you define an additional High or Low alarm trigger point for a particular analog channel, it automatically becomes a new alarm level. The alarm level is assigned based on the comparative values of the trigger settings. For High alarms, greater values have higher levels. The opposite is true for Low alarms. The following table shows how this works:

Type	Value	Level
High	144 VAC	2
High	132 VAC	1
Low	108 VAC	1
Low	96 VAC	2

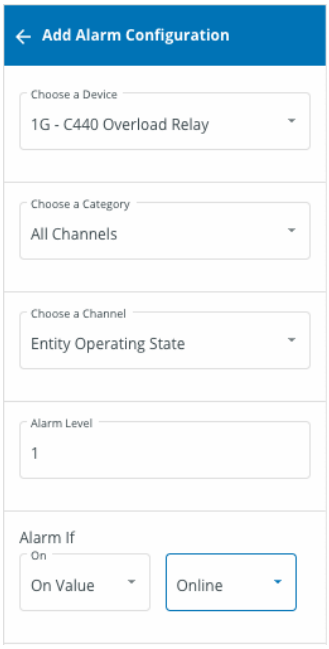
The following figures show the alarm logic and how this is affected by alarm levels. Examples for both latching and non-latching alarms are given.





15.3 Setting Alarm Level for Multi-State Channels

Multi-state channels, such as Status, can't be automatically assigned a level. Instead, you must assign the levels yourself when configuring alarms for such channels. When configuring multi-state channels, the dialog box includes an Alarm level field for this purpose. For multi-state alarms there is no High or Low alarm, just a single set of alarm levels. Note that you cannot assign the same level to two alarms for the same multi-state channel.



16. Settings > Logs

The following audit logs are available to users, depending on the permissions their account has been granted, and provide:

commEvent
Older style audit log format supported by previous versions of the PDMC's firmware. Newly logged data will appear in the more specific log files.

User
Content related to changes made to users or roles, including password changes.

Device
Content related to the addition, removal, enabling, and disabling of connected devices.

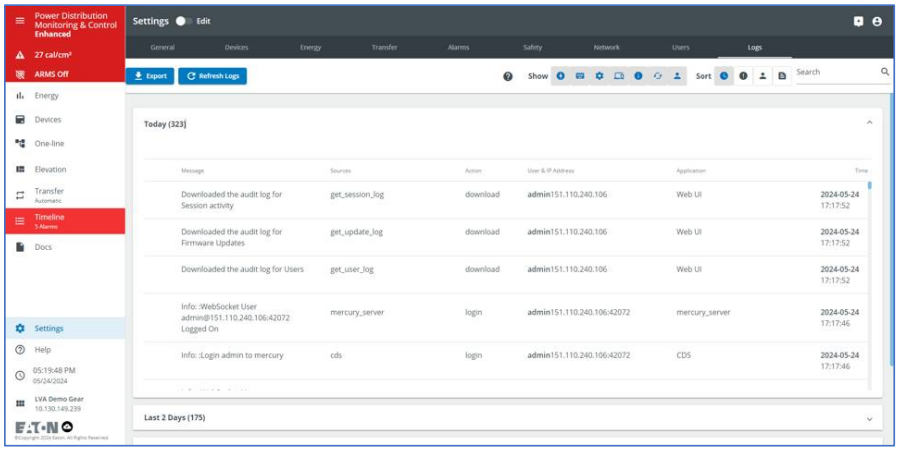
Configuration
Content related to individual changes to system setup; e.g., device or channel settings, time zone, etc.

Session
Content related to logins/sessions, including login failures and invalid (possibly malicious) access attempts. This also includes valid and invalid non-HTTP protocol sessions, including Modbus/TCP and BACnet/IP.

Command
Content related to user actions, such as commands and PDMC reboots.

Update
Content related to the application of firmware updates, configuration file uploads, and factory resets of the PDMC PXG1000.

User can sort these logs on the basis of Time, Status, User and File. Also Export button is provided on the Action bar to allow the download of all Logs entries in .csv format.



Permission Details

Type	Permissions
CommEvent	Troubleshoot Permission is required.
Command	Security Read Permission is required.
Configuration	Security Read Permission is required. Install Permission is required. Data Admin Permission is required.
Device	Security Read Permission is required. Install Permission is required. Data Admin Permission is required. Troubleshoot Permission is required.
Session	Security Read Permission is required.
Update	Security Read Permission is required. Install Permission is required. Data Admin Permission is required. Troubleshoot Permission is required.
User	Security Read Permission is required.

17. Advanced Administration

Several advanced administration actions are available in PDMC. You can access these actions depending on your user permissions when edit mode is active.

Advanced Network Settings
This is a link to the Network Access tab under Settings.

Configuration File Save/Restore
You should always save a configuration file to your PC after setting up the PDMC. You can also reload configurations with this function. If you are replacing an original PDMC PXG1000, this feature may also be used to restore a saved configuration from the older PDMC PXG1000. This is available on the **Settings > General** tab.

Firmware Update
You can upload the new firmware file using this function. When you upload new firmware, the PDMC PXG1000 will reboot after updating itself. This is available on the **Settings > General** tab.

Reboot PDMC
This provides a handy way to reboot a PDMC PXG1000 without accessing the box directly. You'll lose web contact during the rebooting process. This is available on the **Settings > General** tab.

System Inventory
Provides an inventory of all the devices attached to the PDMC PXG1000, including information such as connection port, status, and firmware level. This is available on the **Settings > Device Tree** tab.

Ping Test
Use this to verify that there is an Ethernet connection between the PDMC PXG1000 and another device, NTP server, SMTP server, or Power Xpert Insight software. This is available on the **Settings > Network** tab.

Audit Logs
The audit logs are available on the **Settings > Logs** tab. The logs can be filtered and sorted.

Download Comm Events & Download Comm Statistics
These download diagnostic information about PDMC communications as CSV files. This is available on the **Settings > Device Tree** tab.

18. Waveforms

The PDMC can download waveforms:

- From any attached INCOM device that's capable of capturing waveforms. The following Eaton devices are supported: FP-5000, FP-6000, IQ Analyzer Logger 6400/6600, MPCV Relay, Digitrip 1150.
- If Enable Waveforms in the device configuration for that device is selected.

Waveform support is also available from Eaton's PXM2280 and PXM2290 power quality meters when connected to the PDMC via Modbus RTU.

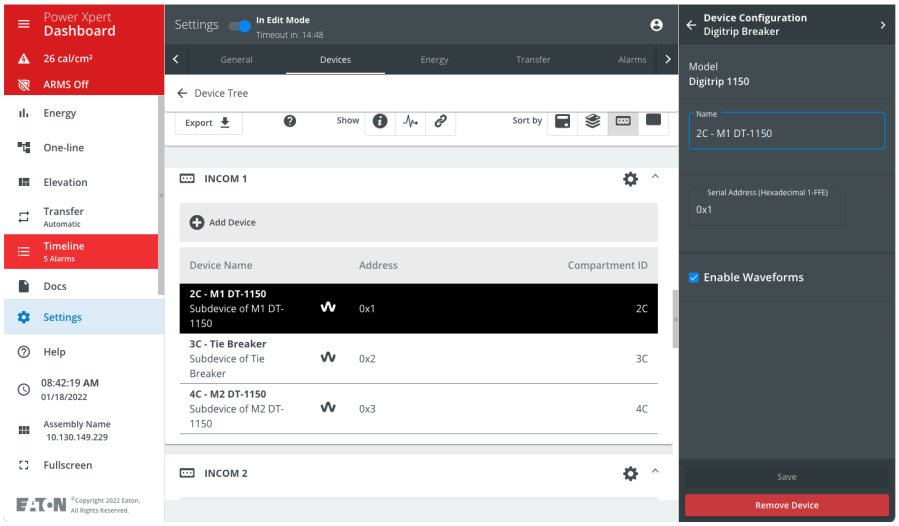
18.1 Waveforms Captured by Devices

Devices have their own waveform capture settings that control what triggers a waveform. You can't set this from the PDMC. Also, what information is captured varies on a device-by-device basis. As such, the first step in using waveforms within the PDMC is to configure each of the devices through their own interfaces, then enable Waveform capture in the PDMC.

18.2 Enabling Waveform Capture in the PDMC

Navigate to Settings à Devices à Device Tree and select a device that supports waveforms to open the Device Configuration sidebar.

- Enable Edit Mode.
- Select Enable Waveforms in the sidebar.
- Click Save Device Configuration.



18.3 Captured Waveforms List

The PDMC maintains a list of all waveforms captured for each device, from which you can download any waveform as a set of Comtrade files. The easiest way to access the list is through the **Settings > Device Tree** tab. Each INCOM device that has captured waveforms shows the following waveform icon. You can click the icon to launch the Waveforms list popout.



To help find the waveform you're looking for you can sort the list by time captured or received, as well as by cause. If you're looking for a specific file, you can also sort by filename. Select the checkbox for each waveform that you wish to download or select the master checkbox (beside the Export button) to select all of the waveforms.

When you click Export, you'll see a file Save As dialog box for each selected waveform so that you can rename each file or save them to various folders. If you have a large number of files selected, you may wish to download these a few at a time;

otherwise, you'll have to deal with an equally large number of Save As dialog boxes.

18.4 Waveform Files

The Waveform files themselves are packaged in a .tgz compressed tar archive file. Unpacking this in Windows

requires special software such as 7-Zip. When you unpack the file, you may need to first decompress it and then unpack the resulting .tar file. Each waveform archive unpacks into a set of four files with the following

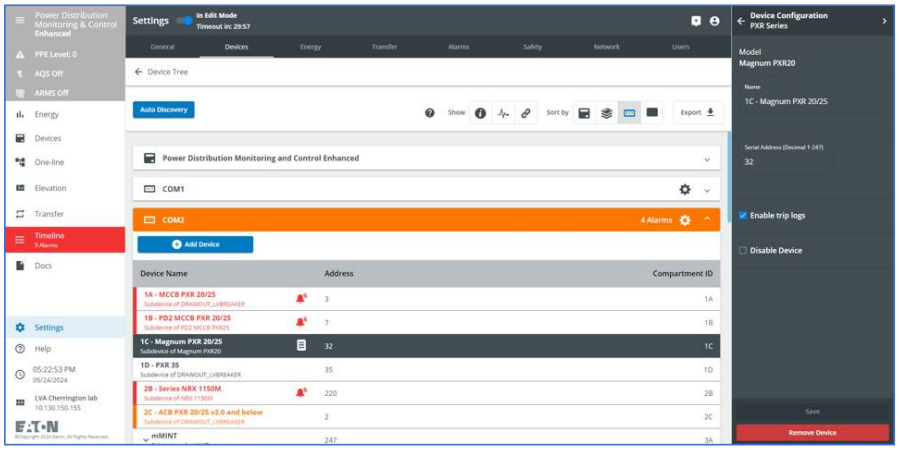
extensions. .hdr, .cfg, .dat, and .inf. Your Comtrade viewer may not require all of these. Export Waveform saves a Comtrade file with waveform data from the chosen captured waveform(s) to your computer file system.

18.5 Manual Waveform Capture

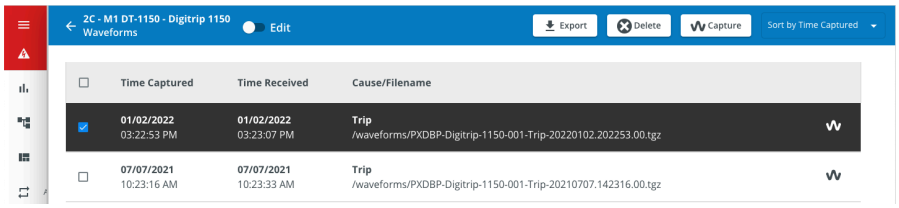
If you wish, you can initiate a waveform capture from an enabled device. This is under the Command section of the Choose an Action menu on the Stack. The stack is available on the **Settings > Device Tree** or Oneline tab when the device is selected. You can also access the menu on the Device Details sidebar.

19. PXR Trip Logs

Trip log contains channel information at the time when Breaker is tripped. This feature is applicable for PXR devices (ACB, MCCB, PXR Magnum etc.). This feature can be enabled by navigating to **Settings > Devices > Device Tree**. Select the Trip Unit and check Enable Trip log check box for PXR and Check Enable Waveform for Digitrip from Sidebar.



Once the feature is enabled then a document icon is shown for Trip units. Clicking on it will navigate to Waveforms stack where the log files are listed. User here can Export the logs and can capture them and can delete them. Also, the data can be sorted based on Time Captured, Time Received, Cause and Filename.



20. Cybersecurity Hardening

The PDMC Eaton recommends that all customers using the PDMC check periodically for updated firmware which is available on Eaton's website at www.eaton.com/PDMonitoringControl. From this landing page and select the "Software, firmware, and applications" button to obtain the latest released firmware. As new PDMC firmware is made available, Eaton recommends that PDMC's be upgraded to the latest available release of firmware to stay up to date with Cybersecurity vulnerability remediations.

For added security, you should always disable HTTP access and enable HTTPS access for the PDMC. To use HTTPS to connect to the PDMC, download the certificate file from the PDMC. Install the certificate in the Trusted Root Certification Authorities store. This is not the default store that the certificate installation wizard will choose, which is why you need permission to manage certificates. Both enabling user management for certificates and installing a certificate require administrative privileges for the PC. If you don't have such privileges, you'll need to contact your IT organization for assistance before proceeding.

Click here for additional details for securing your PDMC.

21. Connecting to the Web Interface

After the PXG1000 is fully configured, users can interact with the web interface. This is best viewed in Google Chrome (current version). Users should have a screen resolution of at least 1280 x 1024 pixels. The connection is https://machine_name where machine_name is the machine name or IP address of the PDMC. You can also use HTTP to connect, if that's enabled, although Eaton recommends that you only connect via HTTPS. For information about enabling HTTPs, see "Cybersecurity Hardening the PDMC".

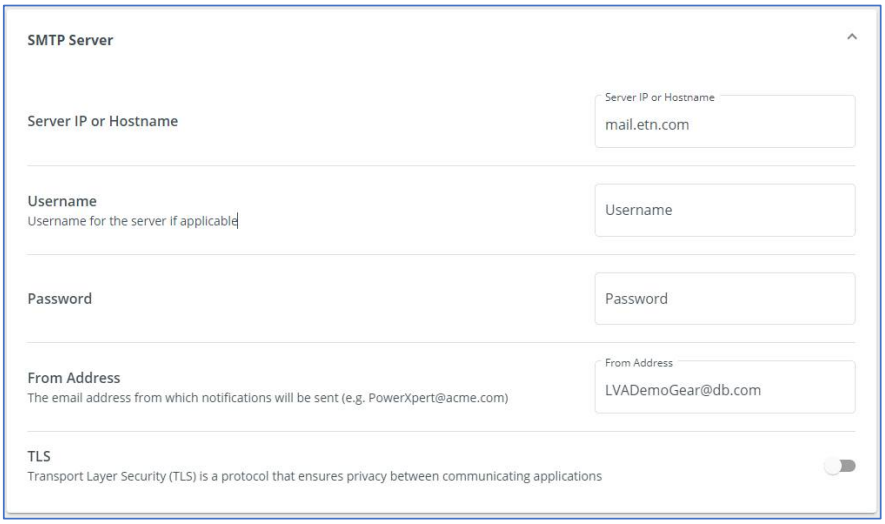
22. Notifications

- Notifications provide email alerts for alarms and related activities, such as acknowledging an alarm, or when an alarm is cleared. Email can include attached event, trend, and/or data logs.
- You set up notifications through Settings. Choose Edit to modify the settings.

22.1. Setting up the Email Server

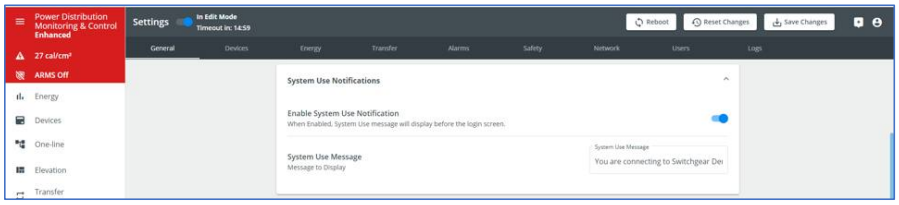
The PDMC can send email for alarm related events. You can set up your email notifications on the Network tab under Settings. Choose edit to access the setup fields. Setup requires that you define:

- The address of the email server.
- The username and password (if required) for the email account used by the PDMC. It's best if you use a username with a password that doesn't change. If the password for that user changes, you'll need to reset it here. Otherwise, Notifications will no longer work.
- The port. You append this to the IP address/machine name by first entering a colon. For example: 10.20.30.1:587. By default, port 25 is assumed (which is typical for unsecured connections, although it could be 587). If you are using SSL, the port is typically either 465 or 587. Always check with your IT group for the proper port number to use.
- The address from which the PDMC sends email.
- Whether SSL/TLS is required.



22.2 System Use Notification

Use the System Use notification to display a system use warning whenever a user first accesses the PDMC web interface. This message appears before logging in and the user must acknowledge it. Companies can use the warning to let the user know who is legally allowed to log into this system and state what are the legitimate uses of this particular PDMC device at this facility. It is available on the **Settings > General** tab.



23. Factory Reset - Part 1

The Factory Reset function is included on the Tools page. It provides the ability to reset to factory default conditions and a way to recover a PDMC PXG1000 under the following conditions:

1. The PDMC password is not known
2. The PDMC IP address is not known
3. Network connectivity issues are preventing connection to the PDMC (configuration issues, etc.)

24. Factory Reset - Part 2

A. Attach a network cable from the Gateway to the Ethernet port on a PC.

B. Configure the IP address and subnet settings for PC's ethernet port from **Network & Internet Settings > Ethernet > Change Adapter Options > Select Ethernet port (1/2/3) > Select protocol as, Internet Protocol Version 4 (TCP/IPv4) > Properties > Edit** the settings.

C. Configure the PC Ethernet port to static IP address 192.168.200.1 and subnet 255.255.255.0.

D. Reboot/Power Cycle your device.

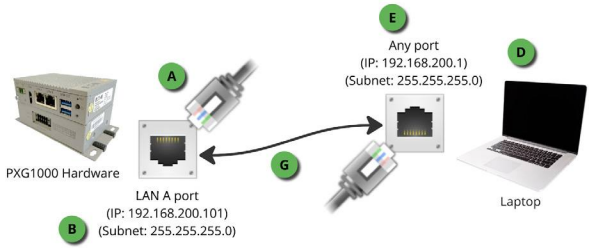
E. Next access the gateway over IP 192.168.200.101 and subnet 255.255.255.0 with recovery user credentials which will be created only in this mode:

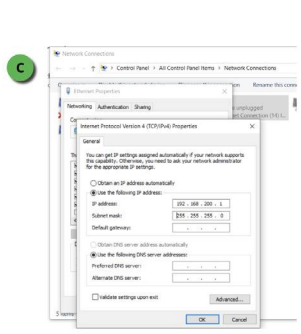
Username: RecoveryUser
Default Admin Password: (See page 6 for default password instructions)

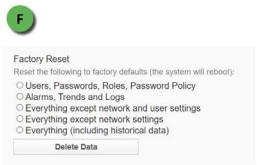
F. Navigate to 192.168.200.101/tools.html to perform following three types of reset:

1. Users, Passwords, Roles, Password Policy
2. Alarms, Trends and Logs
3. Everything except network and user settings
4. Everything except network settings
5. Everything (including historical data)

G. After any type of reset, the gateway will reboot. During reboot user has to remove network cable to perform a normal boot-up sequence.



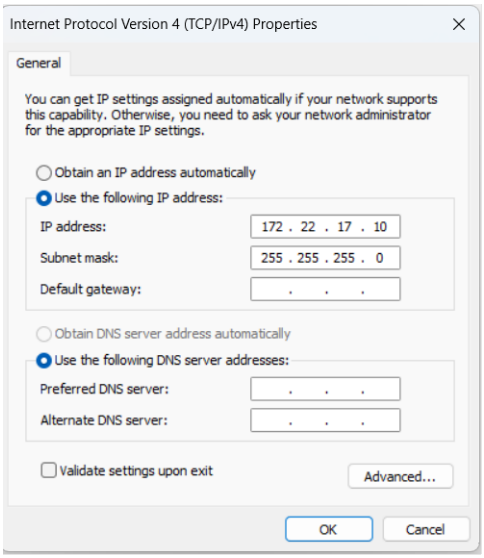




24. Factory Reset - Part 3

A. Configure the IP address and Subnet Settings for PC's Ethernet Port from the **Network & Ethernet Settings > Ethernet > Change Adapter Options > Select Ethernet Port (1/2/3) > Select Protocol as Ethernet Protocol Version 4 (TCP/IPv4) > Properties > Edit the Settings**.

Then Configure the PC Ethernet Port to Static IP address 172.22.17.10 and Subnet Mask 255.255.255.0.



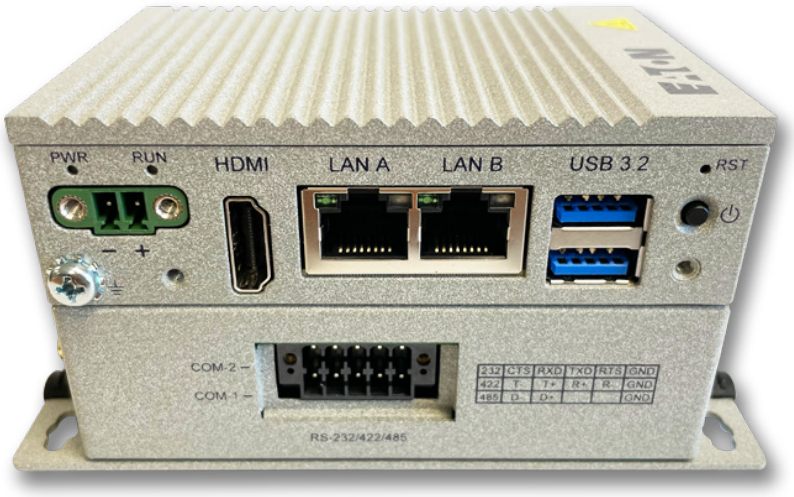
B. Check the run led status on gateway/hardware. (It should be green) and then access the gateway over IP 172.22.17.254 in a web browser and login using the default credentials.

- **Username:** admin
- **Default Admin Password:** (See page 14 for default password instructions)

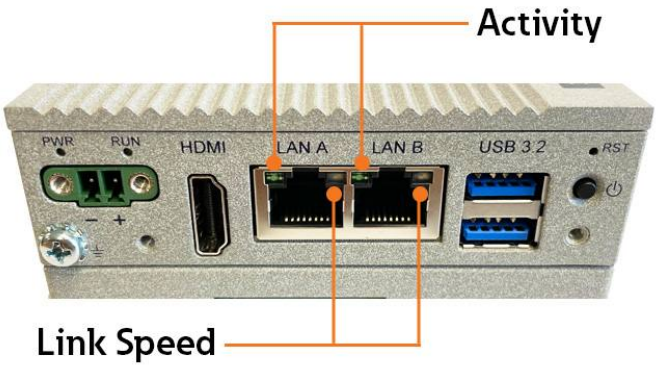


25. PDMC PXG1000 LED Indicators

The PXG1000 is configured to automatically boot when power is applied. The Power button is not functional.



- Power: The Power LED is amber when power is applied to the PXG1000. The PWR LED turns green after approximately 5 seconds.
- Run: The Run LED egins flashing green after ~ 0.5 seconds and changes to solid green after ~ 1 minute to indicate that the boot process has successfully completed. In Recovery mode the Run LED will be flashing fast at a rate ~ 0.2 Seconds.
- Reset Button: Pressing the recessed RST button will power cycle the unit. Configuration information is retained during the power cycle.
- Power Button: The Power button is not functional.



25.1 Link Speed

- Each Ethernet RJ45 connector has a link speed indicator (left) and an activity indicator (right). They have the following meanings:
- **Link speed:** off=no link, yellow=100Mb, green=1000MbStart Guide
- **Activity:** blinks green during network activity. If the activity indicator is blinking but the link speed indicator is off, the connection speed is 10Mb.

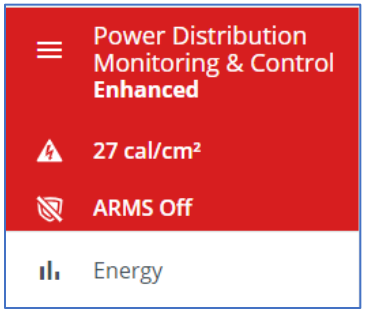
26. PDMC Reveal Panels

The PDMC includes several special Reveal Panels that open to the left of the main navigation menus. The Safety Reveal panel is accessed via the Safety Button at the top of the left navigation menu. The remaining reveal panels are accessed from the buttons at the bottom of the Navigation menu. Available reveal panels include Safety, Help, Assembly, and the Eaton button.

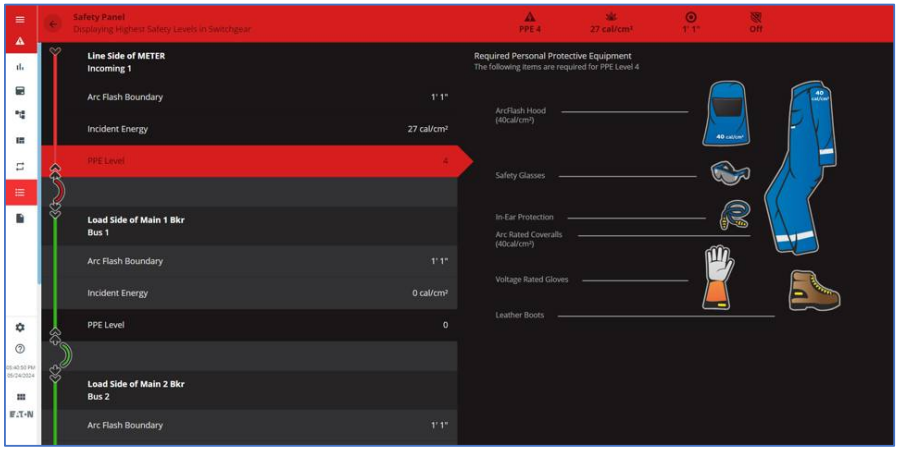
26.1 Safety Reveal Panel

The Safety Reveal Panel opens when the Safety button on the top left corner of the PDMC web interface is selected. The Safety button summarizes the highest PPE level present on the switchgear lineup, the status of the Arc Quenching System (AQS), and an indication if ARMS is enabled for any of the breakers in the switchgear lineup.

An example of the Safety button follows.



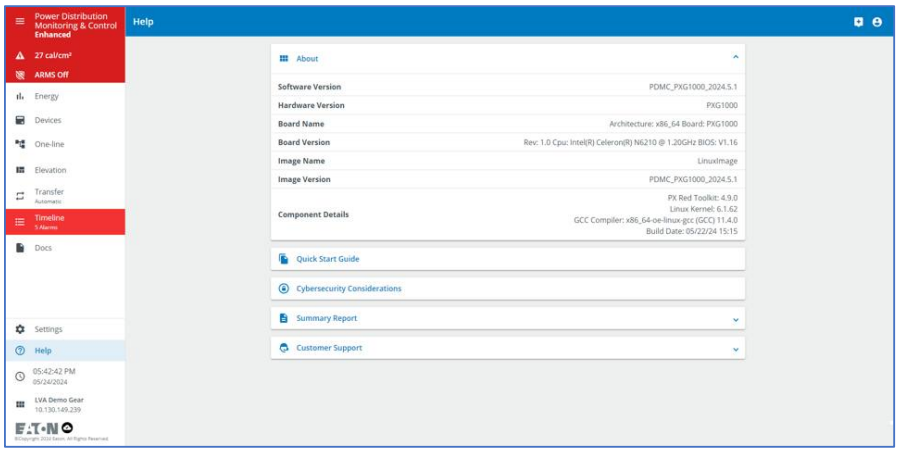
Selecting the Safety button opens the Safety Reveal Panel, which contains more information about incident energy, arc flash boundary, and PPE level requirements for safe access to the switchgear lineup. A linear version of the one-line for the incoming source, main breaker, bus, and if present, the tie breaker, bus, main breaker and second source is shown vertically along the left side of the screen. The required personal protective equipment (PPE) associated with each section of bus is shown based on the incident energy and ARMS status of the switchgear. The required PPE is based on the user configuration entered on the **Settings > Safety** section.



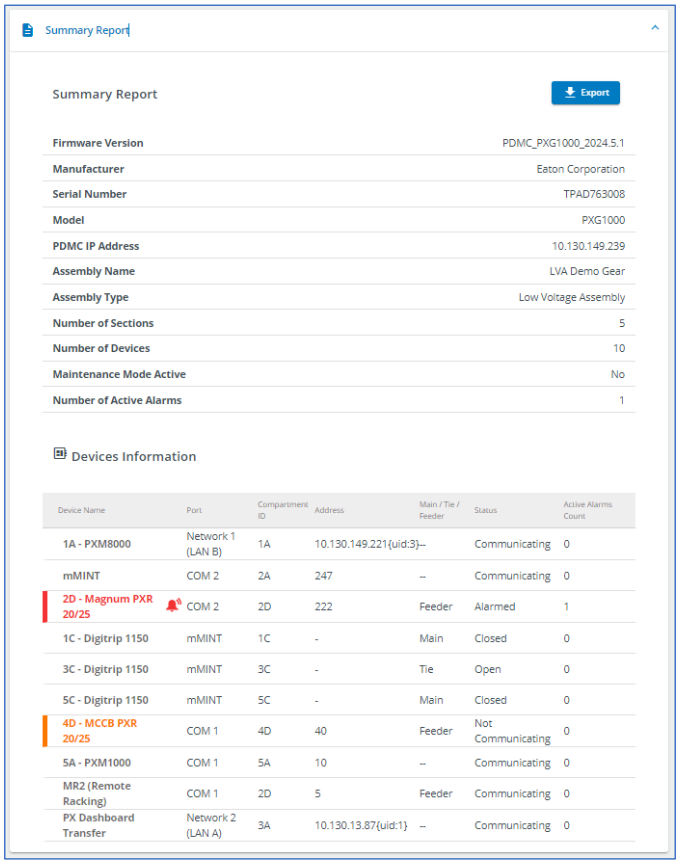
26.2 Help Reveal Panel

The Help Reveal Panel opens when the Help button in the Extras section is selected. The hardware and software versions are displayed at the top of the Help section followed by links to documentation and contact information for our Customer Integrity team.

Help > About displays details of the hardware and software versions.



The Help Reveal Panel also includes links to open the PDMC PXG1000 Quick Start Guide and Cybersecurity considerations documents. The Summary Report captures a summary of the current state of the PDMC PXG1000 and all connected devices.

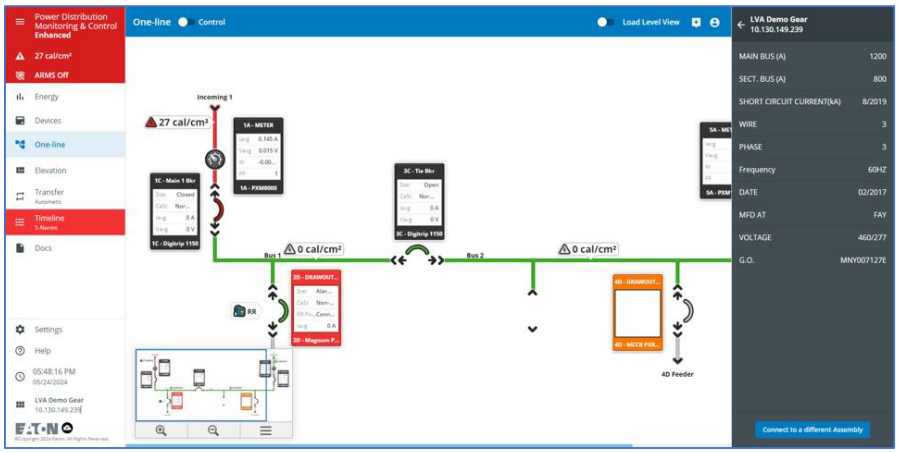


26.3 Assembly Reveal Panel

The Switchgear Reveal Panel opens when the Extras button showing the Assembly Name and IP address is selected.



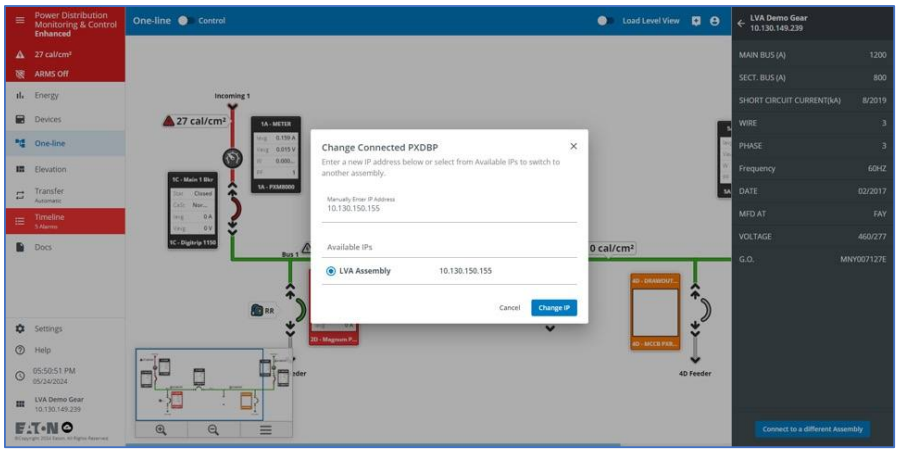
This panel shows the Nameplate information entered during Assembly Setup on the **Settings > General** screen.



The **Connect to a different Assembly** button provides a way to navigate among configured PDMC Enhanced and PDMC assemblies.

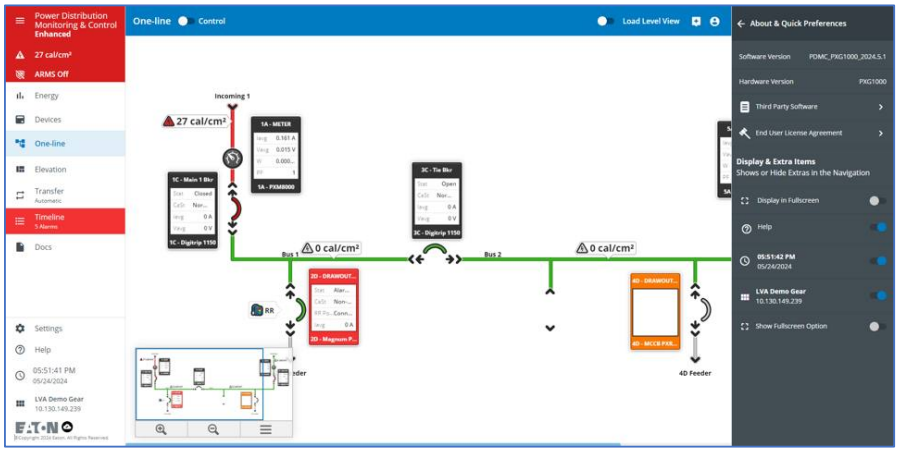
- Connect the HMI and all the PDMC Enchanced and PDMC processors to the same subnet via an Ethernet switch.
- Each of the devices must be configured with a unique IP address on the same Ethernet subnet, i.e. the first 3 numbers of the IP address should match, and the last number should be different for each device.
- Configuration required on each PDMC web interface:
  - Navigate to **Settings > Network** and scroll down to “List of Connected Assemblies” and expand this collapse header.
  - Enable Edit Mode
  - Click “Add New Assembly Connection” and enter the IP address of the another PDMC Enhanced or PDMC
  - Repeat the previous step to add additional assemblies

- To navigate among PXG1000 processors:
  - Click on the IP address near the bottom of the left navigation menu to open the Assembly reveal panel.
  - Click on the “Connect to a different Assembly” button
  - Select the IP address of the PXG1000 processor entered in step 3
  - Click on “Change IP”
  - This will navigate to the selected PXG1000 processor login screen



26.4 Eaton Reveal Panel

The Eaton Reveal Panel opens when the Eaton button in the Extras section is selected. Contents includes the PDMC PXG1000 hardware and software version and access to the Third Party Software Libraries and End User License Agreement. The reveal panel also provides a quick link to show or hide several of the Extras buttons.



27. PDMC Transfer Scheme App

The PDMC provides a standard interface for the most common Main-Main (MM) and Main-Tie-Main (MTM) transfer schemes. The PDMC transfer implementation uses the ELC PLC as the logic controller and the HMI for configuration, monitoring, and manual transfer initiation. A brief description of the key system level parameters used on the transfer screens follows:

Transition Type: The PDMC supports both open and closed transition.

- Open Transition is a break-before-make implementation that results in interruption of power on the bus for the duration of the Dead Bus (Neutral) timer setting when transferring from one source to another.
- Closed Transition is a make-before-break implementation used to maintain power on the bus during the retransfer. A permissive sync check input is required to make sure both sources are synchronized before they are temporarily paralleled.

Transfer / Retransfer Mode: The PDMC supports automatic and manual transfers.

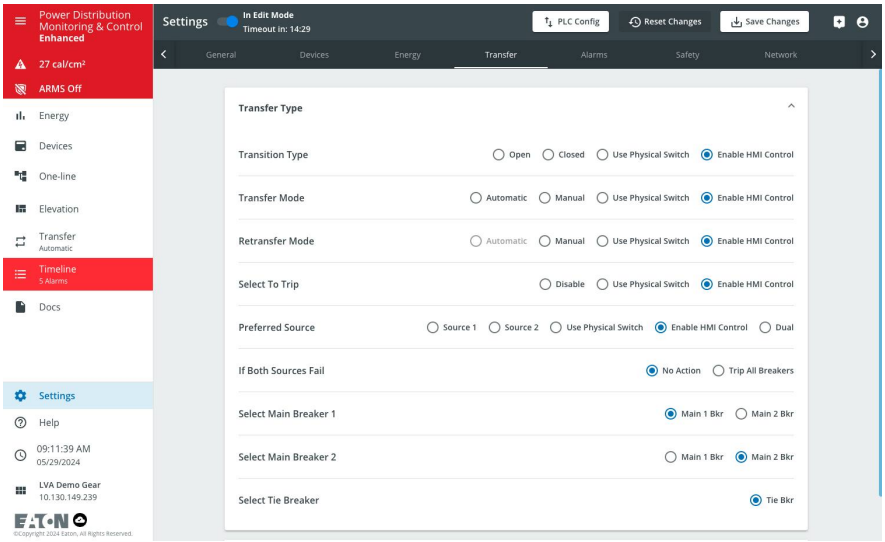
- In Automatic mode the system will automatically transfer to the good source.
- Manual mode requires user intervention to transfer between sources.

Select to Trip: This is a manually initiated closed transition transfer.

Preferred Source: Allows selection of the preferred source of power if both sources are available.

- If both sources are available, the transfer controller will set the preferred source to power the bus.
- Most Main-Tie-Main applications are doubled ended (Dual selection) with the tie breaker open and the Main 1 and Main 2 breakers closed to provide power to their respective bus. Single ended Main 1 gear would have the Main 1 and Tie breakers closed and the Main 2 breaker open.

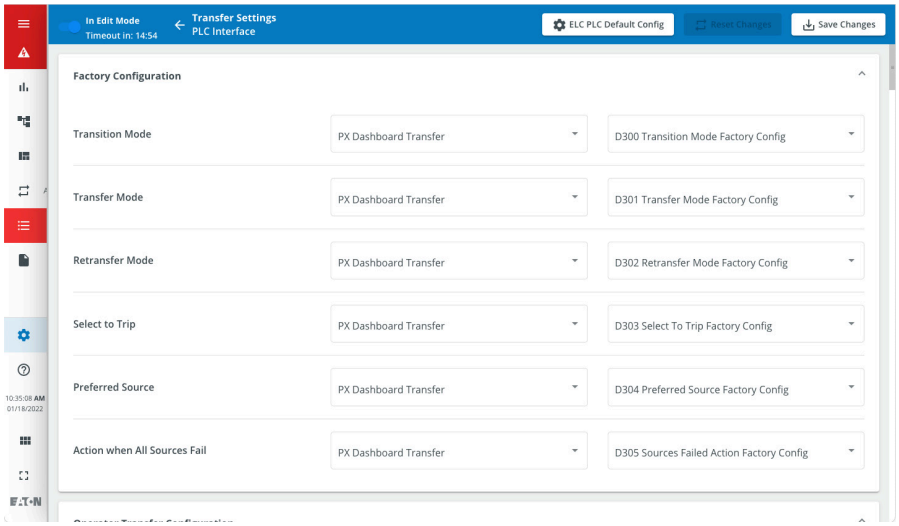
Customer requirements influence the hardware configuration of the assembled switchgear, which impact the transfer screens and parameters available for viewing and configuration. The Transfer Scheme system level settings are configured on the Transfer Scheme Factory Configuration screen shown below by navigating to the **Settings > Transfer** tab.



The PDMC supports open and closed transition, auto and manual transfer and re-transfer, select to trip (for closed transition), and preferred source selection. Each of these options can be controlled by the PDMC HMI, controlled by a physical switch, or set to a fixed value (like Open Transition, Manual Re-transfer, etc.). The hardware components and wiring must be installed at the assembly plant or by field service for a retrofit job.

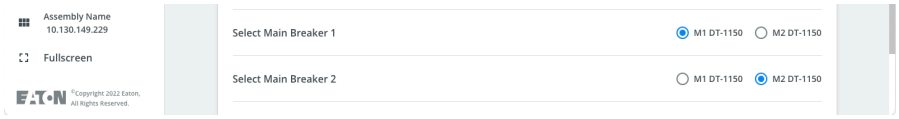
An initial one-time configuration step is required to establish the register mapping for communication between the PDMC PXG1000 and ELC PLC. This is typically done at the factory but would be required if the PDMC PXG1000 or ELC PLC was replaced or if device addresses are changed. First set all the options on the Settings à Transfer screen to “Enable HMI Control”. Then select the **PLC Config button** on the top action bar to open the Transfer Settings PLC Interface screen shown below.

Select the **ELC PLC Default Config button** on the top action bar to load the default configuration for the ELC PLC based transfer scheme. Press the **Save Changes button** to save the configuration to the ELC PLC and press **OK** to confirm.

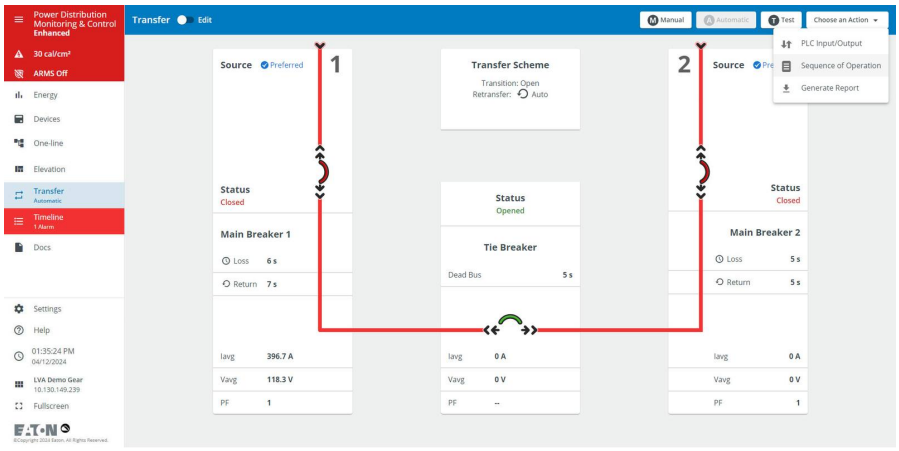


Select the back button <- to return to the **Settings > Transfer** screen. Now set the Transition Type and Transfer and Re-transfer modes to the appropriate settings for the switchgear lineup hardware configuration, then Save Changes.

The lineup can have more than 3 breakers as mains or ties, but at most 2 mains and 1 tie will be involved in the transfer scheme. If a more elaborate transfer scheme is required, contract with PSA to implement a custom transfer scheme. The 3 breakers involved in the transfer are selected at the bottom of the **Settings > Transfer** page.

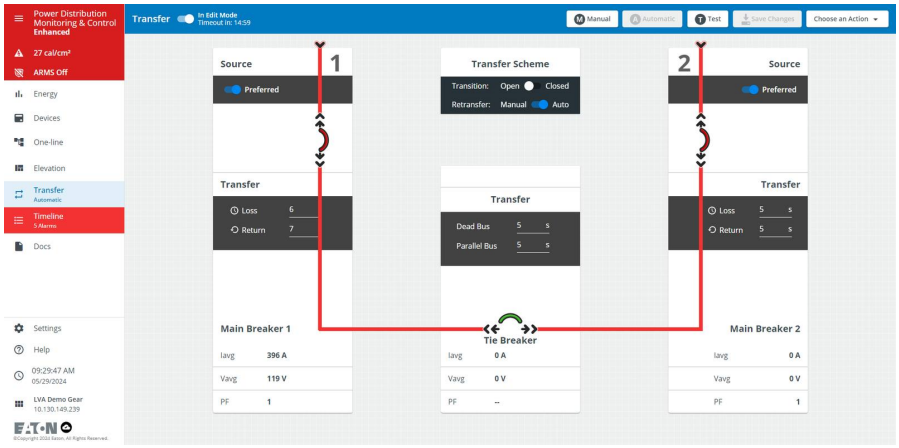


The Transfer schemes and sequence of operation are typically owned by the assembly plants. The PDMC provides the ability to visualize and configure the transfer scheme and to do manual control. This is a screen shot of the Auto Transfer screen.

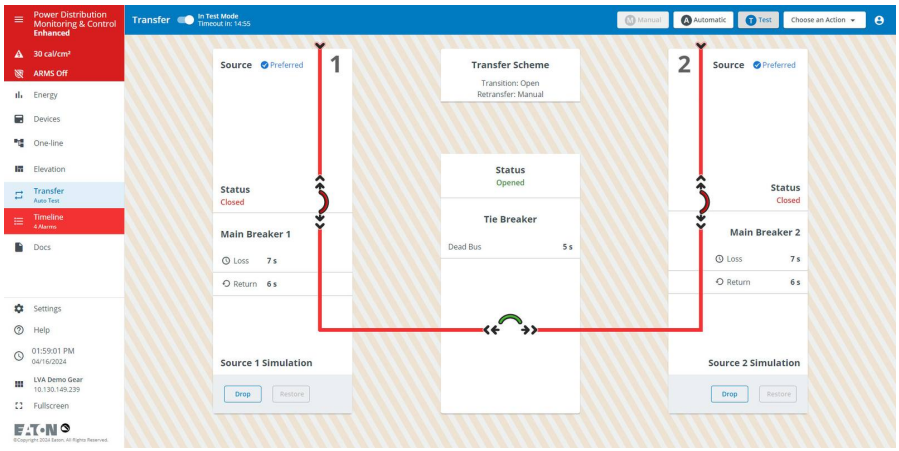


27. PDMC Transfer Scheme App

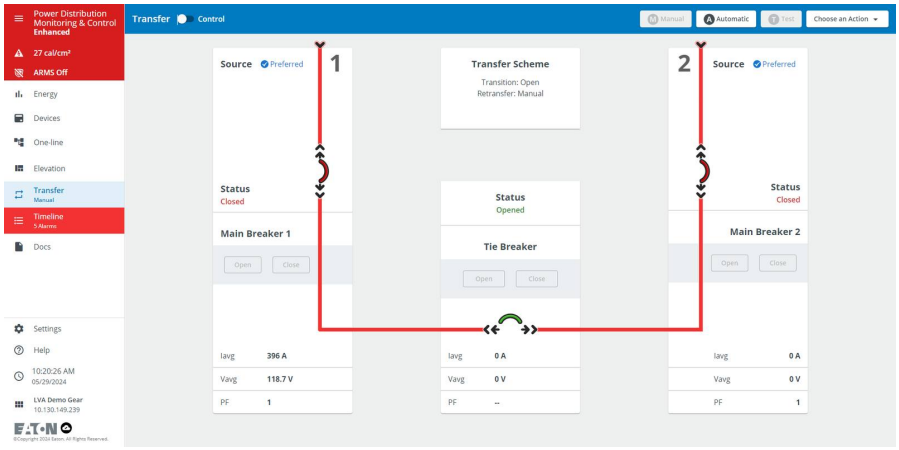
In Automatic Mode the transfer timers can be set when the Edit Mode is enabled. Preferred Source, Transition type, and Retransfer mode may also be edited here if the transfer settings are set to “Enable HMI Control”.



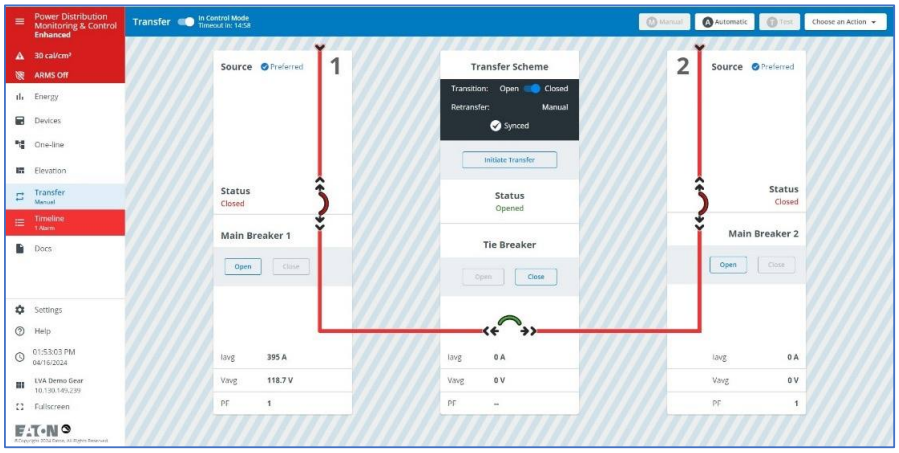
Test mode is provided to simulate dropping the sources to verify proper operation of the transfer scheme during a test scenario. Be careful using the Test mode because it will cause a transfer operation and a brief loss of power during an open transition transfer.



When Manual Transfer Mode is selected the Transfer screen appears as show below:



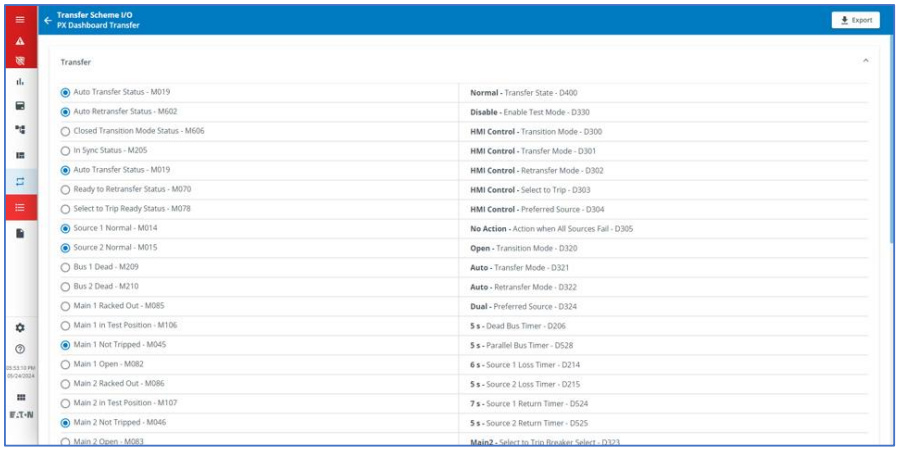
Enabling Control Mode allows the user to manually open and close the main and tie breakers:



What's different with XC303 in Transfer Scheme:

1. Auto hold: Auto-hold mode allows users to perform a Transfer operation on the Main and Tie breakers leveraging enhanced safety associated with the walkaway delay timer. In addition to safety provided by PDMC, the walkway timer provides additional measure of safety by permitting the user to initiate a time-delayed transfer.
2. Support for Generator starting & stopping:
   - In addition to MT & MTM available with ELC-PLC the XC303 supports Main-Generator, Generator-Main, Main-Tie-Generator, Generator-Tie-Main, Main-Tie-Tie- Generator and Generator-Tie-Tie-Main Transfer configurations.
   - XC303 includes additional timers for generator starting and stopping.
   - These timers can be set from **Settings > Transfer**.

Transfer Scheme I/O: PLC I/O and current state of all the registers in the PLC that are being used for control and status can be seen by navigating to **Main Transfer > Choose an Action > PLC Input/Output**. This table assist in troubleshooting Transfer scheme operations.Update screenshot.



28. PDMC Campus Setup

PDMC provides a unique feature to connect with PDMC Campus also known as an assembly of assemblies via MQTT Protocol.

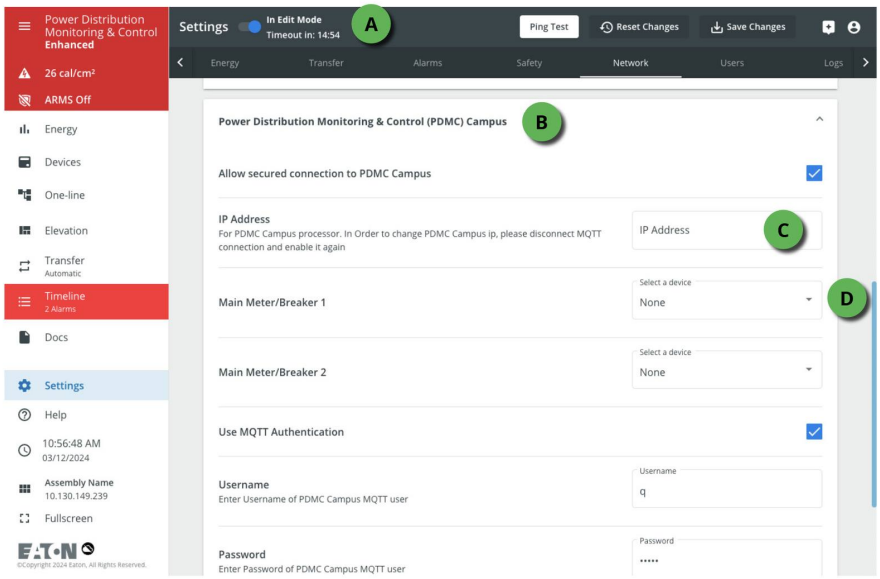
These are the steps to establish the Secure MQTT connection to PDMC Campus:

A. Enable 'Edit' mode using the slider in the Action bar at the top of the screen.

B. Go to **Settings > Network > Power Distribution Monitoring & Control (PDMC) Campus** and select Allow secured connection to PDMC Campus checkbox which will enable the IP address field.

C. Enter the IP address of the Power Distribution Monitoring & Control Campus gateway with which the secure connection needs to be established.

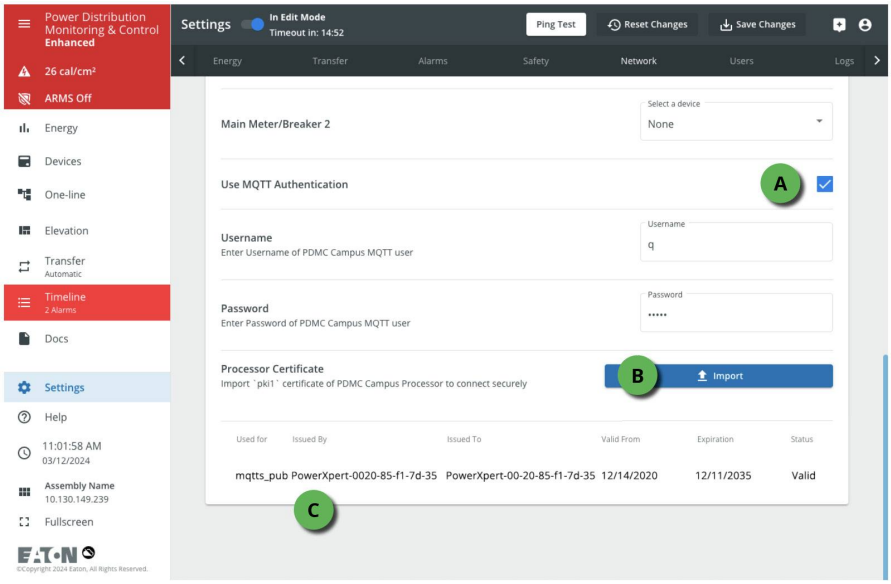
D. After configuring the IP, select a device from the Main Meter/ Breaker dropdown available along with their compartment numbers. The dropdown shown will be based on the type of Lineup Configuration. Similarly, if the configuration consists of single source than user can select one device from Main Meter/ Breaker 1 dropdown.



A. Select “Use MQTT Authentication” checkbox which enables Username and Password input fields which are the Required ones. User needs to enter the username MQTTAdmin and password Admin*1 of the Power Distribution Monitoring & Control Campus MQTT user for Authenticated connection along with secure TLS. The final step towards Secured MQTT Connection is Certificate Management. User needs to Import the “pki1” certificate Exported from the Power Distribution Monitoring & Control Campus (**Settings > Network > Certificate Management**) with which the connection is establishing.

B. To Import the certificate, select the Import button located near the Processor Certificate Label and then select the pki1 certificate of the Power Distribution Monitoring & Control Campus gateway. Then select the upload button. Once the import is successful, the Power Distribution Monitoring & Control Enhanced reloads.

C. The information related to the imported certificate can be seen below the Processor Certificate row like certificated Used for, Issued by, Issued to, Valid From and Expiration Date and Status (valid/invalid).

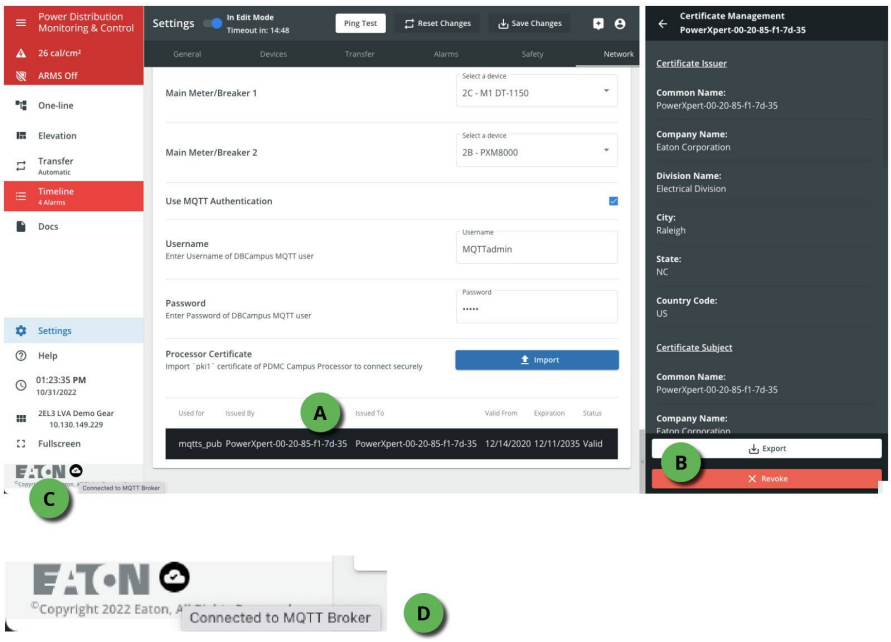


A. When selecting the certificate details, the respected row gets highlighted, and sidebar opens up revealing more information about the Imported Certificated.

B. Also, buttons like Revoke and Export will be enabled. You can find them combined with Import into a Choose an Action menu.

C. After the completion of the above-mentioned steps, select the Save changes button. After saving the configuration, the connection with MQTT broker starts. This connection status can be seen near the Eaton Label present at the bottom left corner of the Application.

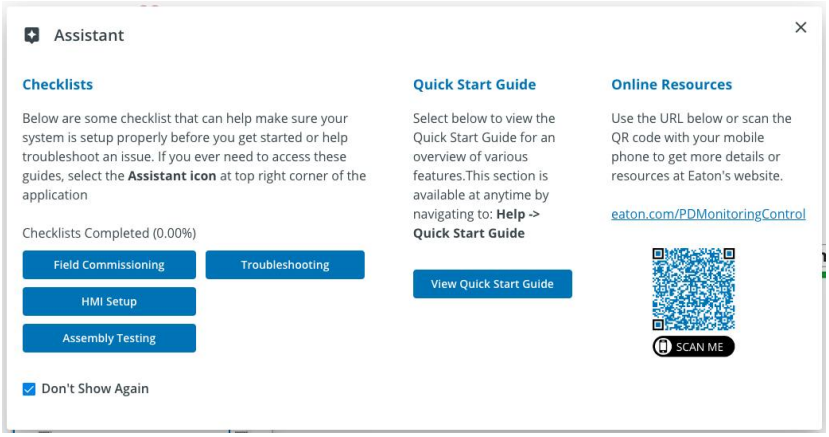
D. When the MQTT connection is successfully established a symbol is shown that displays “Connected to MQTT Broker” when hovering over it.



29. Assistant

The Assistant provides quick access to documentation along with assistance for commissioning and troubleshooting. It opens at each login.

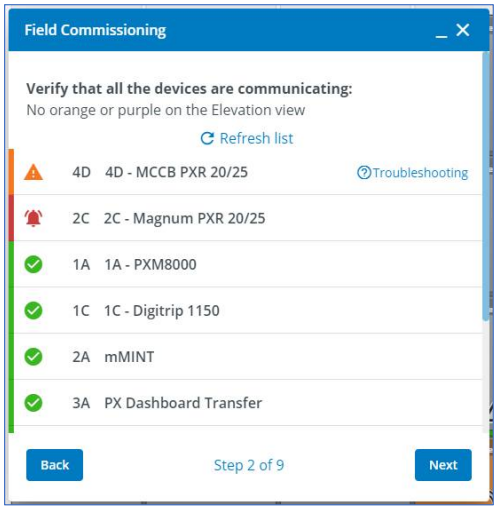
- It is always available for the user at the top right corner of the dashboard for ease of access. It has 3 major sections – Checklists, Quick Start Guide, Online Resources.



- The checklists assist with setup, testing and commissioning.
- Checklists consist of 3 parts which contributes to its 100% completion progress.

1. Field Commissioning

- This section guides the user to complete full testing of the device configuration, communication, and control commands.
- Guidance is provided for connection to customer network and for user's configuration.
- The assistance will navigate to the respective pages where the configuration is required.
- For devices that are not communicating, a quick link to troubleshooting guidance is provided.



- Users get the option to mark the checklist as complete. After successfully completing the checklist user will get option to save field commissioning report.

2. HMI Setup

- User needs to select HMI size to see the list of steps for configuring the selected HMI.
- Once user has read the instruction can click “Done” button which will mark that individual question as completed.
- After successfully completing the checklist user will get option to save the HMI setup report.

3. Assembly Testing

- For Assembly testing, user is instructed to use available Installation instruction to mount wire and setup the PDMC gateway and test it.
- This section guides the user to complete full testing of the device configuration, communication, and control commands.
- For devices that are not communicating, a quick link to troubleshooting guidance is provided.
- After successfully completing the checklist user will get an option to save Assembly testing report.
- All the saved reports for Testing and Commissioning are stored in Docs under Testing and Commissioning report section.
- User will get the option of Troubleshooting, depending on the devices configured and feature license present. To know more about troubleshooting solutions user can click on the list provided.
- Clicking on “Quick Start Guide” will navigate user to Help tab where the quick start guide can be viewed.
- User can directly look up to Eaton’s website to get further details on the PDMC gateway and its features by using below resources:
  - Eaton Website Link
  - QR Code Scanner
- If user do not want to see the Assistant collapse on every login, user can select “Don’t Show” check box. This will be valid until a hard reset.

30. Advanced Graphical Configuration

30.1 Configuration of I/O and Roles

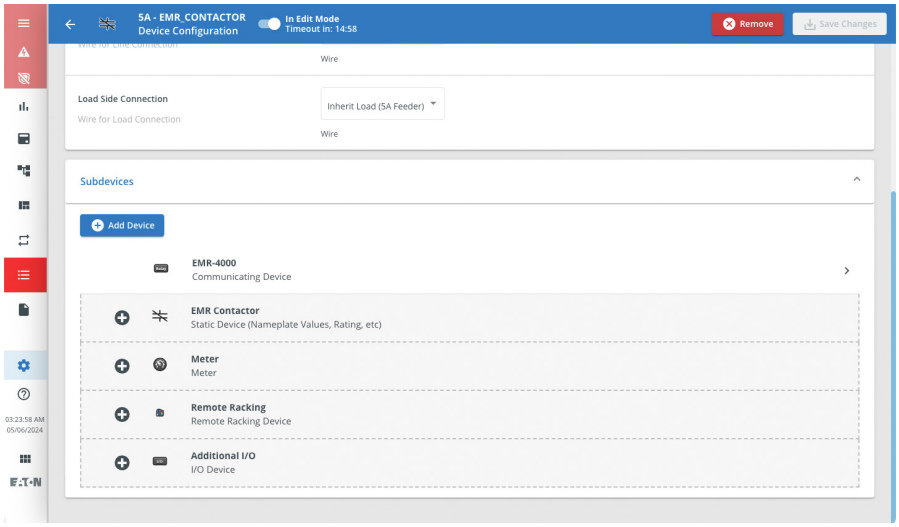
Note: Note: Before editing, be sure to save your configuration file in case you need to restore your system to the previous setup.

To load a configuration file:

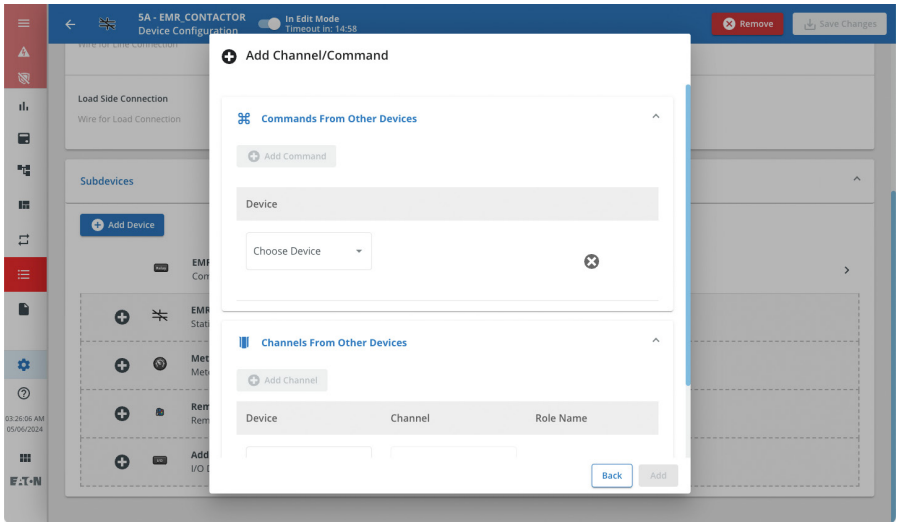
- **Navigate to Settings > General**
- **Enable "Edit" mode** on the top action bar
- Click **Import Settings File** and select the attached file that you have saved to your computer
- Click the double up arrow beside the file name to upload the configuration file
- The PDMC will reboot

Here are the steps to configure the EMR contactor state to change based on the status of Digital Input 1:

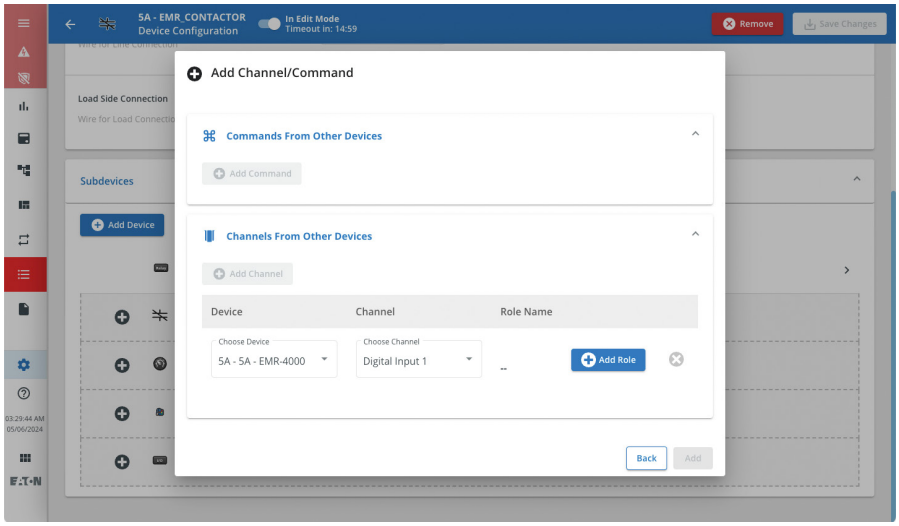
- **Navigate to Settings > Devices > Elevation**
- **Enable "Edit" mode** on the top action bar
- Select the EMR contactor symbol to open the sidebar
- Click the More button at the top of the sidebar
- **Scroll down** and click on the + sign beside Additional I/O



- This will open the Add Channel/Command modal

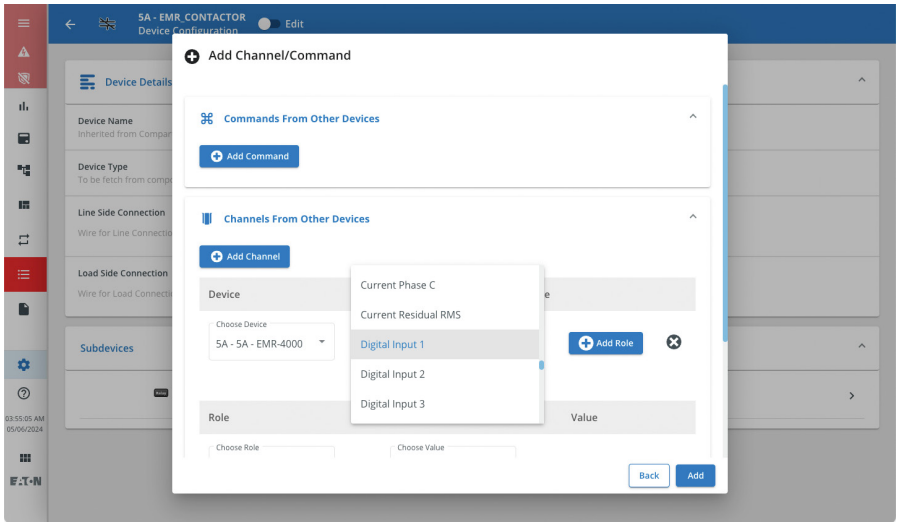


- Select the Add Channel button

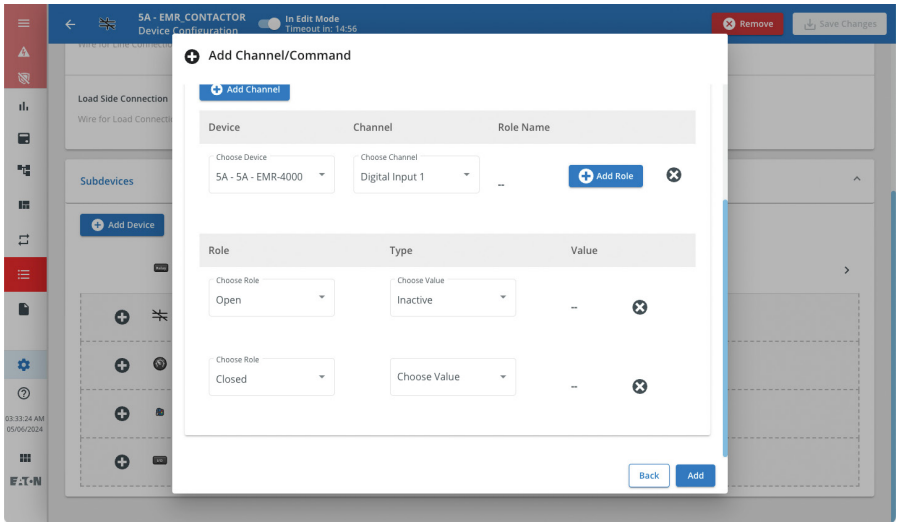


- On the Choose Device dropdown choose the appropriate EMR-4000 for the selected compartment **(Pump 7 is in 1A)**
- On the Choose Channel dropdown choose **Digital Input 1**.

Note: Tip: There are a lot of channels, so you will need to scroll down far or type the first letter of the channel to help sort the list.



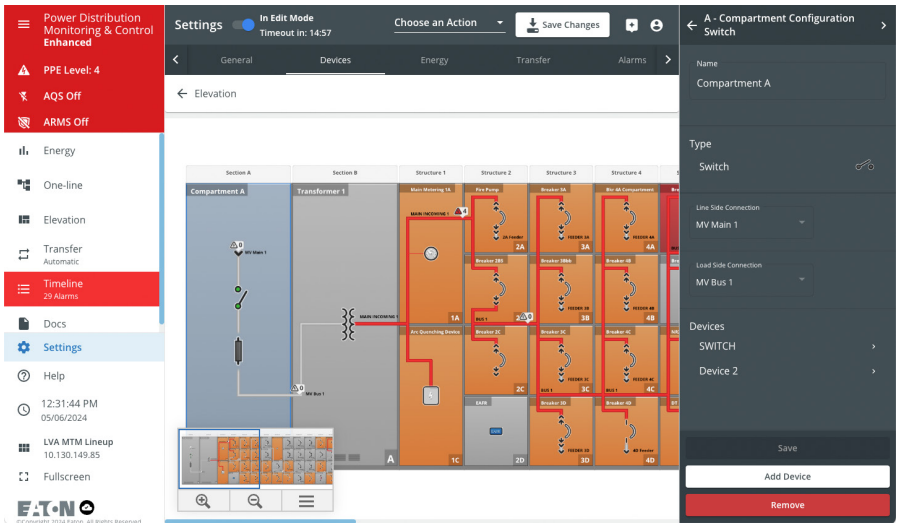
- Next **click the Add Role** button and it will open a new row to select a role.
- To assign the Open indication to the contact:
  - On the Choose Role dropdown **select Open**
  - On the Choose Value dropdown **select Inactive**
  - **Click the Add button** at the bottom of the modal
- **Click the Add Role** button again to assign the Closed indication to the contact:
  - On the Choose Role dropdown **select Closed**
  - On the Choose Value dropdown **select Active**
  - **Click the Add button** at the bottom of the modal
- **Click the blue Add button** at the bottom of the modal to save the changes.



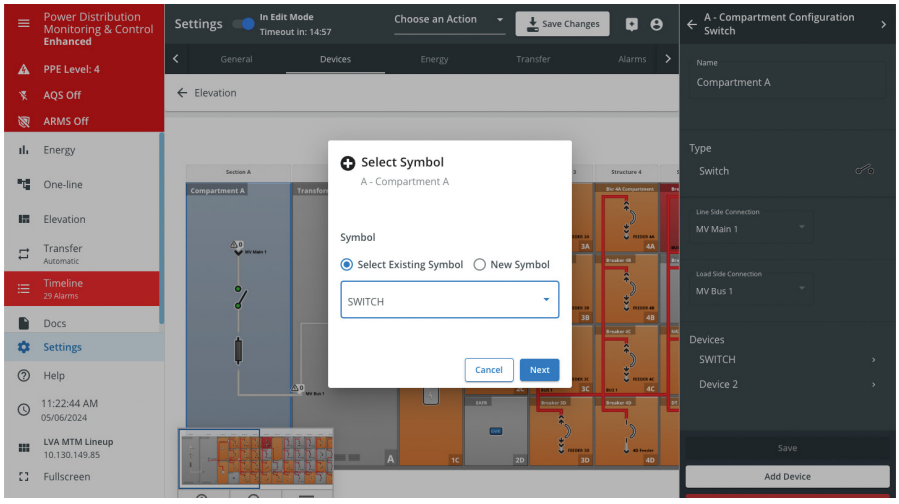
- Now verify that the mapping of the Active and Inactive roles shows the contactor status correctly for Open and Closed.
- If this looks correct select each EMR and add this role configuration.

30.2 Adding Open and Closed Roles to a Device

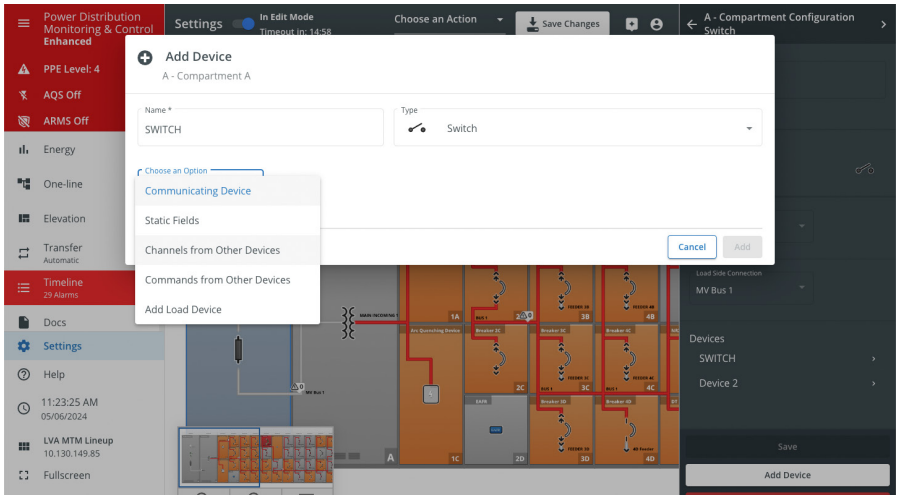
- **Navigate to Settings > Devices > Elevation**
- **Enable "Edit" mode** on the top action bar
- Select a **Compartment** containing a **Switch**
- Select **Add Device**



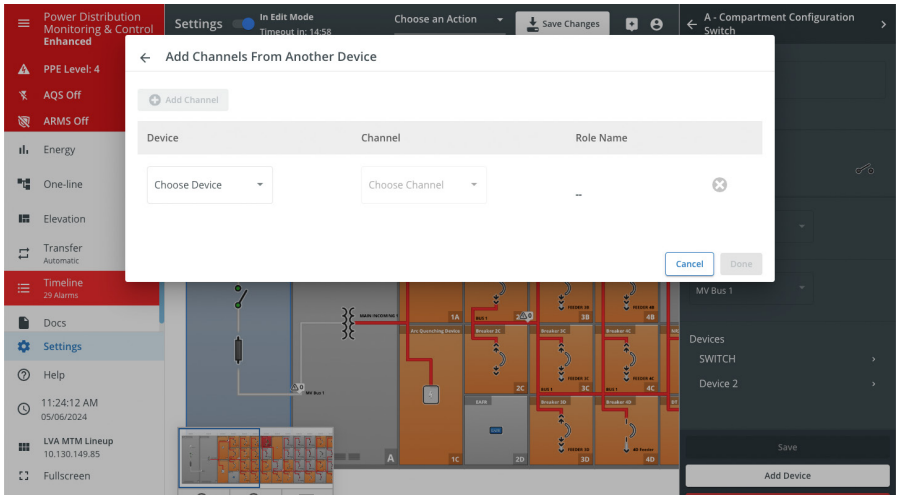
- Choose **Select Existing Symbol**
- Select **Switch** from the dropdown



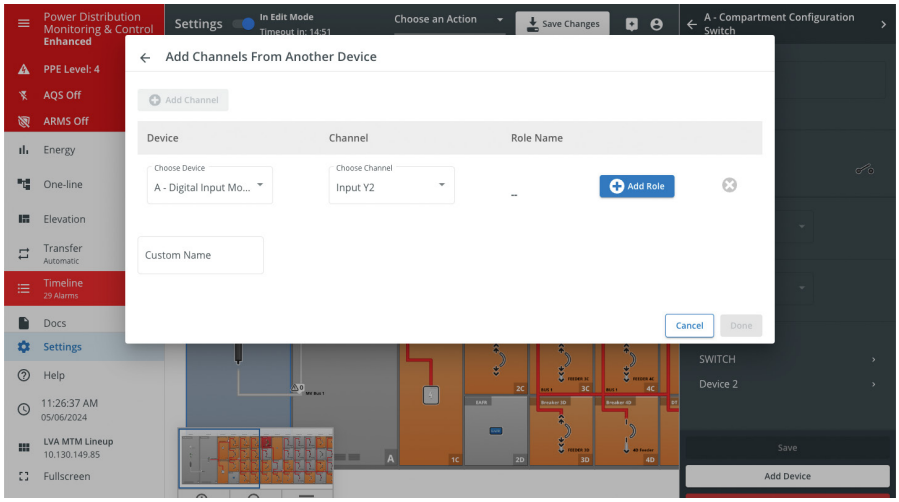
- Check **Channels From Other Devices**



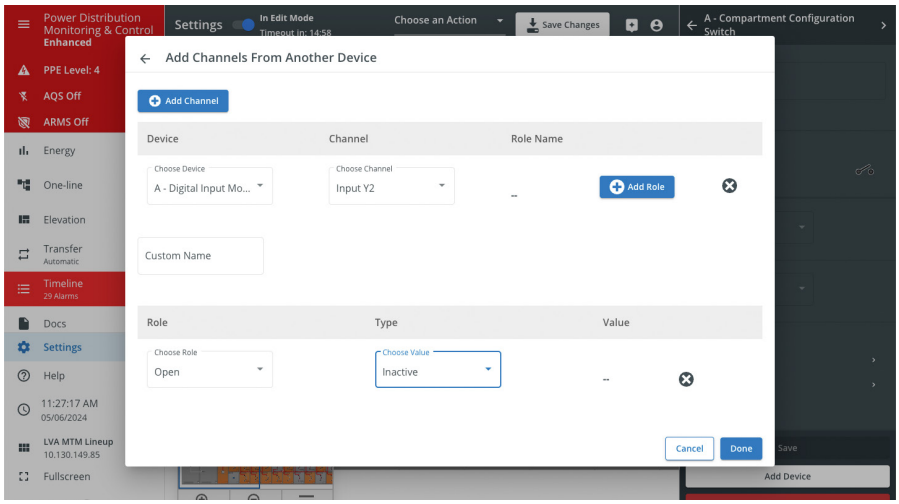
- Next select **Add Channel** under **Channels From Other Devices**



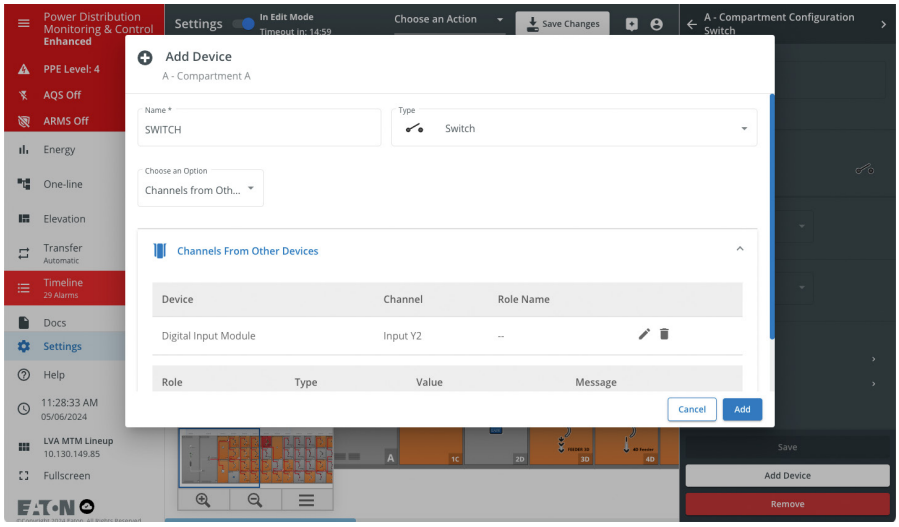
- Next select **Digital Input Module** from **Choose Device**
- Then select an **Input**



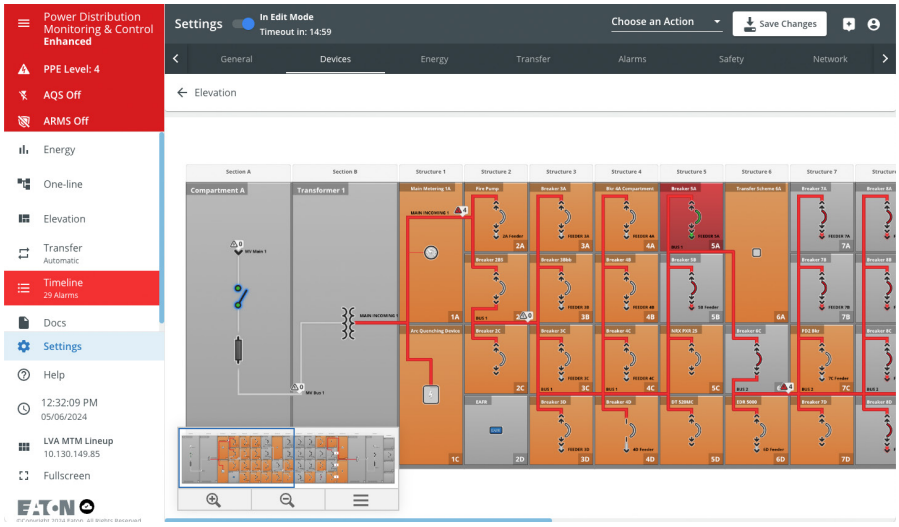
- Next select **Add Role** and choose **Open**
- Then set the Type as **Inactive**
- Select **Done**



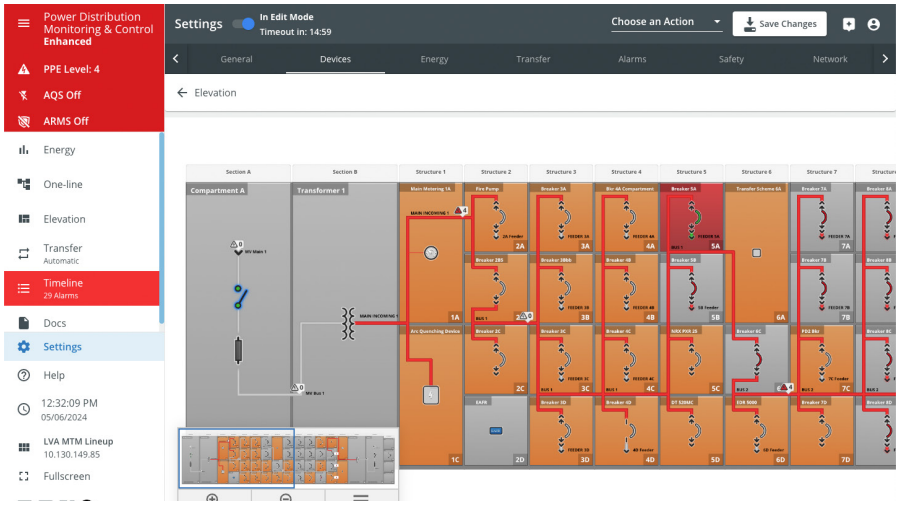
- Now select **Add Device**



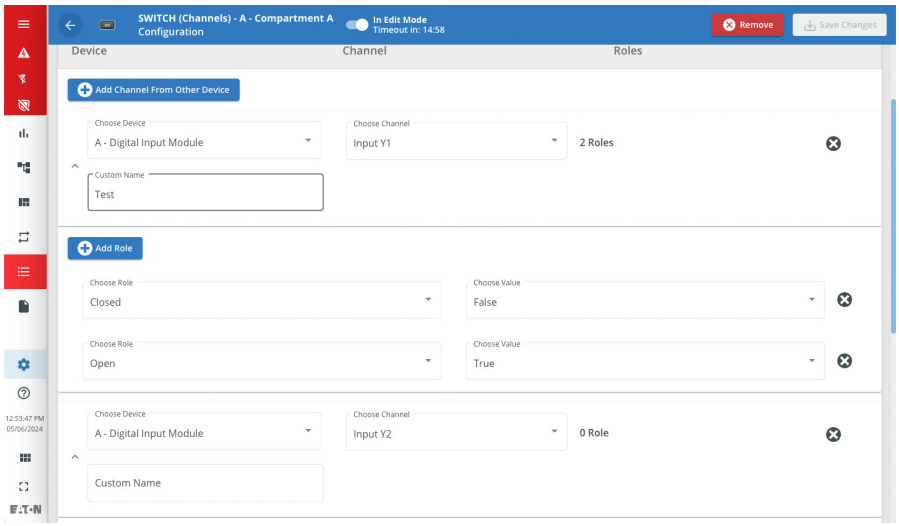
- You should now see your **Switch** lit up green.



- You can now add an additional role, such as Closed, by selecting the device and choosing SWITCH (Channels) under IO Configuration.

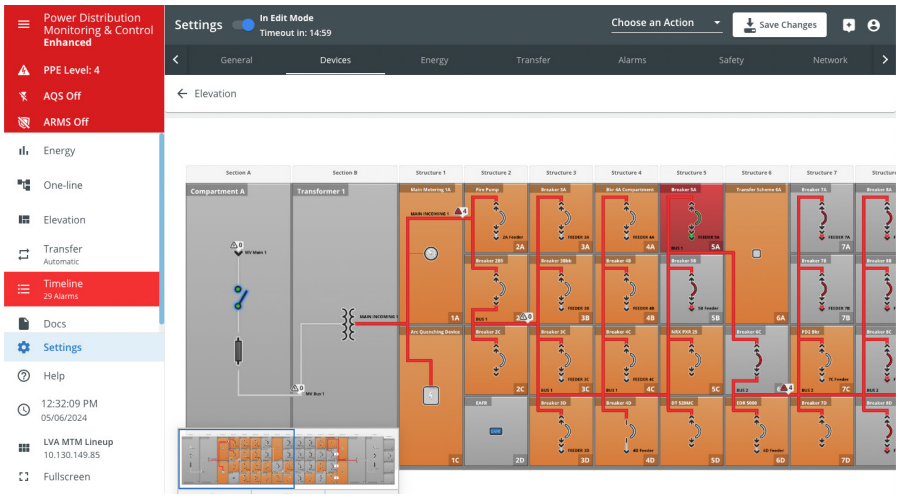


- Select the **Device**
- Then select **Add Role**
- Choose **Closed** and **Active**
- Select **Save Changes** at the top right

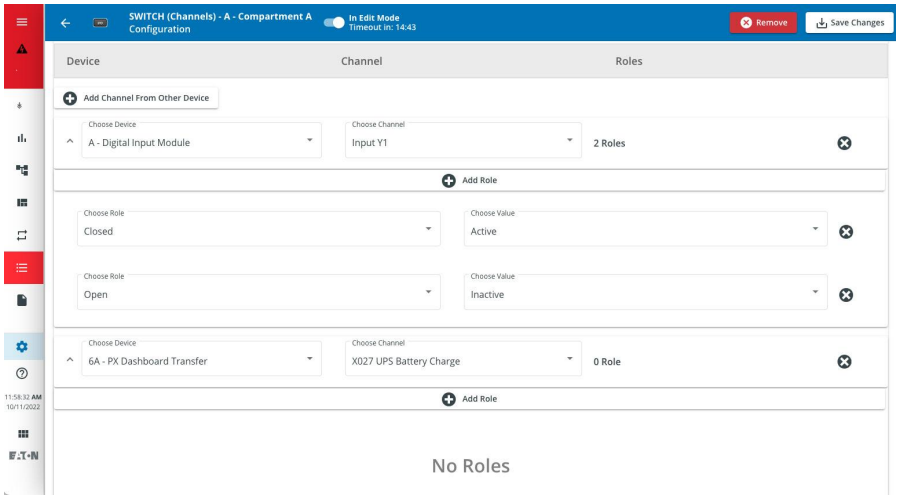


30.3 Adding a Message Based on the State of a Digital Input on a DIM

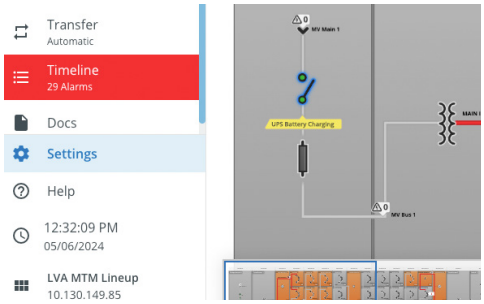
- Select the device and choose SWITCH (Channels) under IO Configuration.



- Select **Add Channel From Other Device**
- Choose **PDMC Transfer** from the **Device** dropdown
- Choose **UPS Battery Charge** from **Choose Channel** dropdown
- Select **Add Role**
- Select Message from **Choose Roles** dropdown
- Select **True** from Choose Value
- **Enter your custom message**
- Select **Save Changes** at the top right

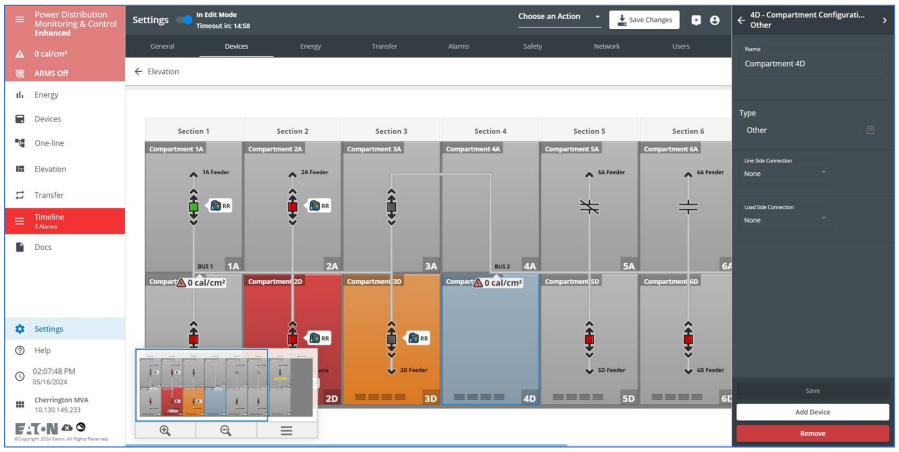


- When the value reads **False**, you will see your message displayed like in the example here.

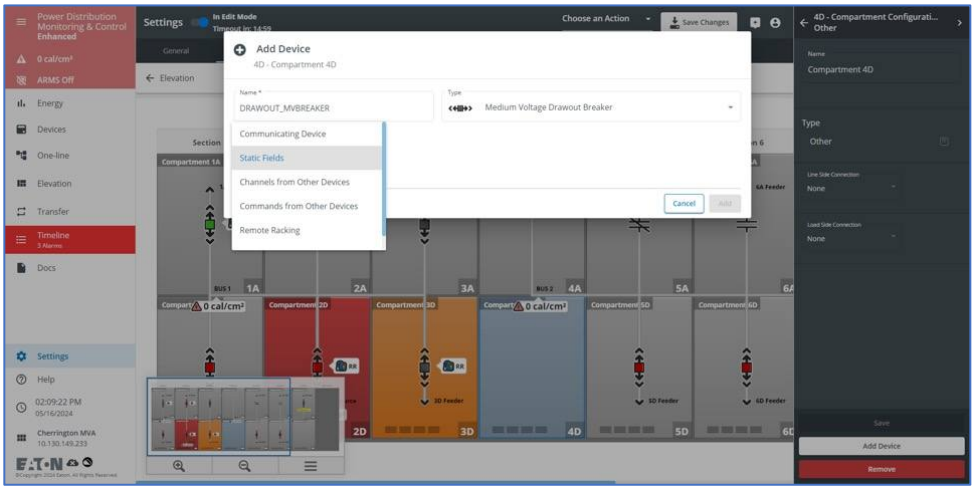


30.4 How to Configure Open/Close for a static Medium Voltage Breaker

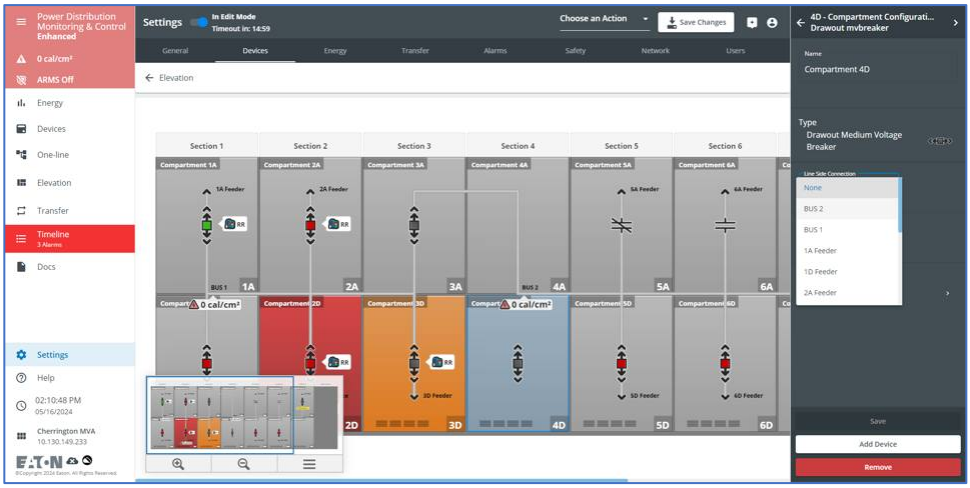
- Choose an empty compartment (in this example 4D) and select Add Device.



- Make sure **Static Fields** is the only box checked.
- Select **Add Device to confirm the device has been added**.

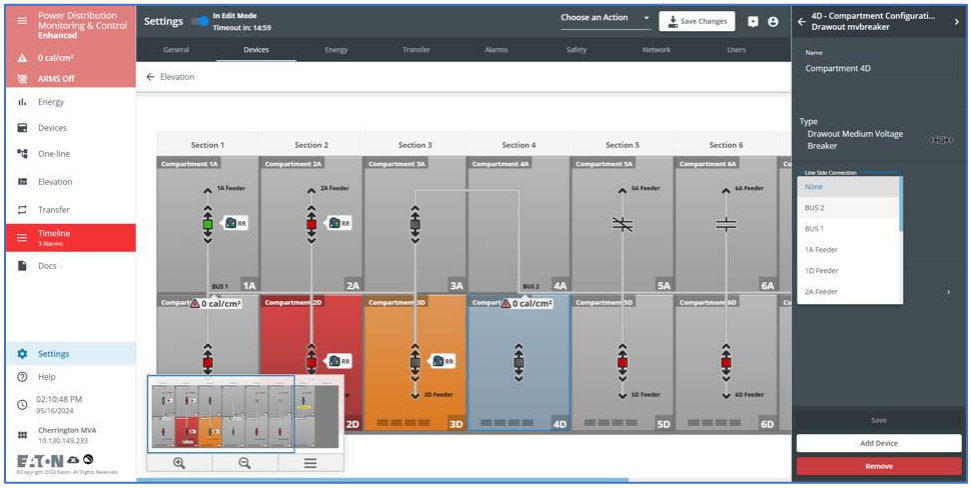


- Select the compartment again
- Choose **Bus 1** from the **Line Side Connection** category
- Select **Save**

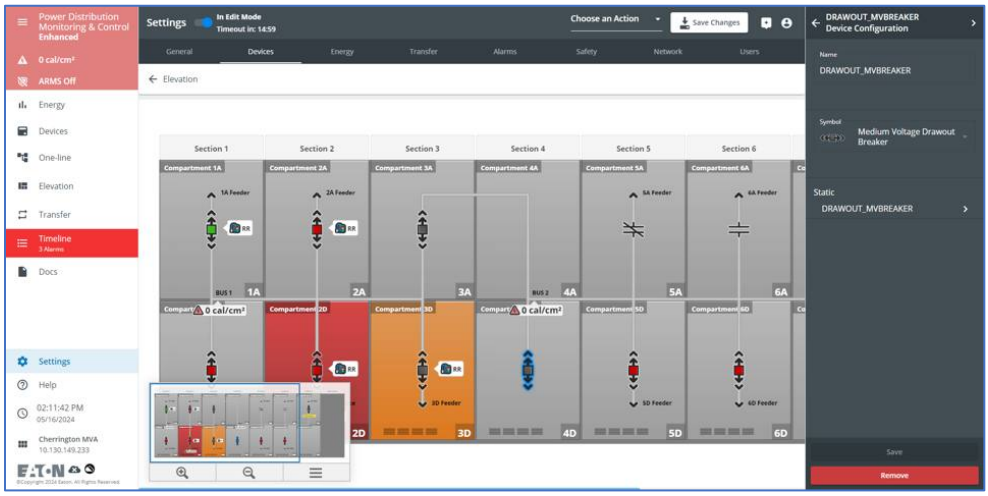


Because there is no protective relay associated with this device, it is not animated. If you want, you can use the input status from another device to see the state of the breaker.

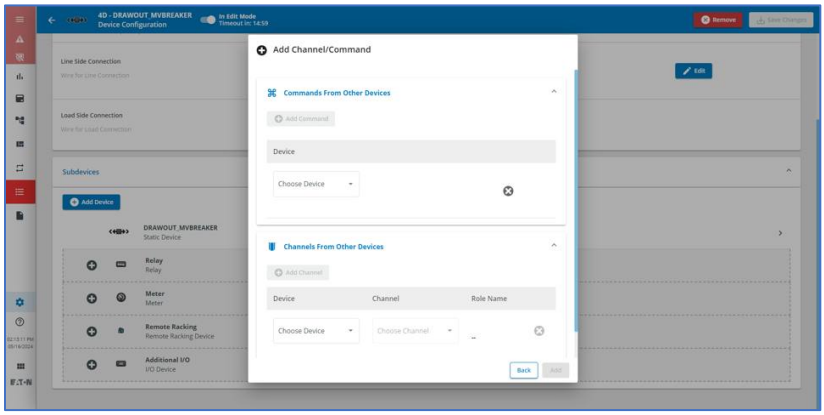
- Start by selecting the device.
- Select the **More** button at the top right.



- From here scroll down and select the (+) button next to **Additional I/O**.
- Select **Add Channel** from the **Channels From Other Devices** category.



This example uses an input from a nearby **EDR-5000** protective relay. The EDR-5000 is selected from the Device category, along with **Digital Input 1** from the **Channel** category.

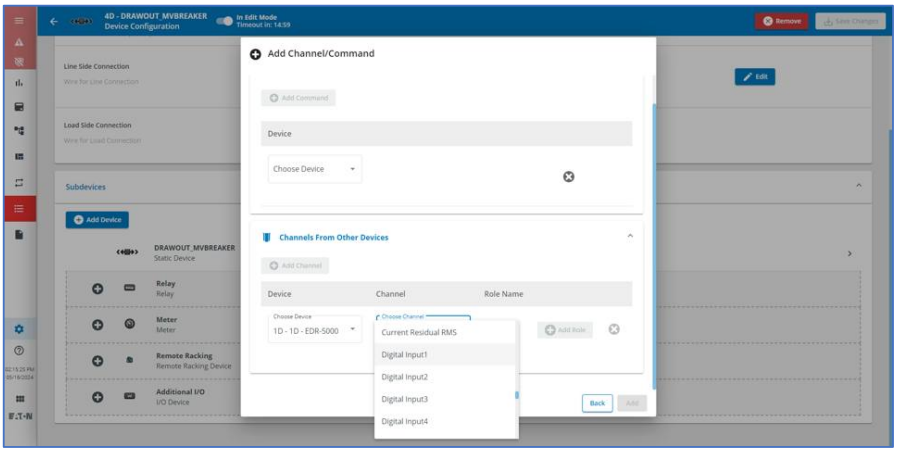


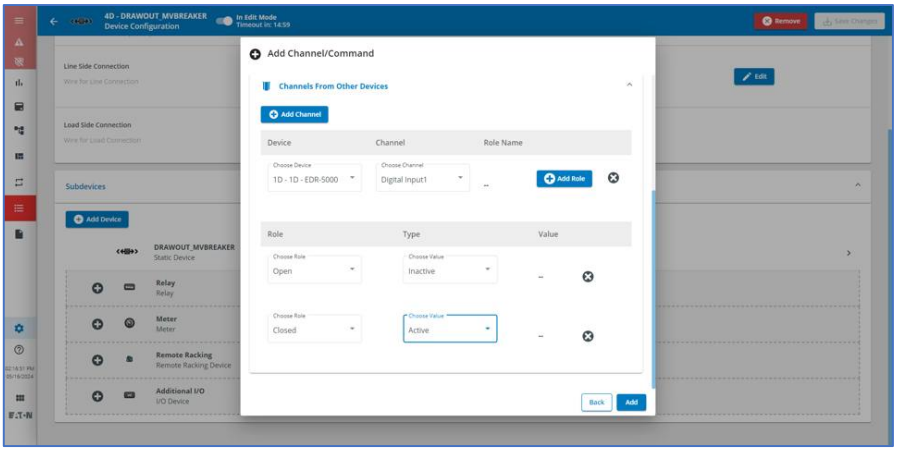
- Next, select **Add Role**.
- Then select **Open** from the **Role** category, along with **Inactive** from the **Type** category.
- Select **Add Role** again.
- Select **Closed** from the **Role** category, along with **Active** from the **Type** category.
- Select **Add**.
- Select **Save Changes** at the top right.

Note: You may need to navigate to another tab, such as One-line, and back to Elevation to see the update.

In the example below the device is now just a static breaker with no active relay or trip unit associated with it. It shows breaker status based on the I/O configuration.

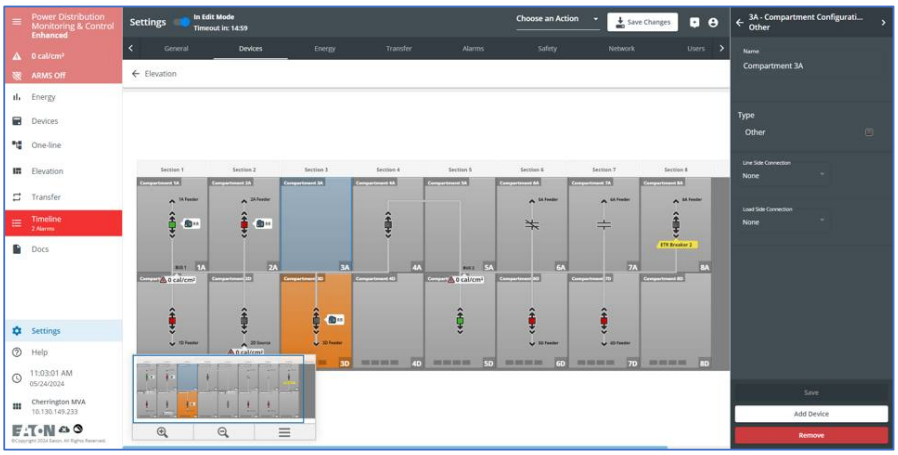
While this example used an Input, an EDR-5000 Output or Logic Equation could have also been used.



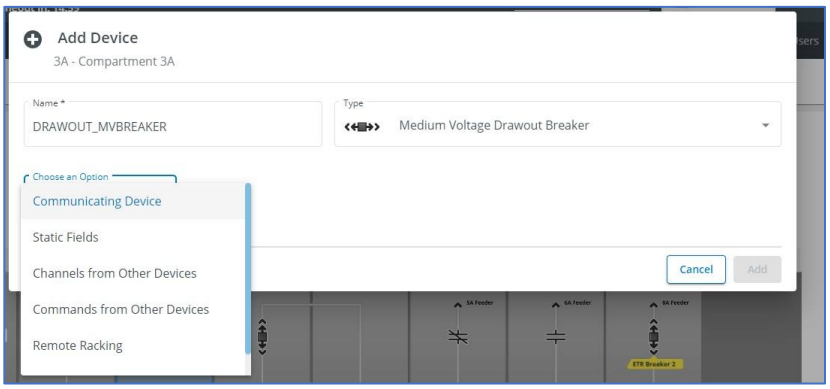


30.5 Adding Remote Racking to Medium Voltage & Magnum Breaker

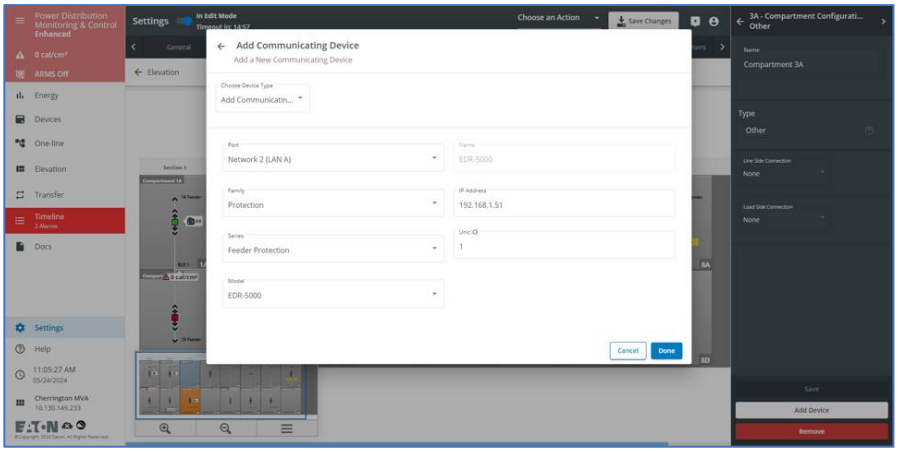
- Remote Racking is supported for Medium Voltage breakers as well as for magnum low voltage breaker.
- Begin by selecting an empty compartment. In this example 3A.
- Select Add Device



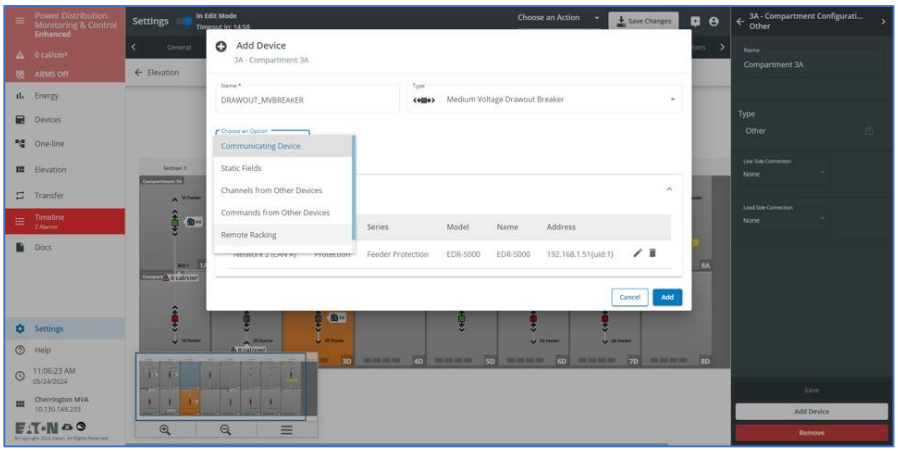
- Choose Communicating Device



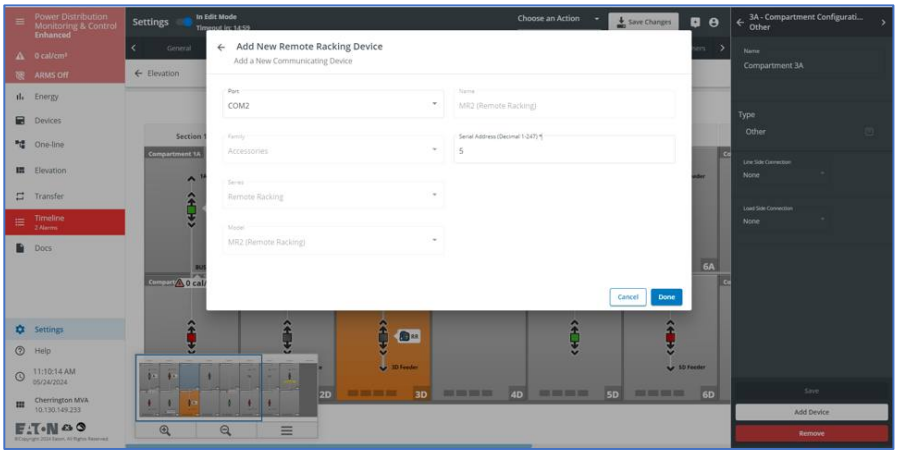
- Next enter the following to add an E Series Relay.
- Network 2 (LAN A) from the Port category.
- Protection from the Family category.
- Feeder Protection from the Series category.
- EDR-5000 from the Model category.
- Name will populate based on the Model selected.
- Enter your IP Address and Unit ID.
- Click “Done”



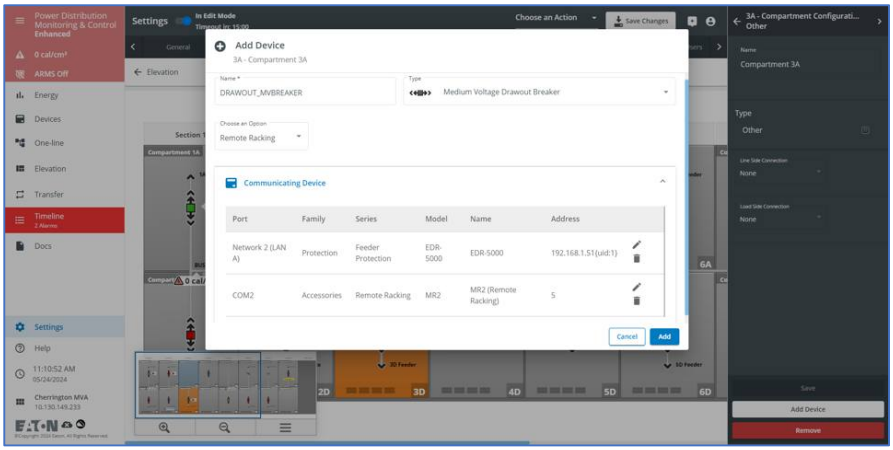
- Choose Remote Racking



- Next choose a COM port
- Most of the remaining fields will already be populated since we already indicated that we are adding a Remote Racking Device.
- Serial Address is the only other field you should need to enter information.
- Click “Done”



- On “Add Device” Modal, click “Add” button to save the configuration.

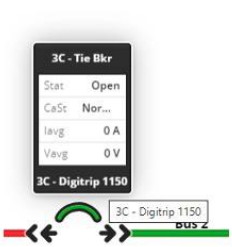


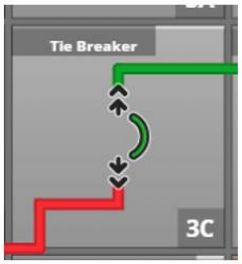
Now you should see that the system has added both the protective relay, that shows the state of the breaker, as well as the Remote Racking symbol.



30.6 Changing Names (Compartments, Symbols & Devices)

- This section will cover how to change names for Compartments, Symbols and Devices on the **One-line view** (below left), **Elevation view** (below middle) or **Device Tree** (below right).

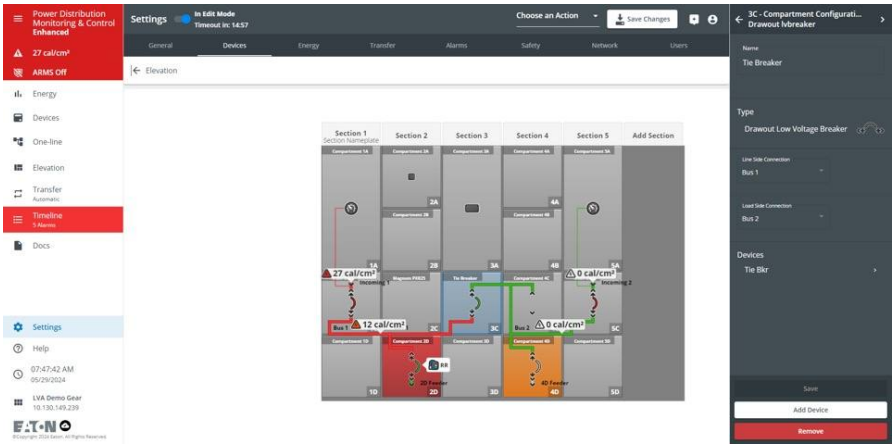




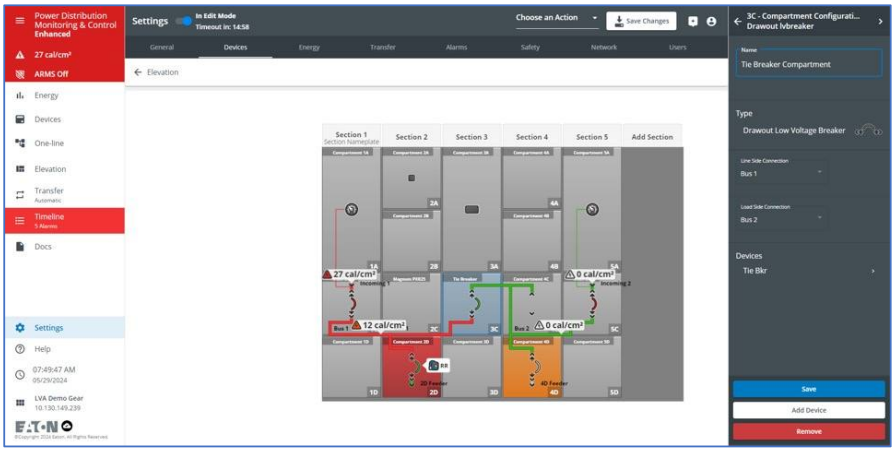


Compartment Name

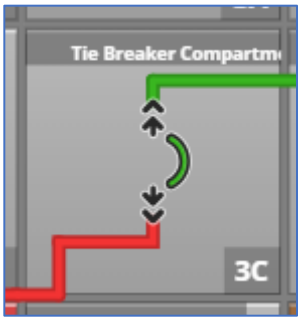
Select a compartment to reveal details in a side panel. In the example below we are selecting compartment **Tie Breaker** via the Elevation view.



Next, we can edit the entry in the **Name** field. In the example below we have updated the name to **Tie Breaker Compartment**.

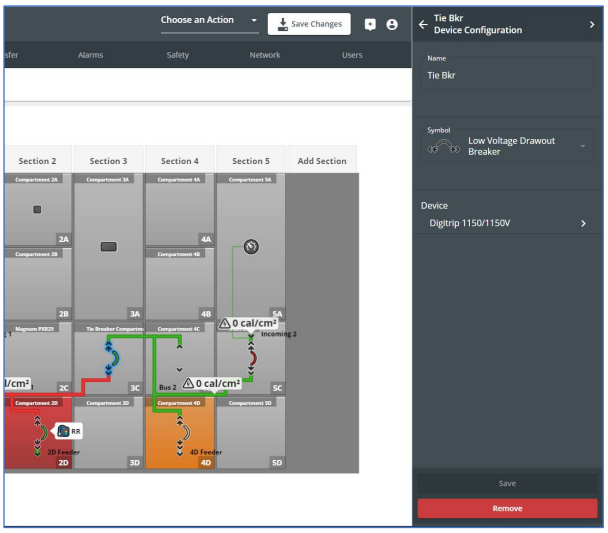


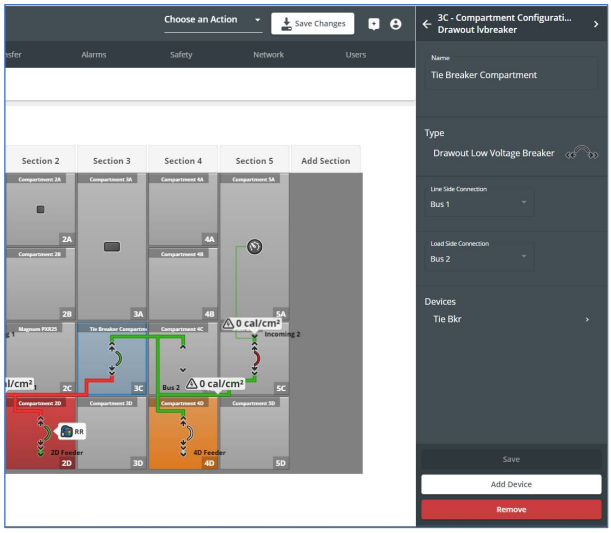
Selecting **Save** will update the compartment name.



Symbol Names

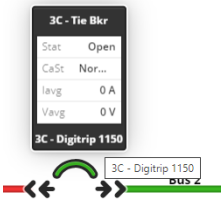
If you selected only the symbol in 3C, you would get a side panel that allows you change the symbol's name. In this example (on the left), the name is Tie Bkr. Changes to the Symbol Name will also show up on the Compartment side panel (on the right).



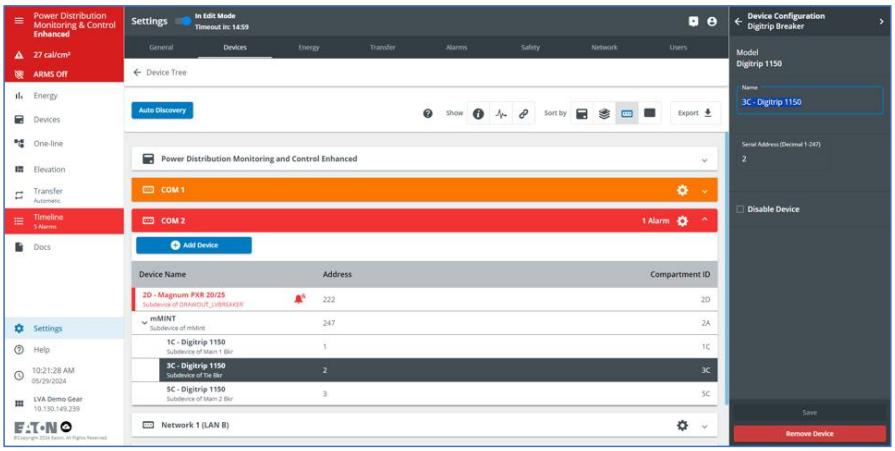


Device Names

On the One-line view, device names are displayed at the bottom of the widget. If you need to update a device name here, navigate to the Device Tree under **Settings > Devices > Device Tree**.

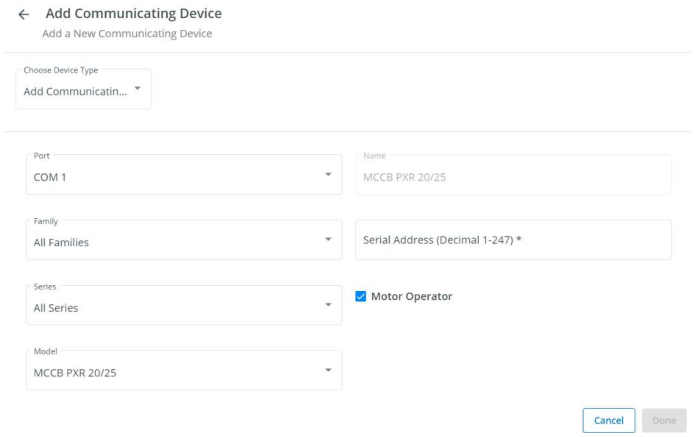


- Use the **Sort By** filter at the top right to select Device Model.
- Then scroll down to locate the device you want to edit.
- Select it to reveal a side panel where you can edit the name.

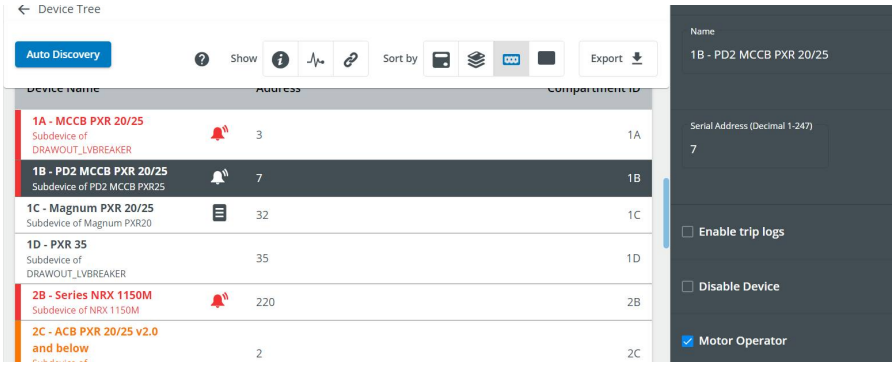


30.7 Motor Operator for Power Defense Molded Case Circuit Breaker

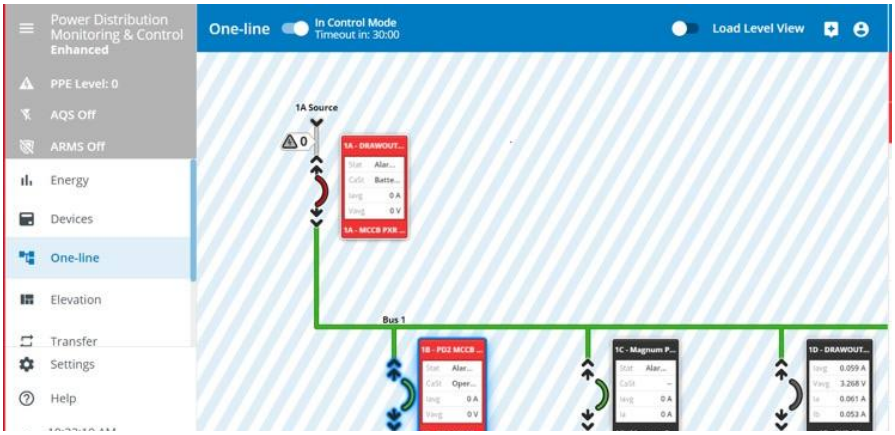
- MCCB remote operator allows the user to add ability of Close breaker operation to the MCCB trip unit.



- Pre-requisite: Remote Motor operator should be physically connected to the PXR MCCB trip unit breaker.
- To enable MCCB Remote operator from PDMC user needs to select the Motor operator checkobx when adding the PXR MCCB device or user can also select the MCCB.
- MCCB remote operator can also be enabled to an already added device from **Settings > Devices > Device Tree** device sidebar.

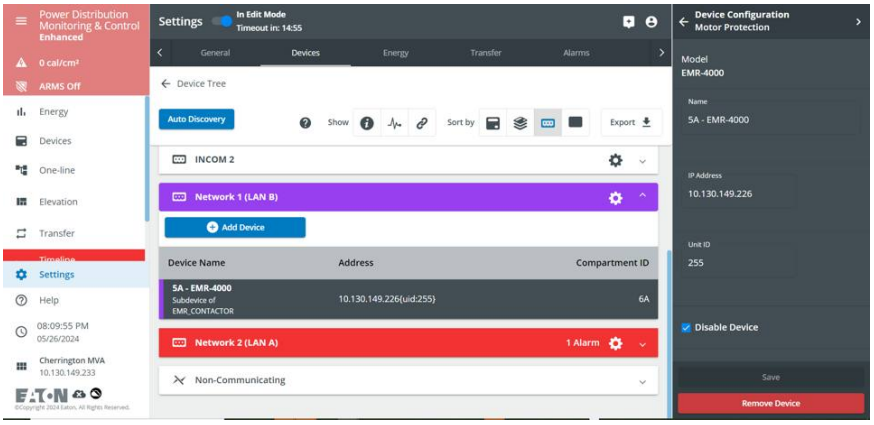


- When MCCB Remote operator checkbox is checked, MCCB will display and support Close Breaker Command operation else Close breaker command won't be available for PXR MCCB.

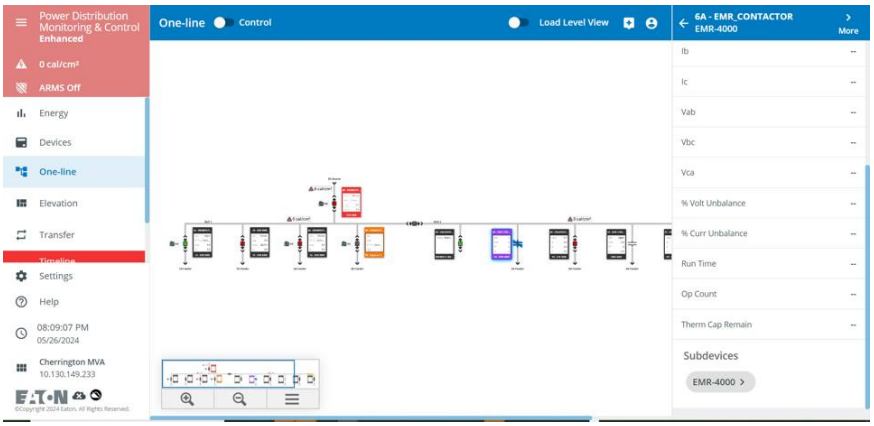


30.8 Disable Device

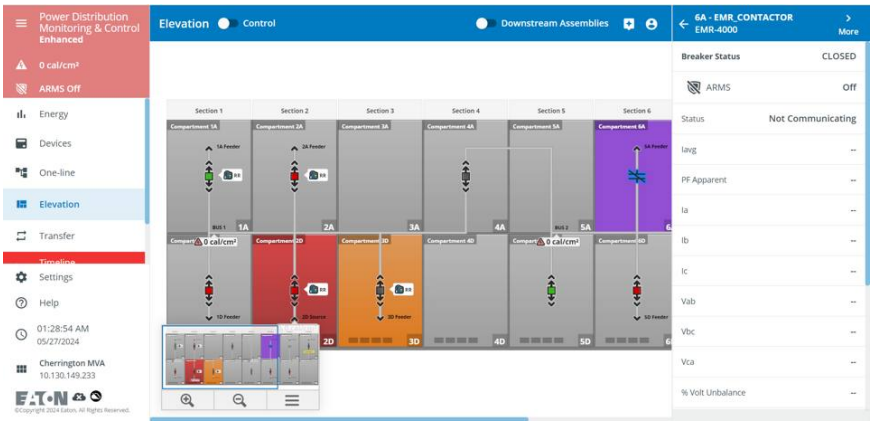
- To disable a device, navigate to **Settings > Device > Device Tree** and enable Edit mode.
- Select a device which needs to be disabled and device configuration sidebar will be opened.
- Select Disable Device checkbox from the device configuration sidebar.
- Purple color stripe will be displayed on the disabled device and port header will also be highlighted in purple.



- On One-line screen, widget color will be purple for the disabled device. On Sidebar, channel values will be shown as "—".



- On Elevation screen, compartment of the disabled device will be purple. On Sidebar, channel values will be shown as "—".

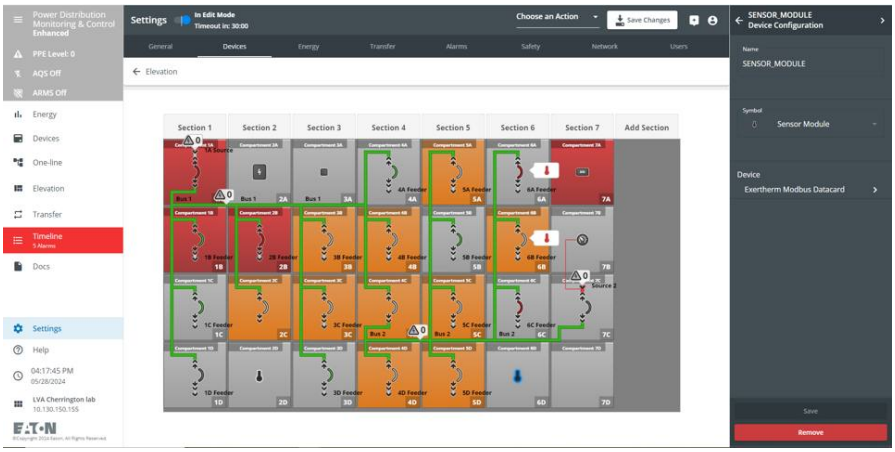


30.9 Assign Exertherm sensors to breaker

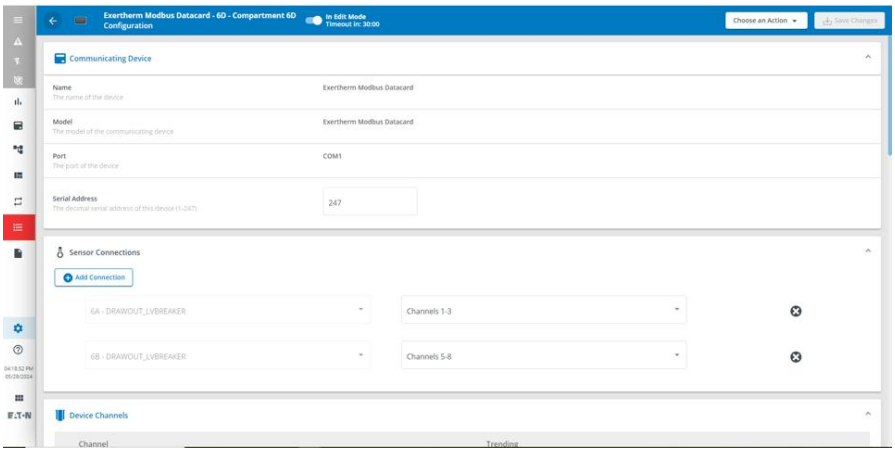
PDMC provides sensor integration for enhanced monitoring of switchgear environmental conditions.

There are two different sensors as below. PDMC supports both of these sensors.

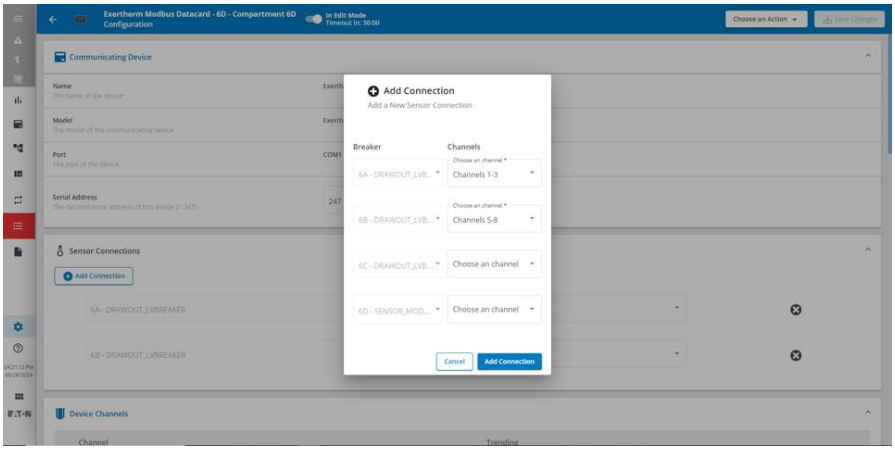
1. Exertherm modbus datacard(LVA) – It has 4 sensors
2. Exertherm MCC datacard (LVMCC) – It has 6 sensors - Incoming & Outgoing connections on Motor control center bucket.
3. Sensor and the breaker should be present in the same section.
4. On **Settings > Devices > Elevation**, select a compartment and add Exertherm Datacard (Refer #8.3 – Adding Devices)
5. Once sensor has been added, click on sensor symbol and sidebar will be opened.
6. Click more arrow under Device in the sidebar.



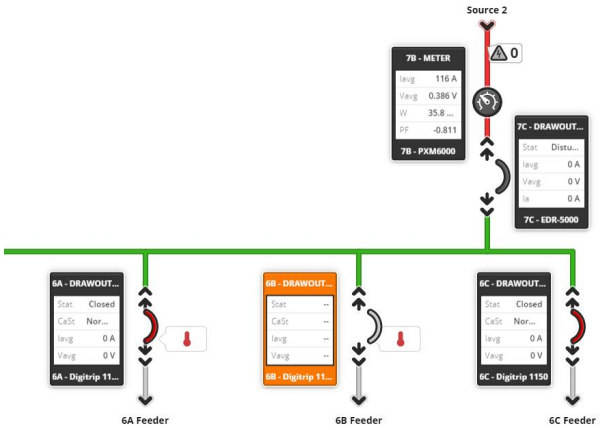
7. On Compartment configuration stack, under Sensor Connection collapse, click on "Add Connections".



8. Click on “Add Connection” button and it will open “Add Connection modal. Select Exertherm channels to be assigned to each breaker. Sensor symbol will be displayed for the respective breaker on One-line and Elevation view.




Section 6
Compartment 6A
6A Feeder
6A
Feeder
5A
Compartment 6B
6B Feeder
6B
Feeder
5B
Compartment 6C
6C Feeder
6C
Feeder
5C
Bus 2
Compartment 6D
Feeder
5D
6D

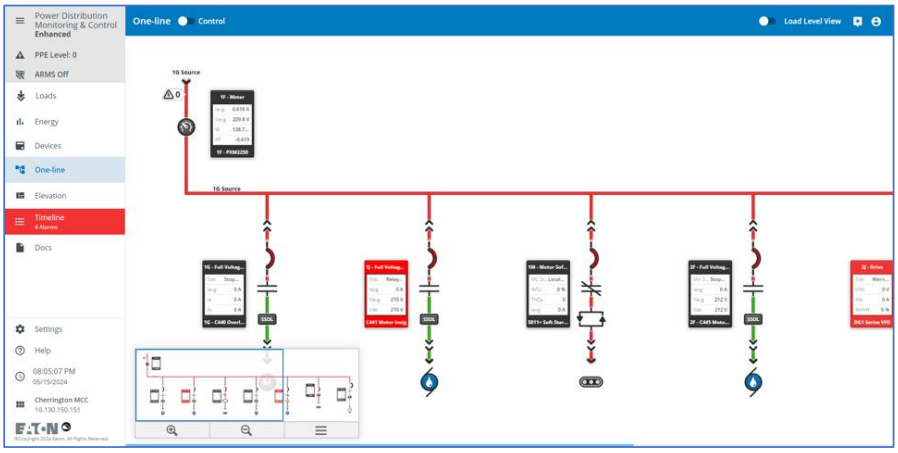

Source 2
7B - METER
7B - PXM6000
7C - DRAWOUT...
7C - EDR-5000
6A - DRAWOUT...
6A - Digitrip 11...
6B - DRAWOUT...
6B - Digitrip 11...
6C - DRAWOUT...
6C - Digitrip 1150
6A Feeder
6B Feeder
6C Feeder

31. Low Voltage Motor Control Centers

The PDMC PXG1000 for Motor Control Centers has many similarities and a few differences from the PDMC PXG1000 for medium and low voltage switchgear. In the typical MCC section the vertical bus connects to all the MCC devices. Except for the one-line, elevation and device configuration, the rest of the settings are the same as above.

31.1 One-line View

- The MCC One-line view is a graphical representation of the power flow to the motor protection and control devices. It shows all the configured devices in the MCC, including motor running status.



31.2 Elevation View

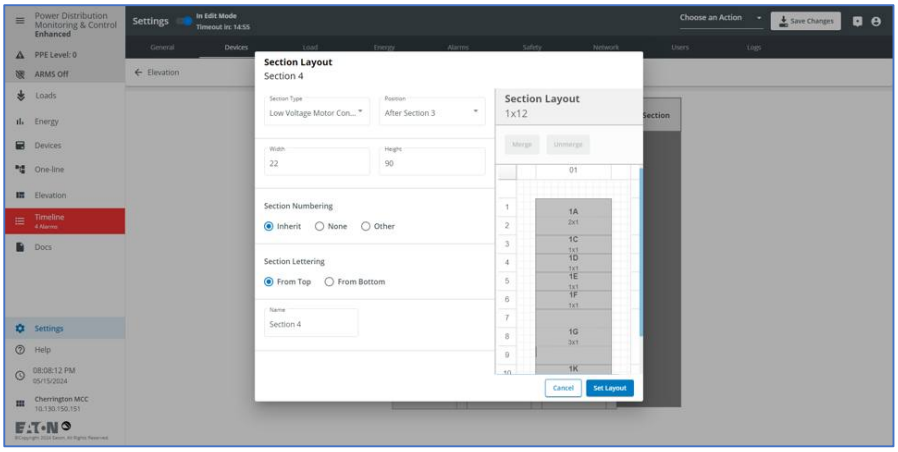
- The Elevation view shows the front view of the switchgear lineup with the status of the MCC devices.
- The compartment Nameplate is hidden by default, but can be configured to be shown.



31.3 MCC Device Configuration

STEP 1: SECTION CONFIGURATION

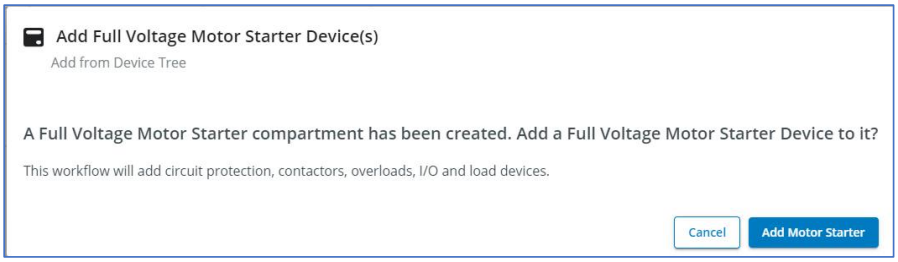
- When creating a lineup manually, click on the 'Edit' mode slider to view the 'Add section' button for adding new sections and compartments.
- Click 'Add Section' and select the desired section type, position, section numbering, section lettering and compartment layout from the Section Layout modal
- A new 'Low Voltage Motor Control' section type has been introduced for Motor Control Centers.



- When the section type is selected as 'Low Voltage Motor Control', the user can customize the compartment layout using the merge/unmerge feature in the Section Layout modal.
- The user can add custom naming of all compartments in a section, using the 'Other' option in 'Section Numbering'.

STEP 2: CONFIGURE MCC DEVICE

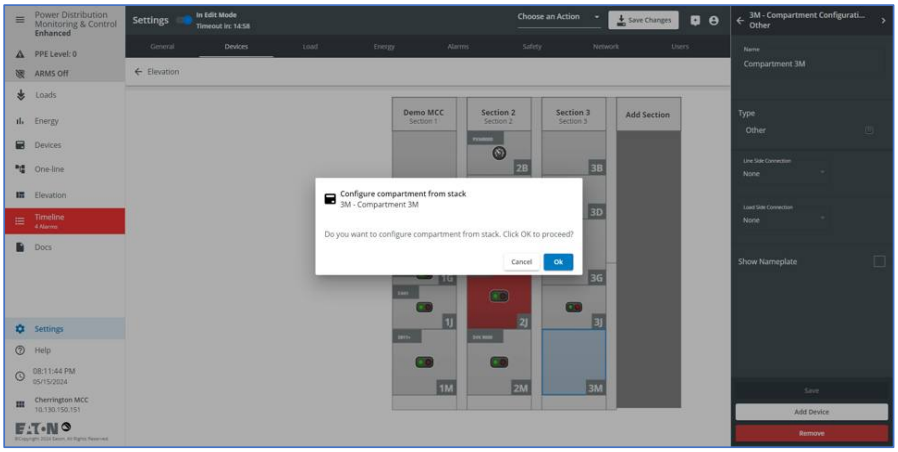
- In Edit mode, on click of the 'More' button in the compartment configuration stack, one can configure additional details like compartment type, static information related to compartment, wiring details and nameplate information etc.
- When you select any compartment type under 'Motor Control', the device wizard gets launched.



- By navigating through this wizard, the user can configure motor control center buckets.

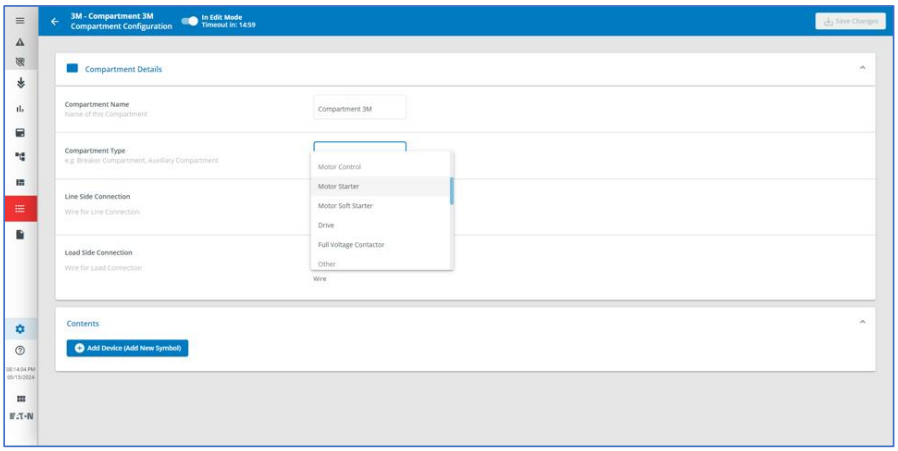
31.4 MCC Wizard Options

User selects a compartment from an existing section. Then selects **Add Device**.

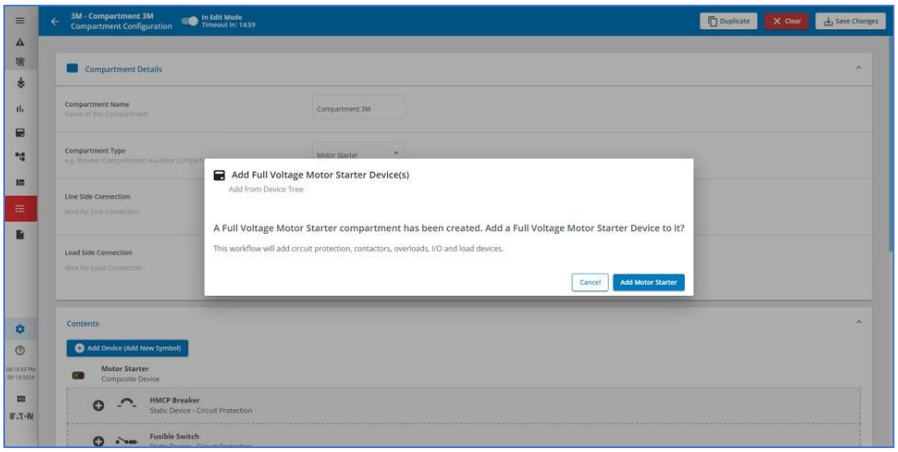


On the next screen you will see a list of **Motor Control** options under **Compartment Type**. These include:

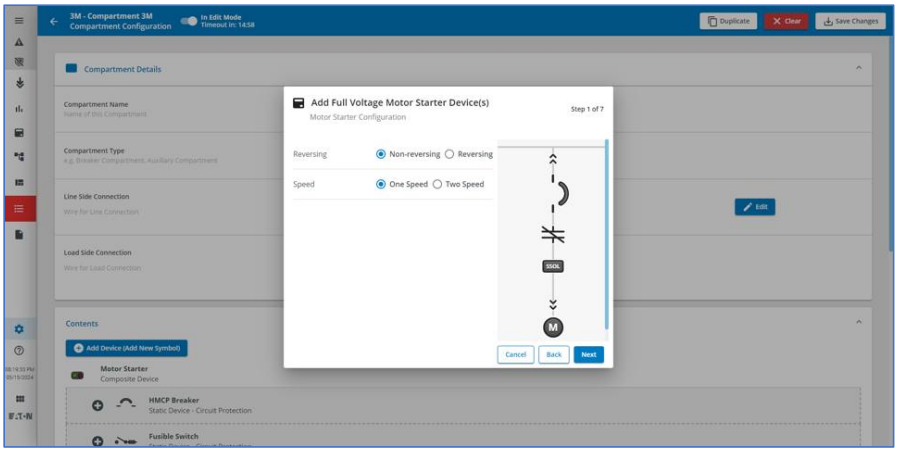
- Motor Starter
- Motor Soft Starter
- Drive
- Full Voltage Contactor



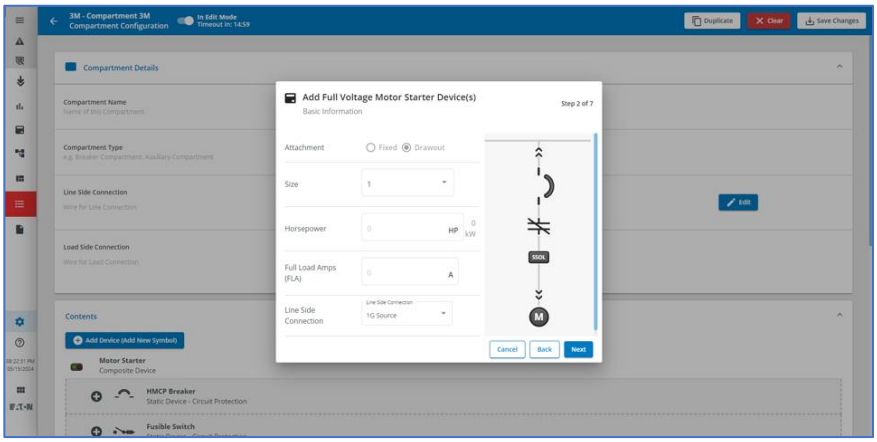
Selecting one of these options will launch the wizard. On the next page we highlight some of the more important parts.



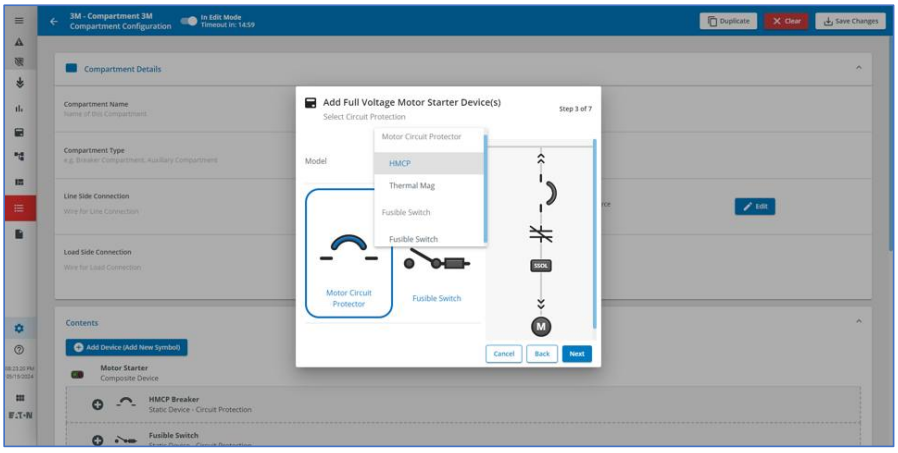
Step 1 provides options for Non-reversing and Reversing and One Speed or Two Speed motors.



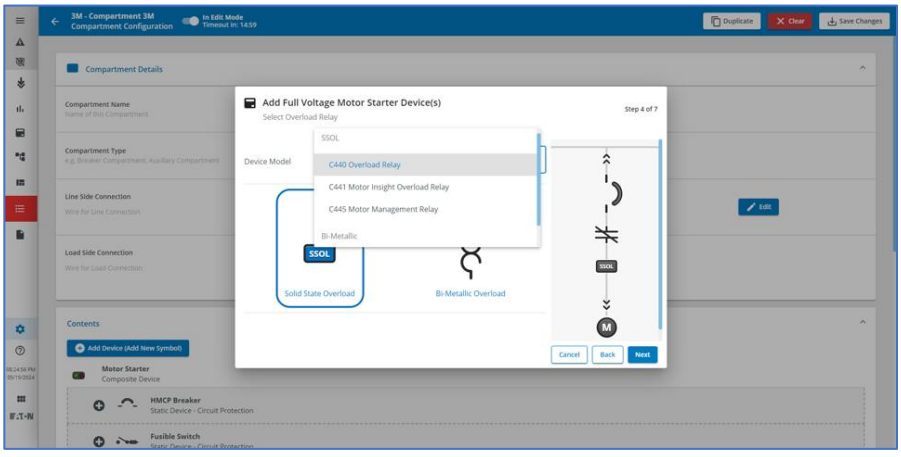
Step 2 While some input is optional, the more details you enter the more information will be displayed for the static fields.



Step 3



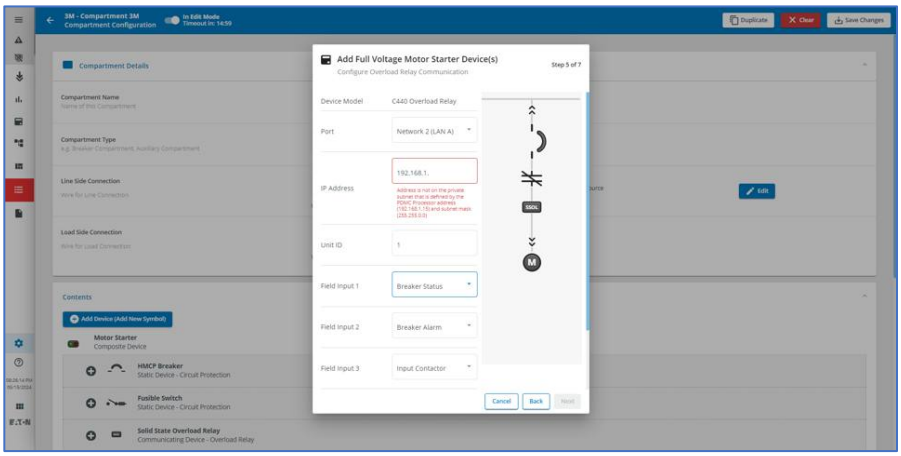
Step 4 allows you to choose the type of Overload, or Bi-Metal, you will be communicating with.



Step 5 provides configuration of the IP addresses and the input functionality. Communication for all MCC Devices is via ethernet. Modbus RTU is not supported.

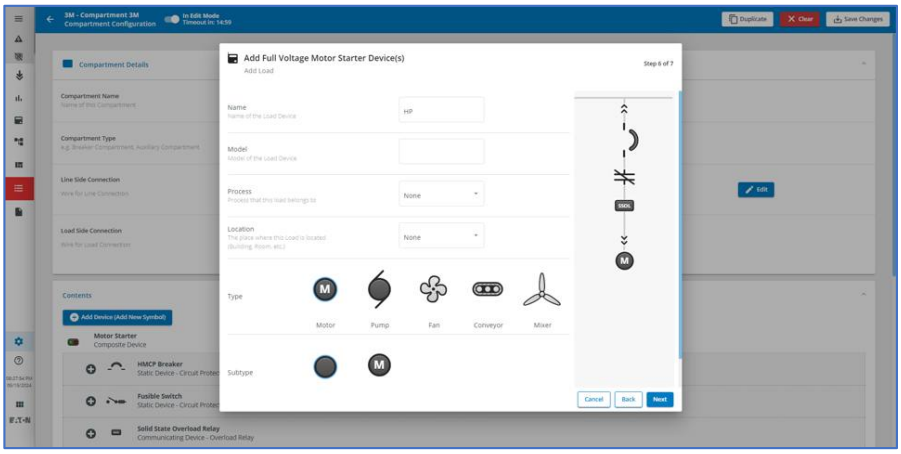
Note: Below are the default settings for the inputs

- Breaker Status will be wired to Field Input 1
- Breaker Alarm will be wired to Field Input 2
- Input Contactor will be wire to Field Input 3

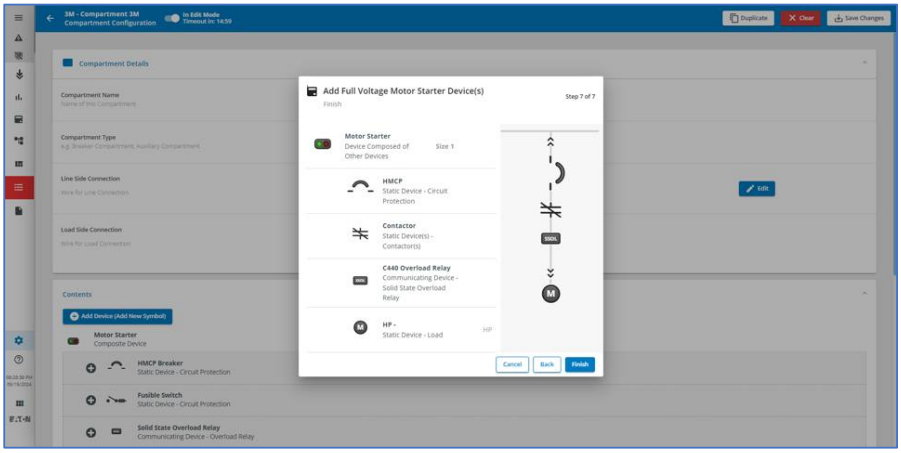


If all the above are provided, the PDMC is then able to provide an animated version of the circuit breaker and contactor seen in the configuration below. If not configured, the symbols for the circuit breaker and the contactor will be displayed as static gray in the open state.

Step 6 you can add a load and process location.

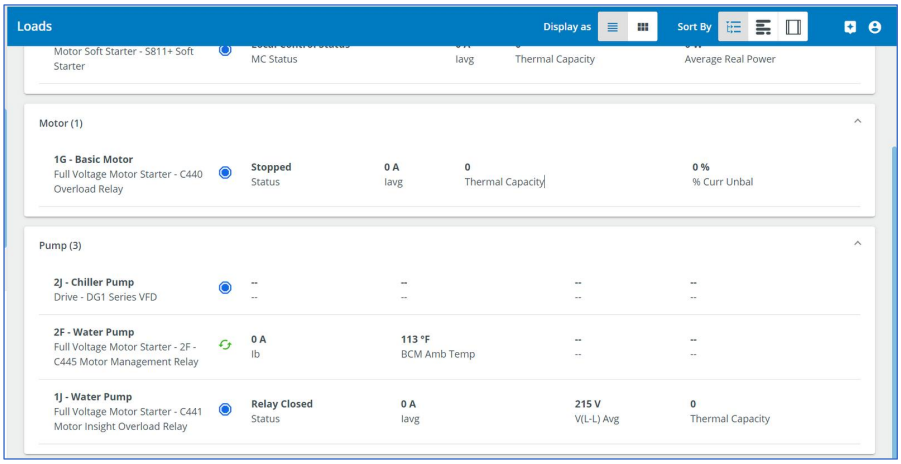


Step 7 allows you to review all your selections.



31.5 Low Voltage Motor Control Centers Load

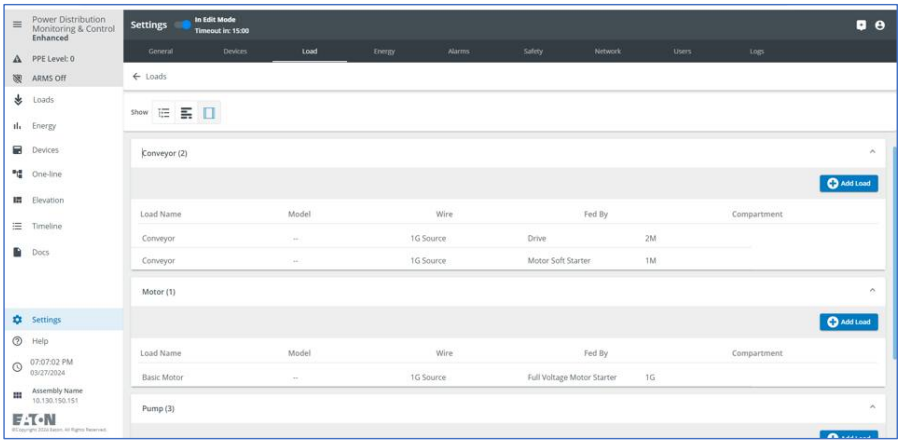
- In LVMCC assembly type, PDMC shows Load grouping and details associated with each load by showing all loads and reading appropriate channel values against each load property.
- Any Load configured will have “Type”, “Sub Type”, “Process (default to None)”, “Location (default to None)” associated with it.

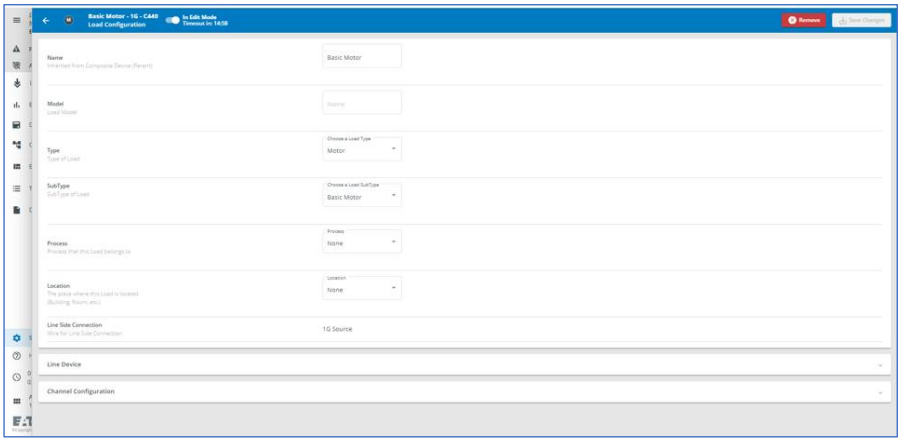


- Action bar holds major two options –
  - Display as (List view / Card view)
  - Sort By (Load Process / Load Type / Location)
- Stop and Running symbol can be seen on the list view.
- When a load type is selected in either Card or List view it provides a device sidebar with the top 16 channels of a device, its ARMs status (in case of breaker), and any active alarms.
- ‘More’ button in the sidebar navigates to device details view with device health, parameters, alarms, trends etc.

31.5 Settings > Load

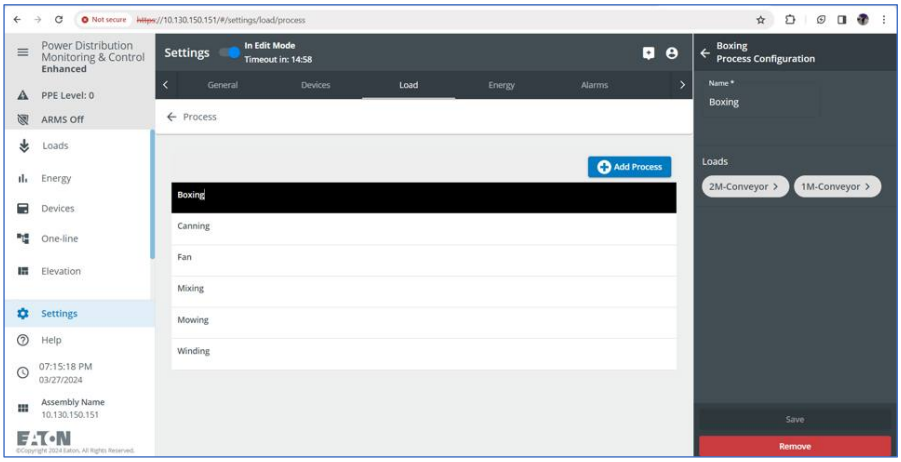
1. **Loads:** List of all the loads already added from **Settings > Elevation** are shown here. If user clicks on any of the load, then load configuration stack gets opened.





- On load configuration stack, user can change Name, Type, SubType, Process, Location.
- Line device connection can also be seen here.
- Channel configuration collapse – User can select up to 4 channels to be displayed for each device.

2. **Load Type:** In this, user can see list of all the load types and sub types added under it for the current assembly. If user wants to add new load type or subtype it can be added from here.
3. **Process:** Predefined process list can be seen on this page. User can add new process or edit the already present process.
   - Click on an item to reveal a sidebar with additional details and ability edit the name of the process or remove it. Also, loads associated with selected process can be seen on the sidebar.



4. **Location:** Predefined location list can be seen on this page. User can add new location or edit the already present location.
   - Click on an item to reveal a sidebar with additional details and ability edit the name of the location or remove it. Also, loads associated with selected location can be seen on the sidebar.

32. USB-INCOM

The PXG1000 USB-INCOM Adapter is an optional accessory used with the PDMC PXG1000 Hardware. Two INCOM ports are available to connect supported devices.

Step 1: Set up USB-INCOM connection

- Power-up the USB-INCOM Adapter using a 24V power source.
- Connect one end of a USB cable (Type A) with PDMC PXG1000 and the other end (Type B -mini) to the USB-INCOM Adapter.
- Connect the downstream devices with USB-INCOM Adapter using 3-pin connector. Each INCOM port support up to 50 devices.



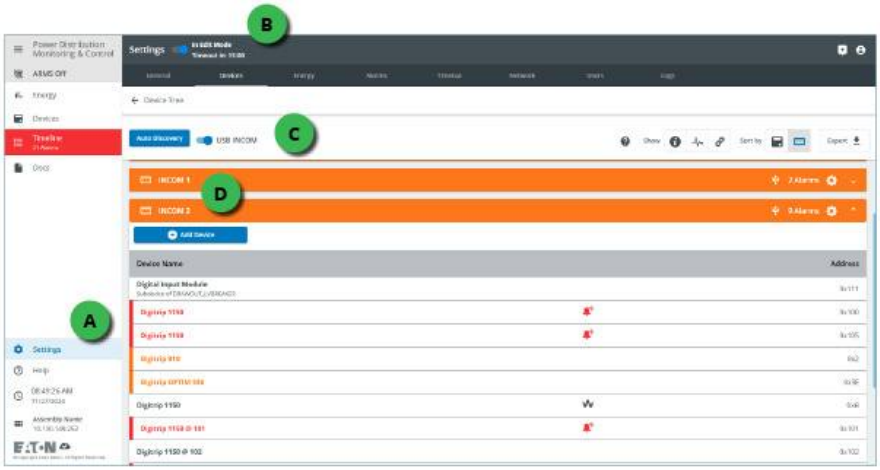
Step 2: Enable USB-INCOM

E. On the PDMC application, Navigate to Settings > Devices > Device Tree

F. Select Edit mode.

G. Enable “USB INCOM” toggle.

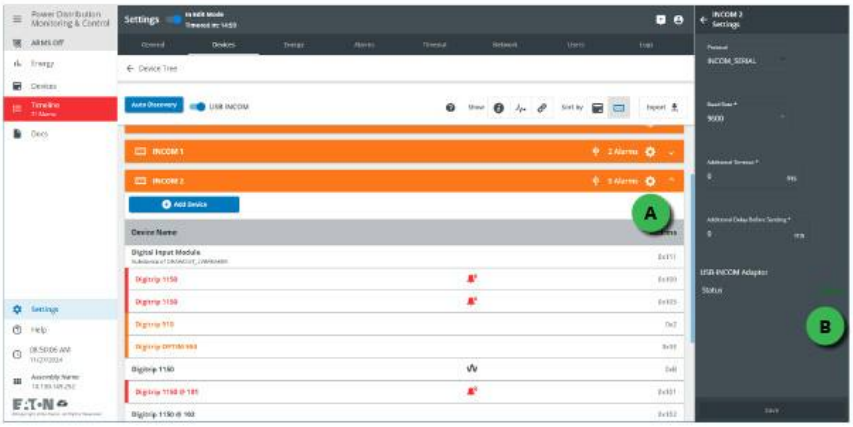
H. Sections for INCOM 1 and INCOM 2 will then be displayed on device tree screen.



Step 3: Verify status

A. To verify USB-INCOM Adapter communication status, click on the gear icon on the collapse section header. This will open an INCOM Settings sidebar.

B. Confirm the USB-INCOM Adapter status is “Online.”



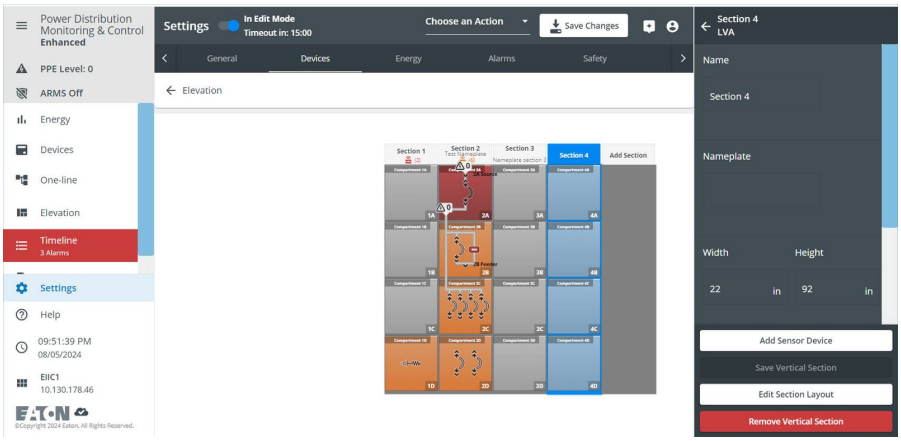
33. INFRASENSING

Infrasensing Base Unit and sensors are devices used to monitor health of the switchgear, which communicates over MODBUS TCP protocol with PDMC PXG1K. PDMC supports five Infrasensing devices:

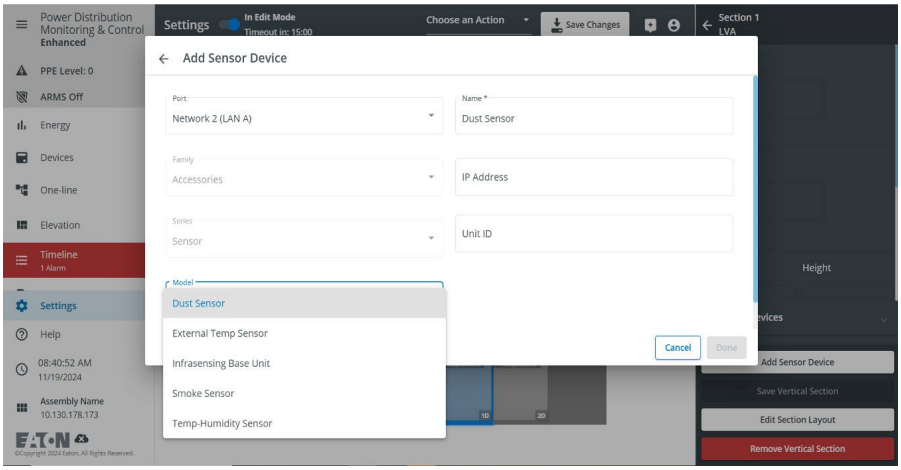
1. Infrasensing Base Unit
2. Dust Sensor
3. External Temperature Sensor
4. Smoke Sensor
5. Temperature-Humidity Sensor.

33.1 Adding Infrasensing devices

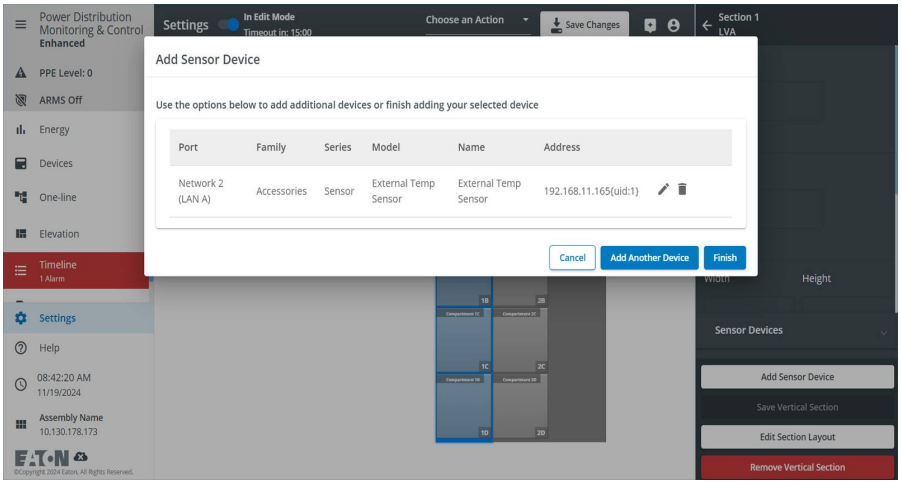
- On PDMC application, Navigate to **Settings > Devices > Elevation**.
- Select Edit mode and select the section where you want to the add the device.



- Click on Add Sensor Device button in the sidebar, which launches a modal for “Add Sensor Device”.
- Enter Device details.



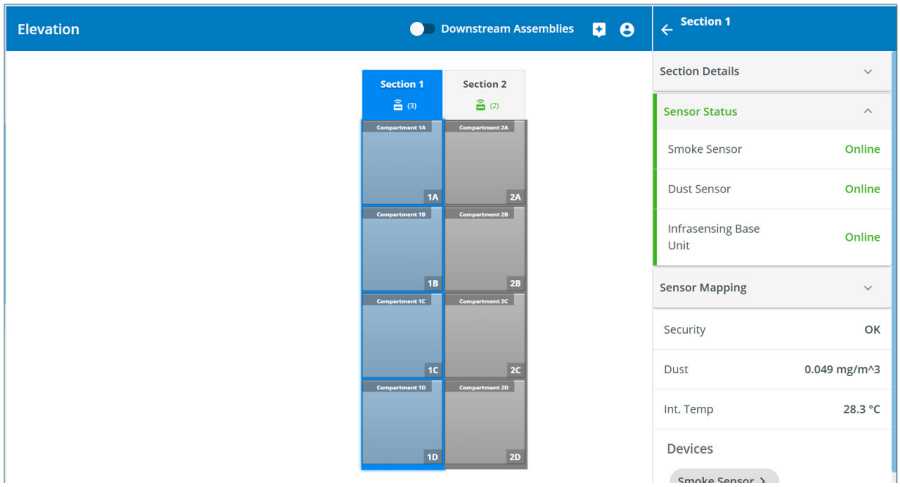
- There is a provision to add another Infrasensing Device in the same instance.
- Click on “Add Another Device”.
- Once all the Infrasensing devices are added, click on Finish button.



33.2 Verify status of Infrasensing devices

After adding the devices, Elevation view will display the Infrasensing device symbol under section header, along with a count of devices added in the respective section.





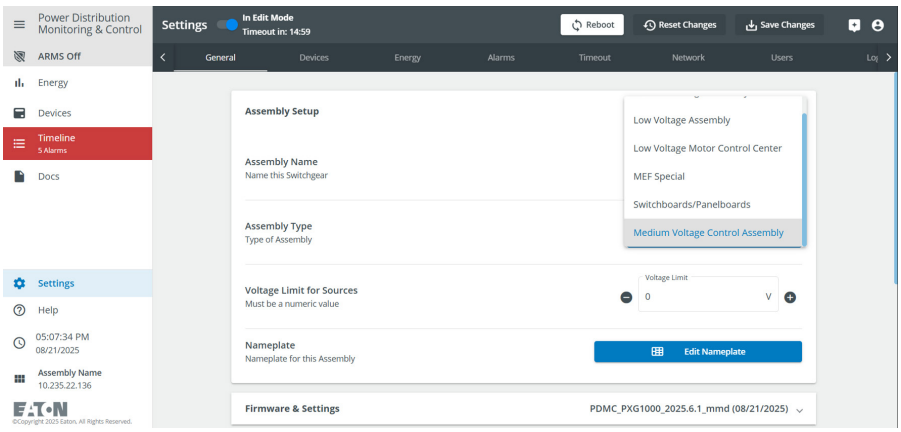
On the Elevation sidebar, device status can be seen for each of the Infrasensing device:

- Online (Green) – indicates that the device is communicating
- Not communicating (Orange) – indicates that the device is not communicating
- Alarm (Red) – indicates that device is communicating and an alarm has been triggered on the device
- Symbol and device count in the section header will be changed to the respective color, based on the overall status of the devices under that section.

34. Medium Voltage Control Assembly (MVCA)

Asset Monitoring for Ampgard Phase 01 uses Exertherm sensors integrated with PDMC PXG1K to monitor switchgear health. It continuously tracks the temperature of key joints and sends targeted alarms based on operating conditions, helping ensure optimal performance.

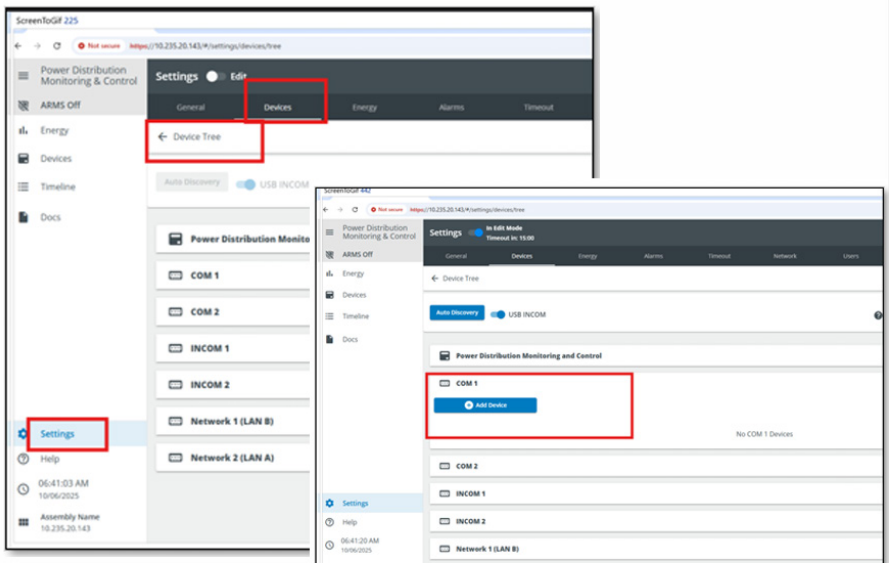
1. **Select MVCA Assembly Type:**
   a. Navigate to **Settings** and select the **General** tab.
   b. Then choose **Medium Voltage Control Assembly** for Assembly Type.



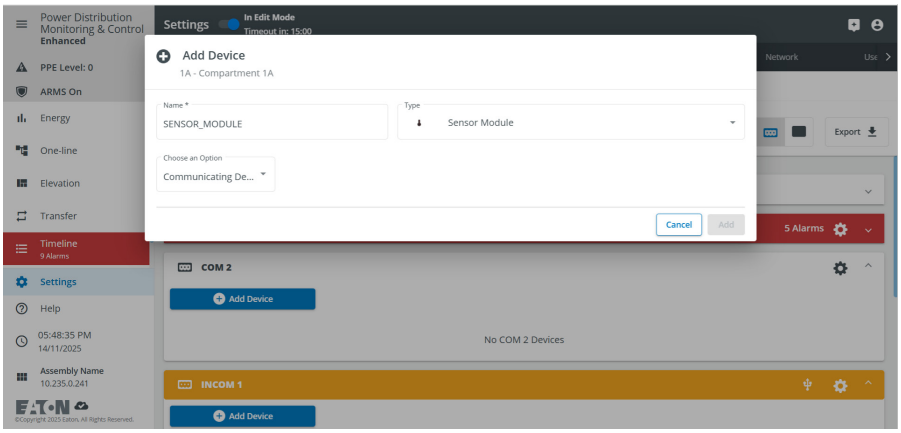
2. **Adding Exertherm MMD-MVCA device:**

Follow these steps to successfully configure the **Exertherm MMD-MVCA**.

   a. Log in to your PDMC.
   b. Navigate to **Settings** and select the **Device** tab.
   c. Enable **Edit Mode** at the top left.
   d. Select **Device Tree** and choose where to add a device (ex. COM 1).
   e. Click on **Add Device**.



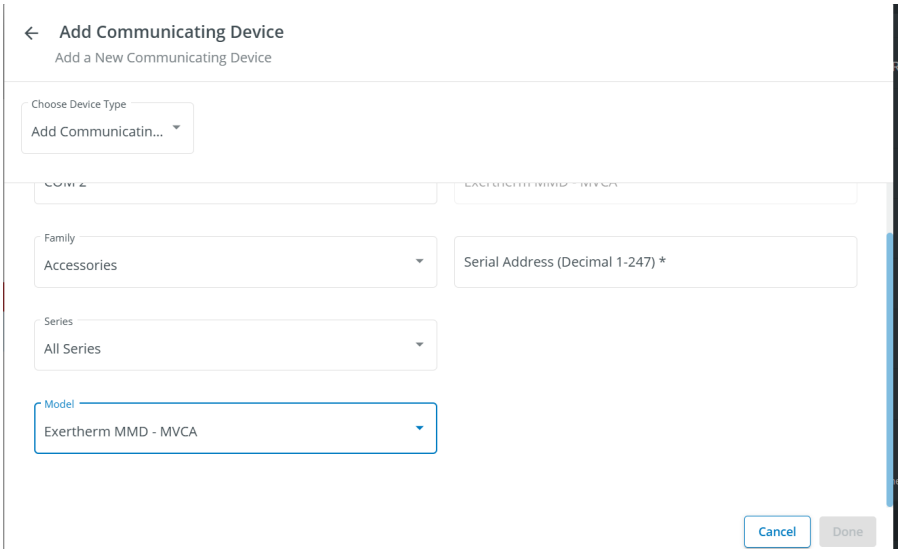
Settings > Device Tab > Device Tree



Add Device Modal

f. Enter device **Name** and select device **Type**.

g. Choose **Communicating device** and select **Add**.

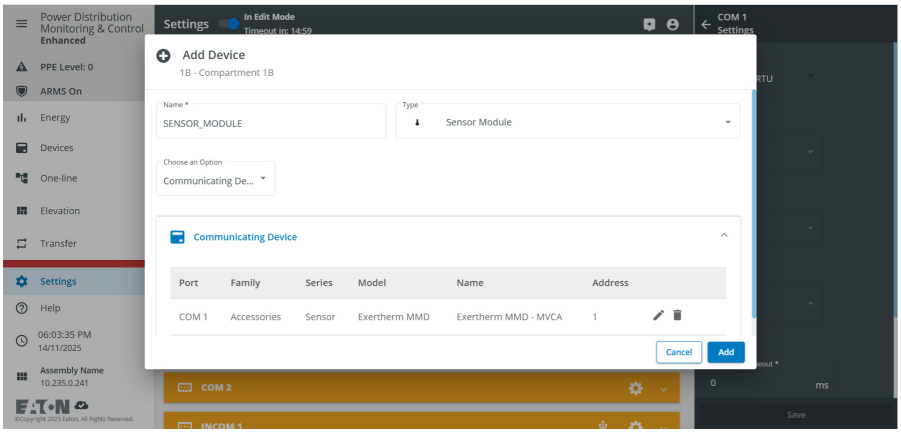


Add Communicating Device Modal

h. Choose **Accessories** from family, **Sensor** from **Series**, and **Exertherm MMD-MVCA** from the **Model**.

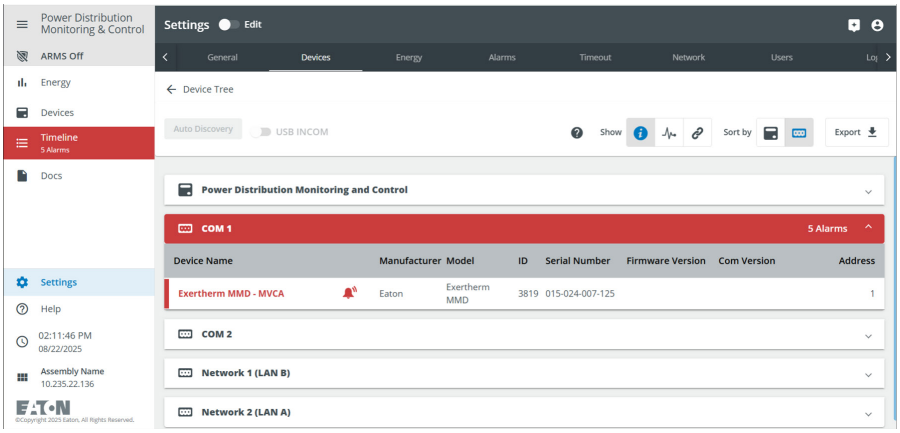
i. Enter the **Serial Address** and confirm the physical address of the MMD prior to entering any data.

j. Select **Done**.



Add Device Review Modal

k. Review selections and click **Add** when ready.



Settings > Device Tree

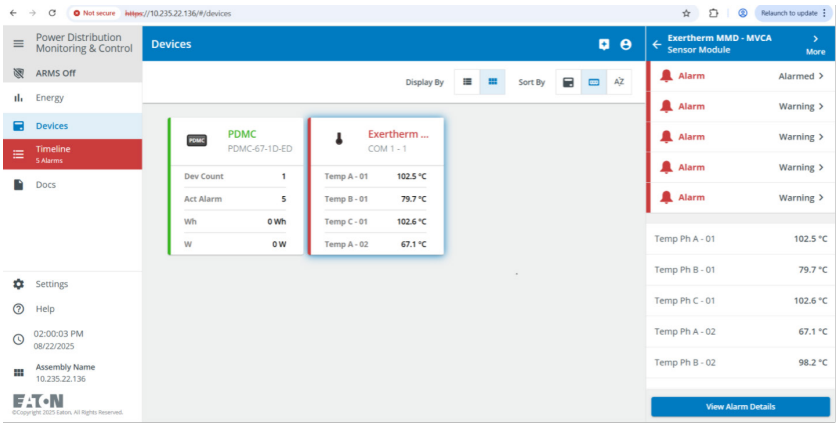
l. Confirm the device has been added to the selected port.

3. **Trigger Alarms on Thresholds:**

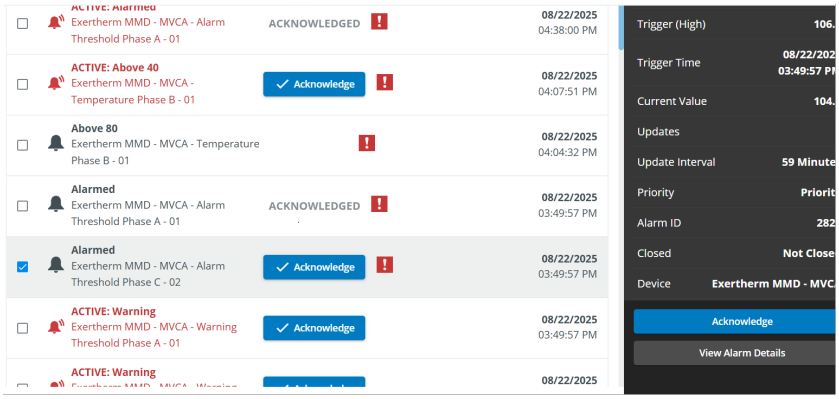
Alarms must be triggered if the temperature of Exertherm sensors crosses thresholds as below:

- 90 < T <= 105 C - Trigger normal alarm with “Warning” message
- T > 105 C - Trigger priority alarm with “Alarm” message

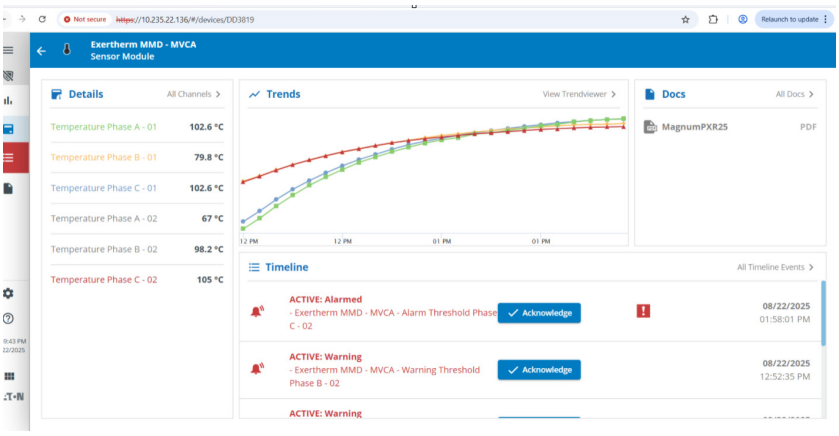
Note: Users can modify the predefined warning and alarm thresholds from Alarms tab under Settings.



Devices



Timeline



Device Details Stack

35. Frequently Asked Questions (FAQ)

This section provides answers to frequently asked questions about the PDMC. Brief answers are provided here with links to manual sections that provides additional information.

Q1. How do I pull data from the PDMC to my BMS or SCADA system?

A1. Refer to **Section 12.2. Modbus TCP Server** and **Section 12.4. BACnet IP Server.**

Q2. How do I upgrade the firmware in the PDMC?

A2. **Settings > General > Import Firmware**; Refer to **Section 10.2. Firmware & Settings**.

Q3. How do I view Trip Log and Waveform data?

A3-1. Select Timeline; select the funnel in the top action bar; sort by Waveform or Trip; see the selected entries at the top of the Timeline.

A3-2. **Settings > Devices > Device Tree**; select the waveform icon to see device waveforms.

A3-3. **Settings > Devices > Device Tree**; select the file icon to see device trip logs.

Q4. How do I access the setpoints on my Trip Units?

A4. Refer to **Section 11.4. Settings -> Devices -> Setpoint Files**.

Q5. How do I configure the bus to show red / green for energized / de-energized?

A5. Refer to **Section 11.1. Settings -> Devices -> Elevation**.

Q6. Why are the compartment and device widgets red / orange / purple?

A6. The normal compartment background color is light gray; the normal widget outline is black.

- **Red:** alarm condition
- **Orange**: not communicating
- **Purple**: disabled (check to be sure the network is configured for communication)

Q7. How do I adjust the time delays for the Transfer scheme?

A7. Enable Edit mode on the Automatic Transfer screen as shown in **Section 27 – Transfer Scheme App.**

Q8. The customer forgot their password. How can I get access to the PDMC?

A8. Follow the instructions in **Sections 23 & 24. Factory Reset**.

Q9. Is it possible to navigate among multiple PDMC processors with a single HMI?

A9. Yes. Dashboard Campus provides the best solution, but it is also possible through the PDMC user interface. Refer to **Section 26.3. Assembly Reveal Panel.**

Q10. How is the incident energy value determined?

A10. The incident energy values from the arc flash study are entered on the **Settings > Safety** tab as described in **Section 13. Settings > Safety.**

Q11. How do I add or upgrade licenses on PDMC?

A11. **Settings > General > Import Firmware**; Refer to **Section 10.4. Licenses for Additional Features.**

Q12.Can I use one PXG1000 for multiple line-ups?

A12. Not recommended. PDMC is designed to support a single assembly.

Q13. How many INCOM, RS-485 and network devices does PDMC support?

A13. Device response time is impacted by the number of devices connected. Refer to **Section 3. Configuring Device Communications.**

Q14.Does PDMC support non-Eaton devices?

A14. Yes. You can find supported devices list here: www.eaton.com/pdmonitoringcontrol

Under Technical data sheets

Q15. Where do I go for updated firmware for PDMC?

A15. To obtain the latest released firmware www.eaton.com/pdmonitoringcontrol, select the "Software, firmware, and applications" button.

Q16.Does PDMC support multiple networks/subnets?

A16. Yes. LAN A port can be configured to support a private subnet and LAN B port mask can be set to communicate with devices on multiple subnets.

Q17.Who do I contact for troubleshooting support for PDMC?

A17. Please contact the Customer Integrity Team (CIT) by calling 1-844-435-8982 or send an email to MRSupport@eaton.com.

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